Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications

A Dissertation Presented

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This thesis is dedicated to my parents, Mrs. Bratati Ghoshal and Mr. Tarun Kanti Ghoshal.
For their endless love, support, and encouragement.

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List of Abbreviations

3GPP	3rd Generation Partnership Project
AR	Augmented Reality
ARFCN	Absolute Radio Frequency Channel Number
AWS	Amazon Web Services
BLER	Block Error Rate
BS	Base Station
BW	Bandwidth
CA	Carrier Aggregation
CAV	Connected Autonomous Vehicle
CC	Component Carrier
CDF	Cumulative Distribution Function
CN	Core Network
CQI	Channel Quality Indicator
CSI	Channel State Information
DC	Dual Connectivity
DCCH	Dedicated Control Channel
DCI	Downlink Control Information
EC2	Amazon Elastic Compute Cloud
eNB	Evolved Node B (LTE base station)
EPC	Evolved Packet Core
FDD	Frequency Division Duplexing

FPS Frames Per Second

FR3 Frequency Range 3

gNB Next-Generation Node B (5G base station)

HO Handover

ICMP Internet Control Message Protocol

KPI Key Performance Indicator

LEO Low-Earth-Orbit

LTE Long-Term Evolution

MCS Modulation and Coding Scheme

MEO Medium-Earth-Orbit

MIMO Multiple-Input Multiple-Output

mmWave Millimeter Wave

NR New Radio

NSA Non-Standalone

NTN Non-Terrestrial Network

OAI Open Air Interface

P5G Private 5G

PCI Physical Cell Identifier

PDCCH Physical Downlink Control Channel

PDN Packet Data Network

PDSCH Physical Downlink Shared Channel

PDU Protocol Data Unit

PHY Physical Layer

PRACH Physical Random Access Channel

PUCCH Physical Uplink Control Channel

PUSCH Physical Uplink Shared Channel

QoE Quality of Experience

RACH Random Access Channel

RAN Radio Access Network

RAT Radio Access Technology

RMSE Root Mean Square Error

RRC Radio Resource Control

RSRP Reference Signal Received Power

RSRQ Reference Signal Received Quality

RSSI Received Signal Strength Indicator

RTT Round-Trip Time

SA Standalone

SCG Secondary Cell Group

SCS Subcarrier Spacing

SINR Signal-to-Interference-plus-Noise Ratio

SN Secondary Node

TBS Transport Block Size

TDD Time Division Duplexing

TN Terrestrial Network

TTT Time-to-Trigger

UE User Equipment

Wi-Fi Wireless Fidelity

XR Extended Reality

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Curriculum Vitae

Education

Northeastern University Boston, USA
Doctor of Philosophy (Ph.D.), Computer Engineering *01/2020 - 08/2026*

Northeastern University Boston, USA
Master of Science (M.S.), Electrical and Computer Engineering *01/2020 - 08/2022*

Experience

Research Assistant 01/2020 - 08/2026
WiNS Lab (Northeastern University, University at Buffalo)

- Worked on understanding the capabilities and constraints of operational 5G/xG networks and their impact on the performance of latency-critical applications (LCAs) such as mobile augmented reality, connected autonomous vehicles, etc. I also demystify different operator policies of cellular networks with respect to handovers, beam management, and resource allocation. My current work also involves system prototyping for 5G/xG networks.
- Worked on experimental evaluation of commercial Wi-Fi 6/6E devices in terms of their performance and interference characteristics.
- Worked on 60 GHz networks and did experimental evaluation of performance and energy consumption of commercially available 802.11ad-enabled smartphones.
- Was involved in porting Multi-Path TCP on commercially available mobile phones to simultaneously use 60 GHz and 5 GHz networks and understand their performance, energy consumption, and challenges associated with it.

Research Associate Intern 05/2025 - 04/2026
Hewlett Packard Labs *Milpitas, CA*

- Worked with the Networking and Distributed Systems Lab (NDSL) to deploy, evaluate and compare Wi-Fi and P5G in terms of their range, coverage, network performance (throughput, latency, etc.), and interaction with latency-critical applications.

Research Intern 05/2024 - 09/2024
Futurewei Technologies *Addison, TX*

- Worked with the wireless standards and research team on the intersection of AI/ML, sensing, and energy-efficiency for next-generation cellular networks.

R&I Intern 05/2023 - 09/2023
InterDigital *Los Altos, CA*

- Worked on 5G PHY layer optimization to reduce end-to-end delay for emerging latency-critical applications like XR, CAV, etc. Implemented the solution in a 3GPP compliant OAI testbed for performance evaluation.

Research Publications

(* indicates co-primary authors and underlining denotes presented at conference/workshop)

Refereed Conference Publications

1. **Too Many Reports, Too Few Handovers: A Large-Scale Study of Configuration Inefficiencies in Operational 5G Networks.** *(Thesis Chapter 5)*
Moinak Ghoshal, Imran Khan, Phuc Dinh, Z. Jonny Kong, Omar Basit, Sizhe Wang, Yufei Feng, Y. Charlie Hu, Dimitrios Koutsonikolas.
Under submission to ACM CoNEXT 2026.

2. **Disentangling the Throughput Contributions of MIMO and Carrier Aggregation in 5G Networks.**
Yufei Feng, Phuc Dinh, **Moinak Ghoshal**, Omar Basit, Sizhe Wang, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the Passive and Active Measurement Conference (PAM 2026), Virtual Conference, March 23-25, 2026. (acceptance rate 31.5%).

3. **Exploring the 5G Digital Divide in the Non-Contiguous US: LEO Satellites to the Rescue?**
Sizhe Wang, **Moinak Ghoshal**, Yufei Feng, Imran Khan, Phuc Dinh, Omar Basit, Zhekun Yu, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the ACM International Conference on Measurements and Modeling of Computer Systems (SIGMETRICS 2026), Ann Arbor, Michigan, USA, June 8-12, 2026. (Winter deadline, acceptance rate 20.7%)

4. **A First Large-scale Study of Operational 5G Standalone Networks.** *(Best Community Paper Runner-Up) (Thesis Chapter 4)*
Moinak Ghoshal, Omar Basit, Sizhe Wang, Phuc Dinh, Imran Khan, Yufei Feng, Zhekun Yu, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the 21st ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT 2025), Hong Kong, December 1-4, 2025.

5. **Replication: Performance of Cellular Networks on the Wheels** *(Thesis Chapter 3)*
Moinak Ghoshal, Omar Basit, Imran Khan, Z. Jonny Kong, Sizhe Wang, Yufei Feng, Phuc Dinh, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the 25th ACM Internet Measurement Conference (IMC 2025), Madison, WI, USA, October 28-31, 2025. (Acceptance rate 23.6%).

6. **5G Metamorphosis: A Longitudinal Study of 5G Performance from the Beginning.**
   Imran Khan, Omar Basit, **Moinak Ghoshal**, Y. Charlie Hu, Dimitrios Koutsonikolas.
   In Proceedings of the 25th ACM Internet Measurement Conference (IMC 2025), Madison, WI, USA, October 28-31, 2025. (Acceptance rate 23.6%).

7. **Vivisecting Beam Management in Operational 5G mmWave Networks.**
   Yufei Feng, Phuc Dinh, **Moinak Ghoshal**, Eduardo Baena, Dimitrios Koutsonikolas.
   In Proceedings of the 21st ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT 2025), Hong Kong, December 1-4, 2025. (Winter round acceptance rate 6.6%).

8. **AI/ML-Based Sensing-Assisted Energy-Efficient Communications in Next-Gen Cellular Networks.**
   Moinak Ghoshal, Abbas Kiani, Amanda Xiang, John Kaippallimalil, Tony Saboorian, Rostand A.K. Fezeu, Nirwan Ansari.
   In Proceedings of the 102nd IEEE Vehicular Technology Conference (VTC Fall 2025), Chengdu, China, Oct 19-22, 2025.

9. **mm-NOLOC: mmWave-based Localization for Mobile Networks without 3GPP Location Service.**
   Phuc Dinh, Yufei Feng, Eduardo Baena, Yunmeng Han, Weiming Qi, Zihan Xu, **Moinak Ghoshal**, Pau Closas, Dimitrios Koutsonikolas and Joerg Widmer.
   In Proceedings of the 26th ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc 2025), Houston, TX, October 27-30, 2025. (Acceptance rate 23%).

10. **Root Cause Analysis of Cellular Network Throughput Degradations under Vehicular Mobility.**
    Eduardo Baena, Zihan Xu, **Moinak Ghoshal**, Imran Khan, Phuc Dinh, Z. Jonny Kong, Y. Charlie Hu, and Dimitrios Koutsonikolas.
    In Proceedings of the 22nd IEEE International Conference on Mobile Ad Hoc and Smart Systems (MASS 2025), Chicago, IL, October 6-8, 2025. (Acceptance rate 29.6%).

11. **A Large-Scale Study of the Potential of Multi-Carrier Access in the 5G Era.**
    Fukun Chen*, **Moinak Ghoshal***, Enfu Nan, Imran Khan, Phuc Dinh, Z. Jonny Kong, Y. Charlie Hu, Dimitrios Koutsonikolas.
    In Proceedings of the Passive and Active Measurement Conference (PAM 2025), Virtual Conference, March 10-12, 2025. (acceptance rate 29.8%).

12. **Enabling low-latency applications through UE-assisted proactive PHY frame reconfiguration in next-generation cellular networks.** *(Thesis Chapter 6)*
    Moinak Ghoshal, Subhramoy Mohanti, Dimitrios Koutsonikolas.
    In Proceedings of the 35th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2024), Valencia, Spain, September 2-5, 2024.

13. **A First Look at 160 MHz WiFi 6/6E in Action: Performance and Interference Characterization.**

Moinak Ghoshal*, Shravan Bhoopasamudram Krishna*, Francesco Gringoli, Joerg Widmer, Dimitrios Koutsonikolas.
In Proceedings of the 23rd IFIP/IEEE Networking Conference (NETWORKING 2024), Recent Results, Thessaloniki, Greece, June 3-6, 2024.

14. **How Mature is 5G Deployment? A Cross-Sectional, Year-Long Study of 5G Uplink Performance?**
Imran Khan, **Moinak Ghoshal**, Joana Angjo, Sigrid Dimce, Mushahid Hussain, Paniz Parastar, Yenchia Yu, Claudio Fiandrino, Charalampos Orfanidis, Shivang Aggarwal, Ana C Aguiar, Ozgu Alay, Carla Fabiana Chiasserini, Falko Dressler, Y. Charlie Hu, Steven Y. Ko, Dimitrios Koutsonikolas, Joerg Widmer.
In Proceedings of the 23rd IFIP/IEEE Networking Conference (NETWORKING 2024), Thessaloniki, Greece, June 3-6, 2024.

15. **Performance of Cellular Networks on the Wheels.** *(Thesis Chapter 2)*
Moinak Ghoshal*, Imran Khan*, Jonny Kong*, Phuc Dinh, Jiayi Meng, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the 23rd ACM Internet Measurement Conference (IMC 2023), Montreal, Canada, October 24-26, 2023. (acceptance rate 25%).

16. **Can 5G mmWave Enable Edge-Assisted Real-Time Object Detection for Augmented Reality?**
Moinak Ghoshal*, Z. Jonny Kong*, Qiang Xu*, Zixiao Lu, Shivang Aggarwal, Imran Khan, Yuanjie Li, Y. Charlie Hu, Dimitrios Koutsonikolas.
In Proceedings of the 31st International Symposium on the Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS 2023), Stony Brook, NY, USA, October 14-18, 2023. (acceptance rate 30%).

17. **Co-located Immersive Gaming: A Comparison between Augmented and Virtual Reality**
Moinak Ghoshal, Juan Ong, Hearan Won, Dimitrios Koutsonikolas, Caglar Yildirim.
In Proceedings of the IEEE Conference on Games (CoG 2022), Beijing, China (Virtual), August 21-24.

18. **Demystifying Resource Allocation Policies in Operational 5G mmWave Networks**
Phuc Dinh*, **Moinak Ghoshal***, Dimitrios Koutsonikolas, Joerg Widmer.
In Proceedings of the 23rd IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM 2022), Belfast, UK, June 14-17. (acceptance rate 31%)

19. **Can 5G mmWave Support Multi-user AR?**
Moinak Ghoshal, Pranab Dash, Zhaoning Kong, Qiang Xu, Y. Charlie Hu, Dimitrios Koutsonikolas, Yuanjie Li.
In Proceedings of the Passive and Active Measurement Conference (PAM 2022), Virtual Conference, March 2022. (acceptance rate 48%).

20. **Throughput Prediction on 60 GHz Mobile Devices for High-Bandwidth, Latency-Sensitive Applications.**

Shivang Aggarwal, Zhaoning Kong, **Moinak Ghoshal**, Y. Charlie Hu, and Dimitrios Koutsonikolas.
In Proceedings of the Passive and Active Measurement Conference (PAM 2021), Virtual Conference, March 2021. (acceptance rate 44%)

21. **802.11ad in Smartphones: Energy Efficiency, Spatial Reuse, and Impact on Applications.**
Shivang Aggarwal, **Moinak Ghoshal**, Piyali Banerjee, Dimitrios Koutsonikolas, and Joerg Widmer.
In Proceedings of the IEEE Conference on Computer Communications (INFOCOM 2021), Virtual Conference, May 10-13, 2021. (acceptance rate 19.9%)

22. **LiBRA: Learning-Based Link Adaptation Leveraging PHY Layer Information in 60 GHz WLANs.**
Shivang Aggarwal, Urjit Satish Sardesai, Viral Sinha, Deen Dayal Mohan, **Moinak Ghoshal**, and Dimitrios Koutsonikolas.
In Proceedings of the 16th International Conference on emerging Networking EXperiments and Technologies (CoNEXT 2020), Barcelona, Spain, December 1-4, 2020. (acceptance rate 24%)

23. **A CFS–DNN-Based Intrusion Detection System.**
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1. **Roads, Radios, and Reality: What 5G Looks Like From the Driver's Seat.**
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2. **How Mature is 5G Deployment? A Cross-Sectional, Year-Long Study of 5G Uplink Performance?**
Imran Khan, **Moinak Ghoshal**, Joana Angjo, Sigrid Dimce, Mushahid Hussain, Paniz Parastar, Yenchia Yu, Xueting Deng, Sumit Hawal, Shirui Huang, Ameya Rane, Yin Wang*, Claudio Fiandrino, Charalampos Orfanidis, Shivang Aggarwal, Ana C Aguiar, Ozgu Alay, Carla Fabiana Chiasserini, Falko Dressler, Y. Charlie Hu, Steven Y. Ko, Dimitrios Koutsonikolas, and Joerg Widmer.
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3. **Demystifying Resource Allocation Policies in Operational 5G mmWave Networks**
   Phuc Dinh*, **Moinak Ghoshal***, Dimitrios Koutsonikolas, Joerg Widmer.
   In IEEE/ACM Transactions on Networking (ToN), Vol. 33(2), pp. 745-760, April 2025.

4. **An Experimental Study of the Performance of IEEE 802.11ad in Smartphones.**
   Shivang Aggarwal, **Moinak Ghoshal**, Piyali Banerjee, and Dimitrios Koutsonikolas.
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5. **An Evolutionary Algorithm Based Parameter Estimation Using Pima Indians Diabetes Dataset.**
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   Accepted in International Journal on Recent and Innovation Trends in Computing and Communication, 2017.

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1. **Private 5G vs. Wi-Fi in the Enterprise: A Measurement Study of Tunability and QoE.**
   Moinak Ghoshal, Shivang Aggarwal, Mohamed Ibrahim, Puneet Sharma, K. K. Ramakrishnan, Dimitrios Koutsonikolas.
   In Proceedings of the 20th ACM International Workshop on Wireless Network Testbeds, Experimental evaluation & CHaracterization (WiNTECH 2026), Virtual, October 30, 2026.

2. **Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications.**
   Moinak Ghoshal.
   In Proc. of the 23rd Annual International Conference on Mobile Systems, Applications and Services (MobiSys 2025) Rising Stars Forum, Anaheim, CA, USA, June 23, 2025.

3. **Beam Management in Operational 5G mmWave Networks.**
   Yufei Feng, Jin Wei, Phuc Dinh, **Moinak Ghoshal**, Dimitrios Koutsonikolas.
   In Proceedings of the 7th ACM Workshop on Millimeter-Wave and Terahertz Networks and Sensing Systems (mmNets 2023), Madrid, Spain, October 6, 2023.

4. **The Power of Asynchronous SLAM in Multi-User AR over Cellular Networks: A Measurement Study.**
   Yuting Guo, Sizhe Wang, **Moinak Ghoshal**, Y. Charlie Hu, Dimitrios Koutsonikolas.
   In Proceedings of the ACM SIGCOMM Workshop on Emerging Multimedia Systems (EMS 2023), New York City, NY, September 10, 2023.

5. **An In-Depth Study of Uplink Performance of 5G mmWave Networks**
   Moinak Ghoshal*, Z. Jonny Kong*, Qiang Xu*, Zixiao Lu, Shivang Aggarwal, Imran Khan, Yuanjie Li, Y. Charlie Hu, Dimitrios Koutsonikolas.
   In Proceedings of the 2nd ACM SIGCOMM Workshop on 5G and Beyond Network Measurements, Modeling, and Use Cases (5G-MeMU 2022), Amsterdam, Netherlands, August 22.

6. **Multipath TCP in Smartphones Equipped with Millimeter Wave Radios.**
   Imran Khan, **Moinak Ghoshal**, Shivang Aggarwal, Dimitrios Koutsonikolas, and Joerg Widmer.
   In Proceedings of the 15th ACM International Workshop on Wireless Network Testbeds, Experimental evaluation & CHaracterization (WiNTECH 2021), Virtual, October 29, 2021.

Posters and Demos

1. **Demo: NextG-UP: A Longitudinal and Cross-Sectional Study of Uplink Performance of 5G Networks.**
   Moinak Ghoshal[*], Imran Khan[*], Qiang Xu, Z. Jonny Kong, Y. Charlie Hu, and Dimitrios Koutsonikolas.
   ACM MobiCom, 2022.

2. **Demo: NextG-UP: A Tool for Measuring Uplink Performance of 5G Networks.**
   Moinak Ghoshal*, Imran Khan*, Qiang Xu, Z. Jonny Kong, Y. Charlie Hu, Dimitrios Koutsonikolas.
   ACM MobiSys, 2022

Patents

1. **WTRU Assisted Proactive PHY Frame Reconfiguration Associated with Low Latency Applications.**
   Subhramoy Mohanti, and **Moinak Ghoshal**.
   Patent No. WO2025111448A1, Application No. PCT/US2024/056872, published May 30, 2025.

Awards

- **Distinguished PhD Teaching Award -** Northeastern University Electrical and Computer Engineering, 2026
- **Best Community Paper Runner-up -** ACM CoNEXT, 2025
- **Outstanding PhD Research Award -** Northeastern University College of Engineering, 2024
- **Travel/Conference Grants -** ACM SIGCOMM (2022), IEEE/IFIP NETWORKING (2024)
- **Best Dataset Award -** Passive and Active Measurement (PAM) Conference, 2021
- **Spot Award -** Keysight Technologies, (2017 (Q4), 2018 (Q1))

Data/Source Code/App Release

- **Dataset** of our PAM 2026 paper - *Disentangling the Throughput Contributions of MIMO and Carrier Aggregation in 5G Networks*.
- **Dataset** of our ACM SIGMETRICS 2026 paper - *Exploring the 5G Digital Divide in the Non-Contiguous US: LEO Satellites to the Rescue?*
- **Dataset** of our ACM CoNEXT 2025 paper - *A First Large-Scale Study of Operational 5G Standalone Networks*.

- **Dataset** of our ACM IMC 2025 paper - *Replication: Performance of Cellular Networks on the Wheels*.
- **Dataset** of our ACM CoNEXT 2025 paper - *Dissecting Beamforming in Operational 5G mmWave Cellular Networks*.
- **Dataset** of our PAM 2025 paper - *A Large-Scale Study of the Potential of Multi-Carrier Access in the 5G Era*.
- Year-long cellular performance **dataset** from 8 cities from our Networking 2024 paper
- **Dataset** of cellular network performance measurements while driving from LA to Boston from our ACM IMC 2023 paper.
- **Dataset** of multi-user AR measurements using Cloud Anchor and Just-a-line application over 5G mmWave, LTE, and WiFi networks from our ACM SIGCOMM EMS 2023 paper.
- **Dataset** of 5G mmWave uplink performance using commercially available mobile devices from our ACM SIGCOMM 5G-MeMU 2022 paper.
- **NextG-UP**: Android application to collect crowd-sourced cellular network data for our MobiSys 2022 demo.
- **Dataset** of multi-user AR measurements over 5G mmWave and LTE networks from our PAM 2022 paper.
- **Dataset** of throughput prediction of 60 GHz mobile devices using high-bandwidth, latency-sensitive applications from our PAM 2021 paper.
- **Dataset** of 802.11ad power consumption, spatial reuse, and application performance on smartphones from our IEEE INFOCOM 2021 paper

Teaching Activities

- **EECE 7374 (NU)**: Fundamentals of Computer Networks (TA → Fall 2025, Fall 2024, Fall 2023, Spring 2022, Fall 2022)
- **EECE 7364 (NU)**: Mobile and Wireless Networks (TA → Spring 2025)
- **CSE 489/589 (UB)**: Modern Networking Concepts (TA → Fall 2020)

Abstract

Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications

by

Moinak Ghoshal

Doctor of Philosophy in Computer Engineering

Northeastern University, August 2026

Dr. Dimitrios Koutsonikolas, Advisor

The rollout of 5G networks, which began in 2019, promised ultra-high bandwidth and ultra-low latency via a combination of physical-layer innovations – higher-order modulation, beamforming, massive MIMO, higher operating frequencies, and wider channels – sufficient to finally support bandwidth-hungry and latency-critical applications such as Augmented Reality (AR), Connected Autonomous Vehicles (CAVs), 360° video streaming, and cloud gaming, often dubbed 5G "killer" apps. These same innovations, however, introduced substantial complexity: a fragmented landscape of coexisting frequency bands, coexistence between LTE and 5G radio access technologies, and two competing deployment architectures (Non-Standalone and Standalone), all governed by operator-specific policies that remain largely opaque and are difficult to replicate in controlled testbed environments. Understanding whether commercial 5G networks actually deliver on this promise therefore requires extensive, real-world measurements.

This thesis addresses this question along three major directions. First, to establish the ground reality of commercial 5G, it presents an in-depth measurement study of cellular network coverage, performance, and application QoE across all three major US operators via a 5,700+ km cross-country driving campaign, and tracks how this ground reality evolves by replicating the same campaign two years later while additionally benchmarking cellular performance against Starlink's LEO satellite network. It further presents the first large-scale study of an operator's 5G Standalone (SA) deployment, comparing it head-to-head with Non-Standalone (NSA) across two driving campaigns 18 months apart. Second, it investigates the operator-controlled policies that remain invisible to end users yet directly shape their experience, presenting the first large-scale measurement study of 5G handover configurations, drawing on four cross-country driving trips, 48,000+ handovers, and 239,000+ handover-related reporting events collected over a 27-month

period. Third, motivated by findings from these measurement studies, it identifies that today's commercial 5G networks rely on static, downlink-heavy resource allocation configurations that are poorly suited to the heterogeneous, often uplink-intensive demands of emerging applications, and designs, implements, and validates a machine-learning-enabled framework that proactively reconfigures physical-layer frames in real time.

Together, these studies show that while 5G coverage and throughput have improved substantially since initial rollout, this progress remains uneven across operators and geographies, and is often undermined by aggressive, overhead-heavy handover configurations whose optimization targets are only weakly correlated with actual user-perceived performance. They further show that closing the remaining gap between 5G's promise and its ability to support next-generation applications requires not only continued infrastructure deployment, but also smarter, application-aware configuration across the network stack – a gap it begins to close with a concrete, standards-compliant solution.

Chapter 1

Introduction

The most recent generation of cellular networks, 5G, promises ultra-high bandwidth and ultra-low latency, far surpassing the performance of 4G LTE, via a combination of PHY layer innovations such as higher modulation schemes, beamforming, (massive) MIMO, higher operating frequencies, and wider channels. For example, several recent studies [1, 2, 3] report downlink data rates of several hundreds of Mbps in the mid bands and up to 3.5 Gbps in the mmWave bands. Such high data rates, combined with low latency, hold the promise to finally support *bandwidth-hungry* and *latency-critical* applications such as Augmented Reality (AR), Mixed Reality (XR), Volumetric Video Streaming, Connected Autonomous Vehicles (CAVs), and the Metaverse, often dubbed as "5G killer" apps, which demand ultra-high network bandwidth and low network latency to support offloading of compute-intensive tasks to the edge cloud.

However, 5G's high frequency bands reduce its range making it difficult to achieve the same widespread coverage as LTE, and also result in high network performance fluctuations. Additionally, with the introduction of multiple frequency bands in 5G, co-existence with LTE networks, and different deployment modes (NSA and SA), the cellular network ecosystem has grown extremely complex. For example, to provide ubiquitous connectivity to users on the move, the network performs frequent handovers, especially in the urban areas. Though handovers ensure that the users are well connected to the network, frequent handovers incur momentary connection loss and result in network performance fluctuations, when the mobile device switches to a different 5G band or to LTE, which in turn cause fluctuations in the end-to-end user QoE. Hence, wide adoption of 5G networks faces a significant new challenge – its actual deployment plays a significant role in the user-perceived performance. Consequently, it is important to study and understand the perceived performance of commercial 5G networks by end users while running various bandwidth-hungry and latency-critical

applications.

5G rollout started in 2019 and the wide-scale deployment has been rapid and aggressively marketed by all mobile network operators. After 7 years of deployments, it is highly anticipated that 5G has significantly improved the performance of mobile networks and the overall user experience compared to LTE. However, the complexities discussed above pose a challenge in realizing the full potential of the 5G networks and hence, it is imperative to understand in detail these intricacies and complexities of the current 5G deployments and their effect on performance and overall end-user QoE.

Answering the above question requires a comprehensive understanding of real-world 5G performance, which can be achieved by conducting out-in-the-wild cellular network measurements. While testbeds designed by academia and industrial research offer valuable insights, they often fail to capture the full range of real-world scenarios, as different network operators implement unique policies and configurations that are difficult to replicate in controlled environments. By conducting extensive real-world 5G measurement campaigns, this thesis aims to bridge the gap between controlled experiments and the complexities of real-world cellular networks and highlight the nuances of current network behavior that can enhance deployment strategies in the future.

1.1 Thesis Statement

This thesis shows that commercial 5G networks have improved substantially since their initial rollout, yet persistent gaps between 5G's promise and its operational reality – in coverage, in the operator-controlled policies that govern mobility management, and in the static resource allocation strategies used across the network stack – continue to limit its ability to support next-generation, latency-critical applications. Closing these gaps requires sustained, real-world measurement to reveal where and why 5G falls short, together with targeted, measurement-informed system redesign.

1.2 Research Questions

We decompose this thesis statement into three concrete research questions that it answers in turn:

- **RQ1:** What is the ground reality of commercial 5G networks today, and how has it evolved over time?

- **RQ2:** What operator-controlled policies – particularly the configuration parameters that govern handovers – shape the experience users actually get, and are they well-calibrated?
- **RQ3:** How can measurement-driven insights be used to design systems that close the gap between 5G's promise and its ability to support next-generation applications?

This thesis tackles these questions along three major directions. First, to understand the ground reality of today's 5G deployments, we perform an in-depth study of cellular networks under driving via a cross-country measurement campaign from Los Angeles to Boston (§2), analyzing coverage, network and application performance and the factors that affect them, handovers, and diversity across geographic locations. Since a single snapshot is not enough to capture this ground reality, we also track its evolution by repeating the same driving campaign along the same route two years later (§3). We further examine 5G's two coexisting architectures – Non-Standalone (NSA) and Standalone (SA) – studying how their deployments and respective performance evolve over an 18-month period (§4). Second, we investigate the operator-specific policies that remain opaque to end users yet directly determine their QoE: we study handover configuration inefficiencies in operational 5G networks, including their overhead, aggressiveness, and impact on application performance (§5). Third, we assimilate the findings and datasets from these measurement studies to identify a real-world problem with the static resource allocation configurations used in today's networks, and design a solution that addresses it (§6).

We elaborate on the motivation, approach, and key findings of each of these studies below.

RQ1: Ground reality (§2, §3, §4).

• **Ground reality via a cross-country drive (§2).** Prior measurement studies of 5G have largely focused on static scenarios or a single aspect of performance under mobility, leaving no complete picture of cellular network behavior under driving – a major real-world use case of mobile networks. To address this gap, we conduct an in-depth measurement study of user-perceived experience via a 5,700+ km driving trip from Los Angeles to Boston, jointly analyzing network coverage, performance, and the QoE of a set of latency-critical 5G "killer" apps across all three major US operators. We find disappointingly low and fragmented 5G coverage, and show that network and application performance remain poor under driving even in areas with full high-speed 5G coverage.

• **Tracking 5G's evolution (§3).** A single measurement snapshot cannot reveal whether operators are actually closing the gaps identified above, especially as 5G reaches what is widely considered its midlife. We therefore replicate our original study two years later along the exact same coast-to-coast route, re-evaluating coverage and network and application performance across the same

three operators, and additionally benchmarking cellular performance head-to-head against Starlink's LEO satellite network. We find that 5G coverage and throughput have improved substantially for all operators, yet two of the three still offer 5G coverage on less than half of the route even five years after 5G's initial rollout.

• **Standalone vs. Non-Standalone 5G (§4).** Most existing measurement studies, including our own above, characterize 5G Non-Standalone (NSA) deployments, which still rely on the legacy LTE core for control signaling; comprehensive real-world evaluations of 5G Standalone (SA) – the architecture required to unlock 5G's full potential – remain scarce. Through two cross-country driving campaigns 18 months apart, we conduct the first large-scale study of an operator's 5G SA deployment, tracking its evolution and comparing it head-to-head with NSA. We find that SA coverage nearly tripled and overtook NSA within this period, and that SA increasingly outperforms NSA as the operator adopts advanced features such as higher carrier aggregation, while NSA's peak performance has plateaued.

RQ2: Operator policy (§5).

• **Handover configuration inefficiencies (§5).** The handover (HO) process is central to seamless connectivity, yet the configuration policies that govern when and how HOs are triggered are set unilaterally by operators and remain opaque to end users and researchers. Analyzing 48K+ handovers and 239K HO-related reporting events collected across four cross-country driving trips over a 27-month period, we show that current HO configurations are overly aggressive – only 13–31% of measurement reports ever trigger a HO – and that this overhead is a direct consequence of configuration choices rather than an inherent property of the events. We further find that the metrics operators directly control are only weakly correlated with actual user-perceived performance, complicating efforts to optimize HO policy.

RQ3: Measurement-driven system design (§6).

• **A solution to static resource allocation (§6).** We found that today's commercial 5G networks rely on static, downlink-heavy PHY frame configurations that cannot adapt to the heterogeneous, bursty, and often uplink-intensive demands of emerging applications such as AR and CAVs. To address this shortcoming, we propose a machine-learning-enabled framework that proactively reconfigures PHY frames in real time by predicting wireless channel metrics at the user equipment. We implement and validate our framework on a 3GPP-compliant open-source 5G testbed, demonstrating that it consistently meets traffic demands better than any fixed configuration used in commercial networks today, while remaining fully compliant with the 3GPP standard.

Finally, we conclude this thesis and discuss the lessons and recommendations that emerge from it in §7, and outline future research directions in §8.

Chapter 2

Performance of Cellular Networks on the Wheels

Understanding the performance of mobile networks under driving is important as they not only provide connectivity to passenger phones [4], but also power the rapidly increasing number of connected vehicles [5], which run a wide variety of apps ranging from basic navigation to AI-driven CAV apps. However, prior measurement studies have characterized various aspects of 5G performance in recent years, either focusing on static performance of 5G networks [6, 7, 8, 1, 2], or on a single aspect (e.g., handovers) under driving conditions [3, 9], leaving no complete picture of cellular network performance under driving.

To gain a complete understanding of 5G network performance and more generally a holistic picture of the cellular network performance as a whole under driving conditions, we conduct – to our knowledge – *the first in-depth measurement study of the mobile networks of all three major US carriers while driving across the continental US (5700km+, from Los Angeles to Boston).* Our study spans all cellular technologies available today (LTE/LTE-A, 5G-low/midband/mmWave), all layers of the protocol stack, and multiple 5G "killer" apps.

Unlike previous measurements in static scenarios or with limited mobility [6, 7, 8, 1, 2], conducting such a cross-country driving measurement faces significant challenges:

• **[C1]** Assessing the end-to-end cellular network and application performance in the wild is known to be difficult and it becomes even more challenging under driving. Many factors can become the performance bottleneck, including the mobile operator's deployed infrastructure and policies, the vehicle's speed, the app design, and the server's location.

Table 2.1: Driving dataset statistics.

Total geographical distance travelled	5711+ km
States/major cities/counties traveled	14/10/100+
Timezones traveled	4
Operators	Verizon (V), T-Mobile (T), AT&T (A)
# of unique cells connected	3020 (V), 4038 (T), 3150 (A)
# of handovers	2657 (V), 4119 (T), 2494 (A)
Total cellular data used	777+ GB (Rx), 83+ GB (Tx)
Total log size	388+ GB
Cumulative experiment runtime	5561 min (V), 4595 min (T), 4541 min (A)

• **[C2]** Lower level key performance indicators (KPIs) and control-plane signaling events are crucial for understanding cellular network performance. However, it is difficult to extract such information from commercial off-the-shelf unrooted smartphones. Furthermore, these messages need to be synchronized with application level logs, which often have different and diverse sampling frequencies and timing formats.

• **[C3]** Designing an accurate and efficient methodology to log the coverage of each cellular technology (4G vs. 5G) while driving across the country is particularly challenging, as operators often deploy complex policies in deciding whether to elevate a UE's service from LTE to 5G. For example, UEs often fall back to LTE or do not switch to 5G in the absence of heavy traffic.

• **[C4]** It is not easy to evaluate the performance of upcoming 5G "killer apps" [10], since several such apps are still not out in the market for public use. For example, various studies show that Augmented Reality (AR) and Connected Autonomous Vehicle (CAV) applications can achieve high QoE by offloading computation to the edge [11, 12], but such applications have not been deployed commercially, and thus need to be custom-built for our study.

To address these challenges, we built a measurement platform consisting of (1) multiple smartphones, laptops, power supplies, and strategically deployed cloud and edge servers in various geographic locations across the US, (2) professional hardware and software tools that collect lower-level cellular network KPIs and control-plane events, (3) a mix of readily available and custom-built 5G "killer" apps, and (4) a custom-built software that includes a suite of tests (throughput, RTT, application QoE) and procedures for synchronization and post-processing of logs from different software and different layers of the protocol stack.

The key findings of our study are summarized as follows.

• In spite of the aggressive deployment efforts over the past few years, 5G coverage while driving is in general disappointingly low and highly fragmented across the country. Our analysis shows very diverse coverage across operators and even for the same operator across different geographic regions and suggests that today's 5G coverage is the result of more complex operator policies and deployment strategies. For example, our data suggest that operators are more likely to upgrade a UE's service to high-speed 5G (midband or mmWave) in the presence of backlogged downlink traffic, while they tend to prefer 5G-low or 4G for backlogged uplink traffic.

• Network performance under driving deteriorates significantly compared to the performance observed under the best static conditions. While downlink throughputs higher than 1 Gbps or of several 100s of Mbps are still possible with Verizon's/AT&T's 5G mmWave and T-Mobile's 5G midband services, the observed throughput with any carrier is below 5 Mbps about 35% of the time. More importantly, network performance is often poor even in areas with full high-speed 5G coverage. We also found that the use of edge servers brings a significant boost to both throughput and RTT compared to remote cloud servers.

• Our data show significant diversity in cellular network performance across operators at a given location and time. Interestingly, an operator that uses high-speed 5G at a given location does not always yield higher throughput than another operator using 5G-low or 4G. These two observations suggest that performance under driving can benefit significantly from multi-connectivity solutions that can aggregate links from multiple operators.

• We examined the impact of a number of cellular KPIs (RSRP, MCS, BLER, carrier aggregation) along with the number of handovers and the vehicle's speed on throughput during driving. Our analysis shows that none of these KPIs has a strong correlation with throughput and the factors that have the highest impact on throughput are different for different operators and even for different traffic directions for the same operator. Somewhat surprisingly, we found that the vehicle's speed and the number of handovers have weak and no correlation, respectively, with throughput, which is mostly affected by signal strength and MCS.

• We found that, while the number of handovers per mile can reach up to 20+, the number is typically low (1-3 handovers per mile in the median case for downlink traffic), and their duration is typically short (61 ms in the median case), resulting in a small throughput drop, which is often counterbalanced by a similar improvement in post-handover throughput. This explains the lack of correlation between throughput and number of handovers.

• We evaluated the QoE of two readily available downlink-centric apps (360° video streaming and mobile cloud gaming) and two future cloud-assisted uplink-centric apps (an AR app and a CAV

app that rely on DNN-based object detection). Our results show that all four apps experience poor performance and large performance variation under driving. While high-speed 5G can improve the worst-case performance of these apps compared to LTE, and the combination of 5G and edge computing can further boost performance, QoE remains disappointingly low and often no better than over LTE. Interestingly, we found that the number of handovers does not affect the average QoE, as such apps have built-in mitigation mechanisms (local tracking in the case of AR and CAV apps, buffering in the case of video streaming, frame rate adaptation in the case of cloud gaming), which help them combat temporary throughput and RTT degradation during handovers.

Contributions. In summary, this study makes the following contributions: (1) We collect a first-of-its-kind large multi-technology, multi-band, multi-carrier, cross-layer dataset of cellular network performance while driving across the country from LA to Boston. (2) Leveraging this dataset, summarized in Table 2.1, we provide the first in-depth study of cellular networks under driving, analyzing coverage, network and application performance and the factors that affect them, handovers, and diversity in terms of geo-location. Our dataset and scripts are publicly available [13].

2.1 Methodology

Drive Tests. We drove 5700+ km from Los Angeles to Boston covering all major cities in between (Las Vegas, Salt Lake City, Denver, Omaha, Chicago, Indianapolis, Cleveland, Rochester) over 8 days – 08/08/2022 to 08/15/2022. All measurement results reported in this study were obtained while driving on inter-state highways, in suburban areas, or inside cities, except for a few baseline measurements, which were conducted in major cities under static conditions.

2.1.1 Testbed.

Experimental Methodology. We now describe our methodology to address the four challenges mentioned in §2.

To address **[C1]**, we built a testbed (Fig. 2.1) consisting of multiple smartphones, laptops, Accuver XCAL Solo devices [14], application cloud and edge servers, and multiple AC power supplies to constantly charge all the devices. XCAL Solo is a standalone commercial tool that is attached to a smartphone via the USB-C port and it taps into the diagnostic interface of the smartphone to log all the PHY-layer KPIs and signaling messages. We used 6 Samsung S21/SM-G998U phones as our user equipment (UE), a state-of-the-art 5G smartphone model at the time when the measurements were done. We purchased multiple unlimited data plans from all three major U.S.

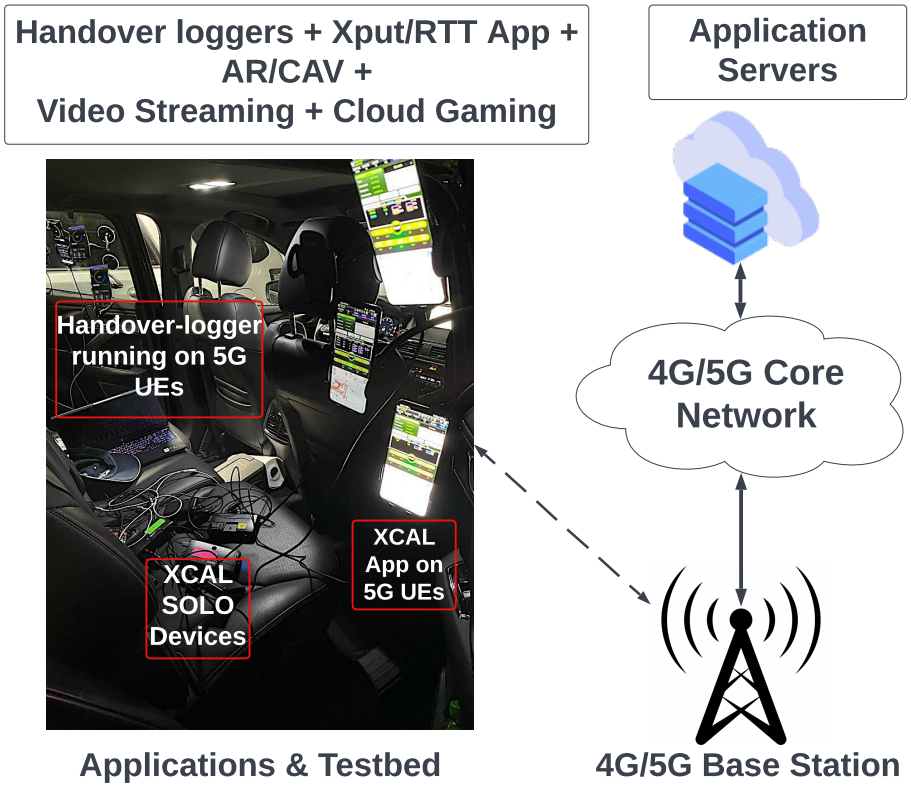


Figure 2.1: Overview of our measurement testbed.

carriers: Verizon, T-Mobile, and AT&T. All three carriers have low-band, mid-band, and mmWave services deployed.

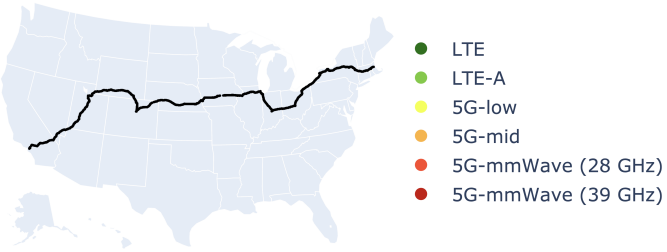
To test end-to-end network and application performance, we deployed multiple AWS EC2 instances – two in California for the tests done in the Pacific and Mountain time zones, and two in Ohio for the tests done in Central and Eastern time zones. Additionally, to assess the benefits of edge computing on 5G apps, we deployed 5 Amazon Wavelength [15] edge servers in Los Angeles, Las Vegas, Denver, Chicago, and Boston. Wavelength servers are located *inside* Verizon's network in selected cities and specially designed for edge computing. For tests over the Verizon network, we used the deployed Wavelength server in each of these five cities and the cloud servers in the rest of the trip. For the other two operators, we only used the cloud servers.

To address **[C2]**, we collected app-level information from three unrooted smartphones (each connected to a different carrier) and the cloud/edge servers, and lower layer KPIs and control-plane signaling events from the XCAL Solo devices. An associated challenge here was to sync and post-process all the data from different layers, as the applications and XCAL logged information

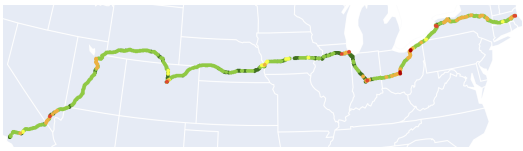
using different time formats. To address this challenge, we wrote sophisticated software that maps each app-layer log to the corresponding XCAL file taking into account the different timestamp types and the timezones we crossed, loads all the segregated XCAL files of a particular type of test to the XCAL parsing software (XCAP-M), and creates a consolidated database, which includes both the XCAL and the app layer data.

We used the other three unrooted phones ("handover-loggers" for the rest of this study) to constantly log the handover events over the three carriers throughout the 8-day trip. We wrote an Android app to constantly send ICMP-based ping traffic with 38 bytes of data at an interval of 200 ms to prevent the cellular radio from going to sleep mode, and we logged key pieces of information related to mobility management: GPS information, cell IDs, cellular technology, etc. via Android APIs. However, we observed that often times the app would not switch from 4G to 5G even though we were in a 5G coverage area. We conjecture that the operators might be conservative and do not upgrade to 5G when the network traffic demand is low. To mitigate this challenge **[C3]**, we also extracted handover and technology information from the XCAL data logs generated during the network performance and application tests. A comparison between the two approaches to handover logging is described in §2.2.1.

We performed two different types of tests: (1) TCP bulk data transfers in both downlink and uplink directions, and RTT measurements. The experimental setup for these tests is described in §2.3. (2) Mobile app measurements. We evaluated the performance of 4 5G "killer" apps, two downlink-centric apps – 360° video streaming and cloud gaming – and two uplink-centric apps – AR and CAV. The experimental setup for each app is described in §2.5.1.1, §2.5.2, §2.5.3.1. While video streaming and cloud gaming apps are readily available, edge/cloud-assisted AR or CAV apps [11, 12] that offload data to an edge/cloud server, which performs DNN-based object detection, are not yet available on the market. To address this challenge (**[C4]**), we built a canonical edge-assisted AR/CAV app running on an Android phone that offloads dummy camera frames or LIDAR point clouds to an edge server. Details are provided in §2.5.1.1. We ran the bandwidth, RTT, and four mobile app tests in a round robin fashion on the three smartphones connected to XCAL Solo devices (each using a different carrier).



(a) Cross-continental drive route.



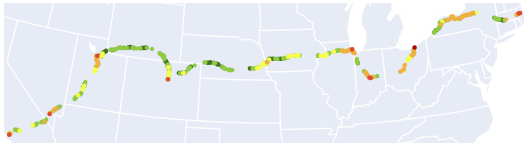
(b) Verizon, handover-logger.



(c) T-Mobile, handover-logger.



(d) AT&T, handover-logger.



(e) Verizon, XCAL logs.

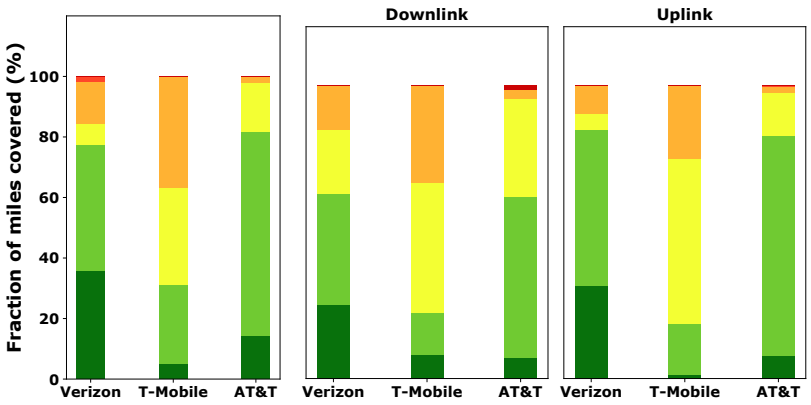


(f) T-Mobile, XCAL logs.

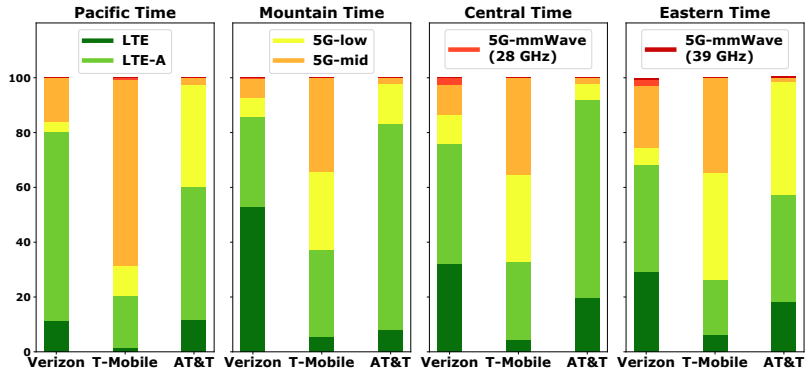


(g) AT&T, XCAL logs.

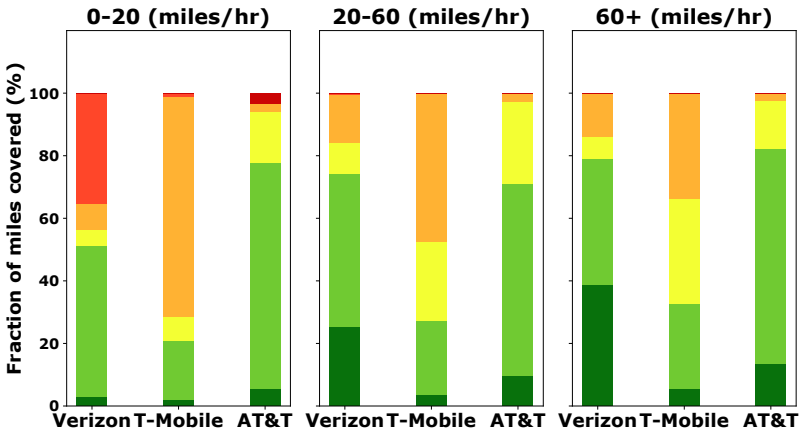
Figure 2.2: Comparison of the two approaches used to log cellular network coverage.



(a) Overall. (b) By traffic direction.



(c) By timezone.



(d) By vehicle speed.

Figure 2.3: Cellular technology breakdown per operator.

2.2 Network Coverage

2.2.1 How to Measure Coverage

Fig. 2.2 compares the two approaches we described in §2.1 to log cellular network coverage for all three operators. We observe that the cellular technology spread obtained by the two approaches is very different. The handover-logger data (Figs. 2.2b, 2.2d, 2.2c) present a very pessimistic view in terms of coverage, suggesting LTE/LTE-A as the dominant technologies along the route for all three operators. In the most extreme case (AT&T), Fig. 2.2d shows that LTE/LTE-A are the only two technologies along the whole driving route. In contrast, the XCAL logs during throughput/application tests (Figs. 2.2e, 2.2g, 2.2f) show a very different view, with multiple areas of 5G coverage. This disparity in the results with the two approaches suggests that *passive approaches (e.g., [16]) that simply log the cellular network state in the absence of heavy traffic are not reliable due to conservative operator policies that do not upgrade a UE from LTE to 5G in the presence of low traffic rates*. In fact, these policies may vary by location even for the same operator; for example, in the case of T-Mobile (Figs. 2.2c and 2.2f), our results with the two approaches agree for the east half of the country but show a very different view for the west half. Overall, a lesson learned from this study is that *obtaining a representative coverage view is challenging, as it requires high-overhead (in terms of bandwidth usage for operators and monetary cost for clients) active probing approaches using heavy traffic*. Based on our findings, in the following, we only discuss the data reported by XCAL during the throughput/application tests, even though this approach unavoidably relies on a smaller dataset.

2.2.2 Results

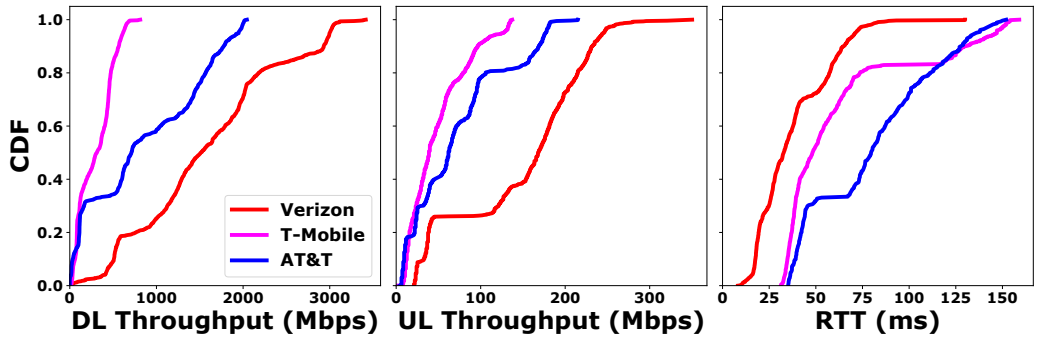
Fig. 2.3a shows the technology coverage for all three operators as a percentage of miles driven during the entire trip. We observe that, in spite of the aggressive deployment efforts over the past few years, *5G coverage is in general disappointingly low and highly fragmented across the country*. Our data confirm (the well-known fact) that T-Mobile has the highest percentage of 5G coverage (68%); in contrast, with Verizon and AT&T, we were able to connect to 5G only for only around 18-22% of the total distance traveled. The percentage of high-speed 5G (midband and mmWave) is even lower ranging from 38% (T-Mobile) to as low as 3% (AT&T). Verizon offers the highest mmWave coverage; in contrast, we rarely experienced mmWave connectivity with the other two carriers throughout the trip. Our results suggest very different deployment strategies by the three

carriers. Verizon has prioritized the deployment of 5G mmWave (in downtown areas of major cities), while T-Mobile has focused on expanding the coverage to larger geographical areas by prioritizing low/mid-band deployments. In contrast, AT&T offers better 4G coverage (a much larger percentage of LTE-A vs. LTE) compared to the other two carriers.

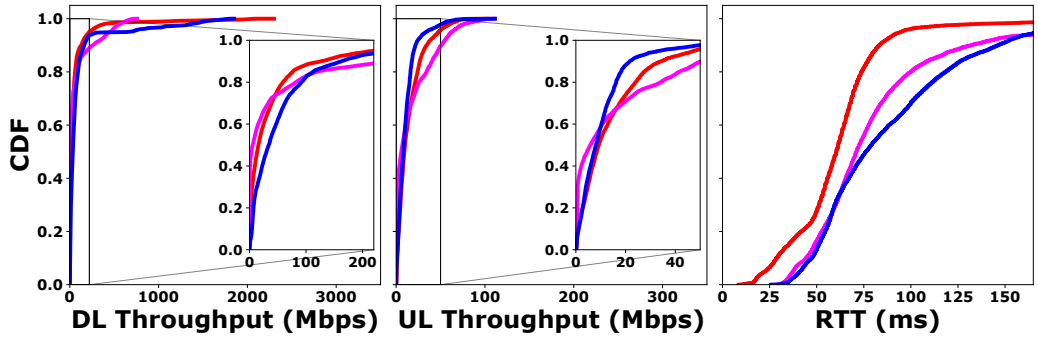
In Fig. 2.3c, we break down the technology coverage into four geographic regions. This figure shows *very diverse deployment strategies even for the same operator across different geographic regions*. For example, T-Mobile has a much higher percentage of 5G mid-band connectivity in the Pacific timezone compared to the rest of the country. AT&T has a very low percentage of 5G connectivity in the Mountain and Central timezones compared to the other two timezones. Verizon exhibits higher 5G coverage in the eastern half of the country (Central and Eastern timezones).

In Fig. 2.3d, we break down the technology coverage into three different speed bins: low (0-20 mph), mid (20-60 mph), and high (60+ mph). Note that the speed bins act as a proxy to the type of regions where the measurements were performed. For example, the low speed coverage samples are mostly from cities whereas the high speed ones are from the inter-state highways. We observe that high-speed 5G (mid band and mmWave) coverage decreases as we move from low to high speed bins for all three operators, suggesting that operators prioritize the deployment of high-speed 5G, in particular 5G mmWave, in urban areas. This is particularly true for Verizon, where the high-speed 5G coverage reduces from $\sim$43% in the low speed bin (cities) down to $\sim$13% in the high speed bin (inter-state highways). Among the three carriers, T-Mobile is the only one that maintains a significant high-speed 5G (midband) coverage under medium and high speeds (47% and 33%, respectively).

Up till now, we have examined the technology coverage regardless of the traffic direction (downlink vs. uplink). We now investigate if the technology a UE connects to at a particular location depends on the traffic direction. Here, we use only the data from our throughput tests, which always backlog the network in one direction, in order to maximize the possibility that a UE is serviced by 5G, and plot in Fig. 2.3b the technology distribution separately for downlink and uplink traffic. Interestingly, Fig. 2.3b shows that (1) for Verizon and AT&T, the 5G coverage overall is significantly higher in the downlink direction compared to the uplink, and (2) for all three carriers, the high-speed 5G coverage (midband and mmWave) is higher in the downlink direction compared to the uplink. Given the high asymmetry of downlink vs. uplink bandwidth in 5G mmWave and midband [2, 1], we conjecture that *operators are more willing to upgrade UEs to high-speed 5G in the presence of heavy downlink traffic, while they often prefer 5G-low or LTE/LTE-A for uplink traffic*.



(a) Static measurements done in major US cities.



(b) Driving measurements.

Figure 2.4: Overall throughput and RTT performance during static and driving measurements.

2.3 Network Performance

In this section, we take a detailed look at the throughput and RTT of cellular networks while driving. To measure throughput, we used nuttcp [17] with the default TCP congestion control algorithm, CUBIC, to generate downlink and uplink backlogged traffic from/to an edge/cloud server. We used the default nuttcp configuration with a single TCP connection, as our intention was to measure the performance that would be experienced by applications such as file downloads/uploads, video streaming, CAV/AR cloud-assisted apps, etc., instead of measuring peak performance measured with tools such as SpeedTest. Each test lasted for 30-35 s and logged throughput every 500 ms. To measure the RTT between the UE and an edge/cloud server, we used the ICMP-based ping utility. Each test ran for 20 s and sent one ICMP packet every 200 ms.

2.3.1 Static vs. Driving Performance

To compare the driving performance with a baseline, we measured the downlink/uplink throughput and RTT in each of the major cities we visited while driving from LA to Boston. In each

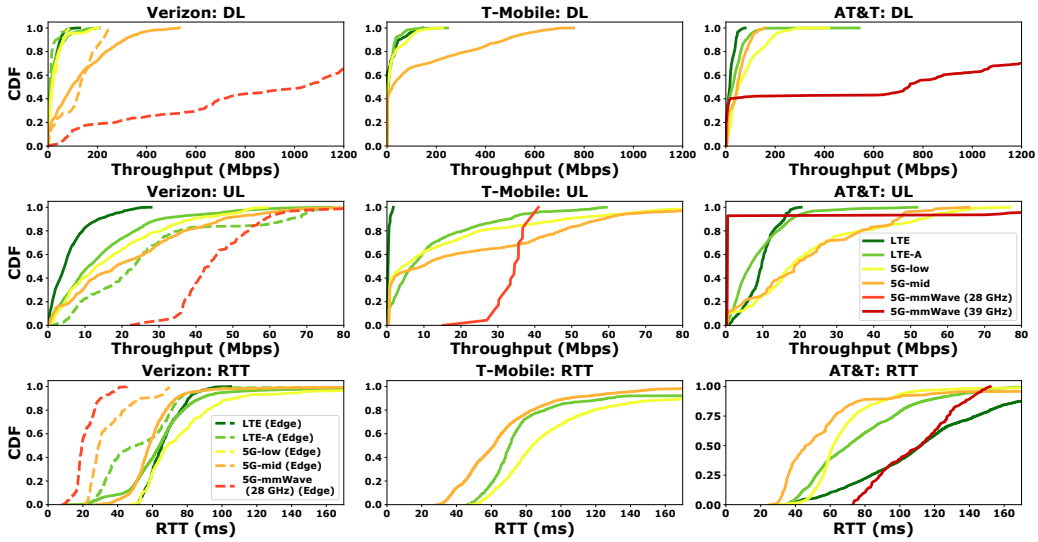


Figure 2.5: CDFs of throughput and RTT per technology while driving.

city, we tried to find a 5G-mmWave BS for each operator and performed the static measurements facing the BS. In cases we failed to find a mmWave BS, we measured the 5G mid-band performance. We omitted the static tests for those operator-city combinations for which we were not able to get 5G-mmWave or mid-band connectivity. Fig. 2.4 shows the CDFs of the static and the driving performance. The throughput CDFs consist of all the application-layer 500 ms throughput samples logged by XCAL and the RTT CDFs of all the individual RTT samples.

Fig. 2.4a shows that all three operators can provide very high downlink throughput in static, urban scenarios via their 5G mmWave or midband services, with median values of 1511/710/311 Mbps and maximum values as high as 3415/2043/812 Mbps for Verizon/AT&T/T-Mobile, respectively. The uplink throughput is an order of magnitude lower, as expected [2], with median values of 167/62/39 Mbps and maximum values of 350/215/137 Mbps for the three operators. Surprisingly, we observe a non-negligible fraction of low throughput values, especially for AT&T and T-Mobile, showing that 5G midband and mmWave can yield unpredictable performance even in ideal scenarios. We also observe that the RTT values show a very large variation, ranging from 8 ms to 100+ ms for Verizon and from 30 ms to 150+ ms for the other two carriers. Interestingly, Fig. 2.4a shows that AT&T with its mmWave coverage yields higher throughput than T-Mobile (which uses almost exclusively 5G midband) but also higher RTT. Upon closer inspection of our XCAL traces, we noticed that most of the RTT tests over AT&T were run over LTE/LTE-A even though the phone's screen showed 5G, suggesting that, in most cities, AT&T does not upgrade a UE to 5G under very low ICMP traffic.

Fig. 2.4b shows *a drastic drop in throughput during driving compared to static scenarios*. While the maximum downlink throughput is still higher than 1.8 Gbps for Verizon and AT&T and 600 Mbps for T-Mobile, the median/75-th percentiles are between 6-34 Mbps and 47-74 Mbps for all three operators, 1-5% of the corresponding values in static scenarios. The situation is similar in the uplink case, with median/75-th percentiles between 6-9 Mbps and 14-24 Mbps, 5-15% of the static values. Additionally, *we observe a significant fraction (35%) of very low throughput values (below 5 Mbps) in both directions for all 3 operators*. Further, Fig. 2.4b shows *a significant inflation to the RTT values under driving compared to the static scenarios*, with median values between 60-76 ms and maximum values as high as 2-3 s for all 3 operators.

2.3.2 Network Performance While Driving

To analyze the performance while driving, we plot the throughput and RTT CDFs for each cellular technology in Fig. 2.5. Additionally, for Verizon, we show separately the performance using an edge server (dashed lines) and a cloud server (solid lines). Overall, the results in Fig. 2.5 show that *cellular performance, in terms of both throughput and RTT, is often disappointingly low while driving, regardless of technology coverage*. 5G (including 5G-low) achieves higher throughput than 4G (LTE, LTE-A), but the performance with any technology varies dramatically and is often limited to a few 10s of Mbps or lower, even in the downlink direction. Similarly, RTTs are several 10s of ms with all technologies (with the exception of Verizon's 5G mmWave) and can exceed 100 ms.

When we look at the throughput performance in Fig. 2.5, we make the following observations: (1) In the downlink, 5G mmWave often achieves throughput higher than 1 Gbps even under driving, but it can also yield extremely low throughput; e.g., 40% of the AT&T downlink throughput with 5G mmWave samples are lower than 16 Mbps. (2) In the uplink, 5G mmWave offers the highest median throughput with Verizon and T-Mobile, but its maximum throughput is similar (Verizon, AT&T) to or lower (T-Mobile) than that of other technologies; in addition, with AT&T, 90% of the mmWave uplink throughput samples were lower than 0.5 Mbps. (3) T-Mobile's 5G midband service provides throughput up to 760 Mbps in the downlink, significantly higher compared to Verizon's or AT&T's midband, and its uplink throughput is often higher than T-Mobile's 5G mmWave uplink throughput. However, T-Mobile's midband service also exhibits the largest fluctuation among the three carriers with 40% of its throughput samples falling below 2 Mbps in both directions.

When we look at the RTT performance in Fig. 2.5, we observe again that 5G mmWave achieves the lowest RTTs among all technologies in the case of Verizon, and 5G midband achieves

lower RTTs than 5G-low and the 4G technologies for all three carriers. Interestingly, for both Verizon and T-Mobile, LTE-A achieves lower RTTs than 5G-low, demonstrating a tradeoff between throughput and RTT for these two technologies.

Finally, our results for Verizon show that *the use of an edge server brings a significant improvement to both throughput*[1] *and RTT compared to a cloud server*; for example, LTE-A with an edge server achieves higher uplink throughput and lower RTT than 5G midband with a cloud server; and the 5G mmWave RTT with an edge server is kept below 40 ms with a median value of 18 ms. This suggests that *edge computing is critical to boosting the performance of 5G killer apps (§2.5)*.

2.3.3 Geo-diversity in Cellular Network Performance

We next analyze the performance of cellular networks from a geographical angle. Our discussion of the diversity in network coverage across the country in §2.2.1 revealed different (and often complex) deployment strategies employed by different carriers. However, the real question of interest is whether higher 5G coverage actually translates to higher performance during driving in different parts of the country.

Fig. 2.6 breaks down the downlink and uplink throughput of each carrier across the four time zones. We make the following observations: (1) with the exception of AT&T in the downlink direction, where the throughput has higher values in the Eastern timezone, the throughput is clearly higher in the Pacific timezone in comparison to the other timezones, for all three carriers and both directions; (2) Verizon exhibits the lowest performance in the Eastern timezone among all four regions; (3) The performance in the Mountain timezone is low for all three carriers. Interestingly, *it is not always possible to explain the performance diversity across timezones based on the coverage results from Fig. 2.3*. The high performance for T-Mobile in the Pacific timezone and for AT&T in the Pacific (uplink) and Eastern (downlink) timezones can be possibly explained by the higher 5G midband coverage (compared to 5G-low) for T-Mobile and higher 5G-low coverage (compared to LTE/LTE-A) for AT&T in those timezones (Fig. 2.3a). On the other hand, Verizon has the highest 5G mmWave and midband coverage in the Eastern timezone, where it exhibits the worst performance.

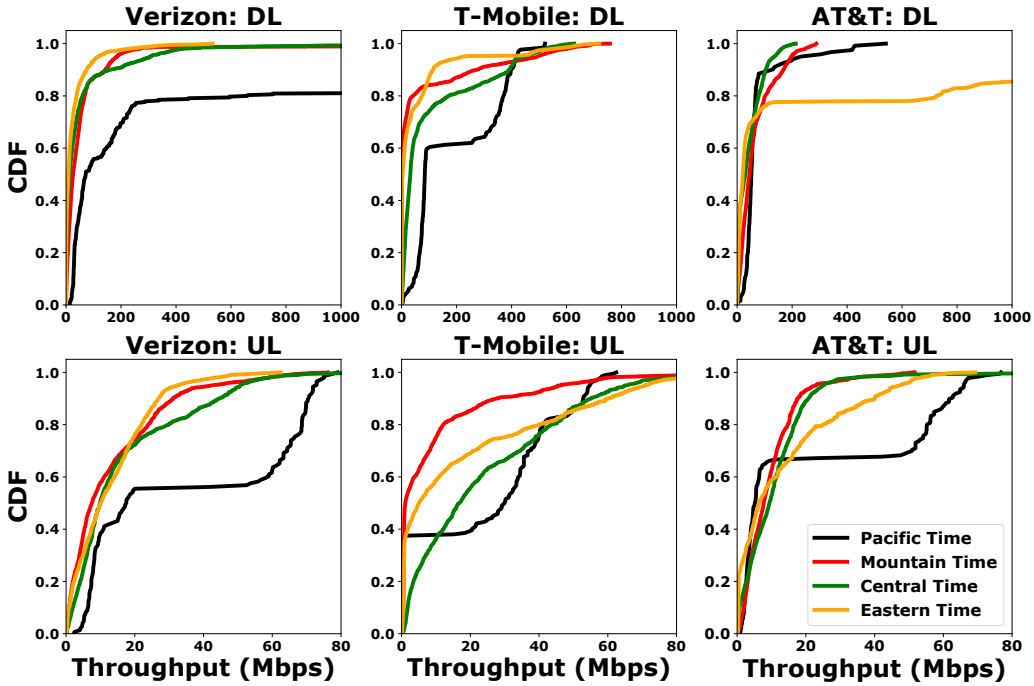
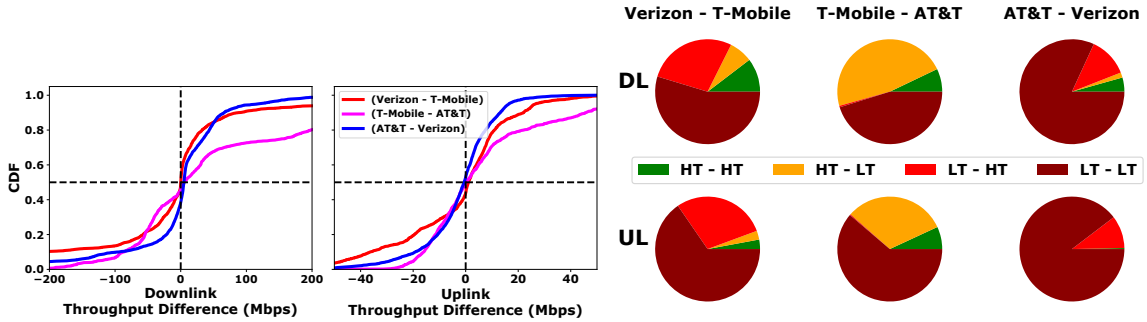
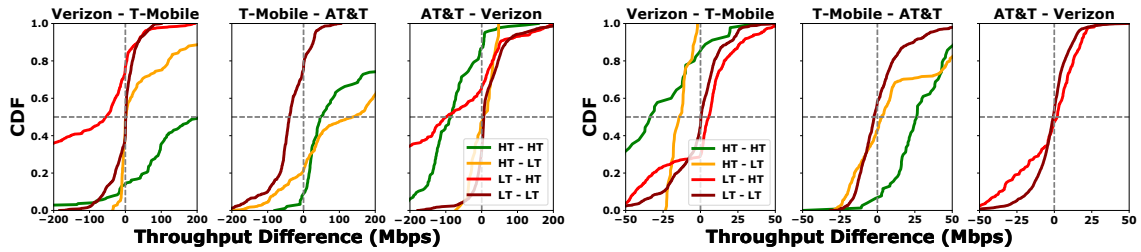


Figure 2.6: CDFs of throughput in different timezones.



(a) Overall CDF.

(b) Technology-breakdown.



(c) Break down by technology (downlink).

(d) Break down by technology (uplink).

Figure 2.7: Operator-wise throughput difference for tests done in parallel.

2.3.4 Operator Diversity in Cellular Network Performance

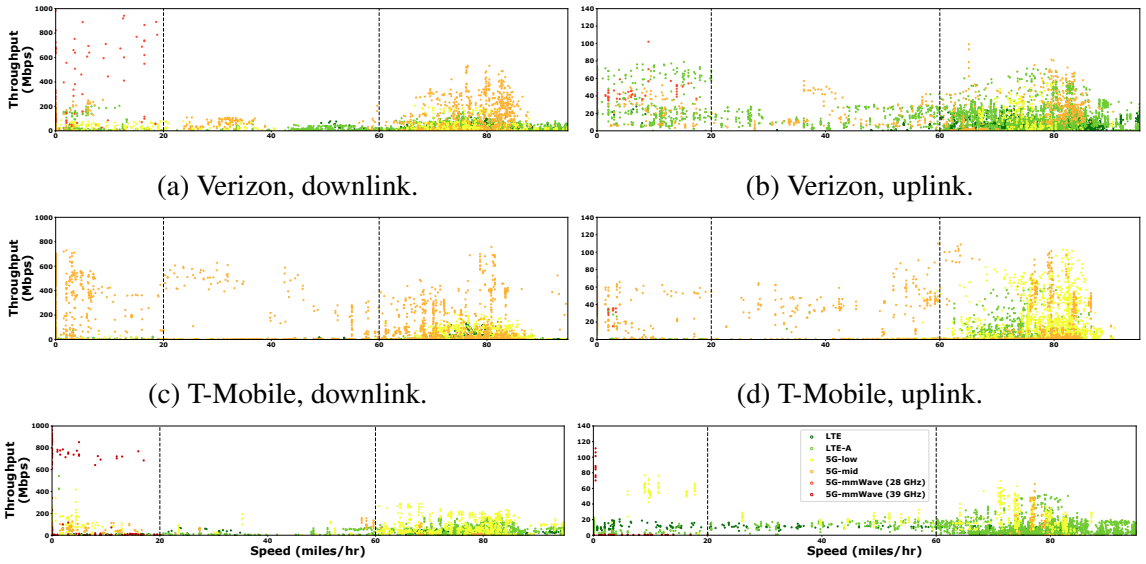
We now explore operator diversity in cellular network performance at a given location and time. For all throughput samples collected concurrently for any pair of operators, we plot in Fig. 2.7a the CDF of the throughput difference for each pair of operators. The results show that performance at a given location can be highly diverse across operators in both directions. This observation suggests that *performance under driving can benefit significantly from multi-connectivity solutions, e.g., over Multipath TCP, that can aggregate links from multiple operators [18, 19].*

We next investigate what cellular technologies contribute to such diverse performance across operators at the same location and time. For example, a large throughput difference could be due to the fact that one operator is using a high-throughput (HT) technology (5G mmWave or midband) while the other one is using a low-throughput (LT) technology (LTE/LTE-A/5G-low) at a given location. We break down the points of each CDF (corresponding to a pair of operators) in Fig. 2.7a into 4 bins, based on the technology used by the two operators: HT-HT, HT-LT, LT-HT, LT-LT. Fig. 2.7b plots the distribution of the 4 bins for each operator pair in the downlink and uplink directions and Figs. 2.7c, 2.7d plot the CDFs of throughput differences for each bin. We make the following observations:

First, the majority of the samples for all three operator pairs in the uplink direction and for Verizon - T-Mobile and AT&T - Verizon in the downlink direction come from the LT-LT bin, which also contributes a large fraction of samples for T-Mobile - AT&T in the downlink direction. Figs. 2.7c, 2.7d show that the LT-LT CDFs for Verizon - T-Mobile and AT&T - Verizon in both directions and for T-Mobile - AT&T in the uplink directions are symmetric around 0, indicating roughly equal probability for one operator to outperform the other one. In contrast, in the case of T-Mobile - AT&T in the downlink direction, AT&T outperforms T-Mobile in $\sim$80% of the locations thanks to its superior LTE-A and 5G-low services (Fig. 2.5).

Second, the HT-HT bin contributes a very small number of samples in all 6 cases, ranging from 0.3% (AT&T - Verizon, uplink) to 10% (Verizon - T-Mobile, downlink). Interestingly, the HT-HT CDFs in Figs. 2.7c, 2.7d are not symmetric around 0, unlike the LT-LT CDFs. In the downlink direction, Verizon outperforms both the other operators thanks to its strong mmWave coverage in cities; and T-Mobile outperforms AT&T thanks to its superior midband service and the low mmWave coverage for AT&T. On the other hand, in the uplink direction, T-Mobile outperforms the other two

[1]The first graph in Fig. 2.5 shows that 5G midband throughput is higher with a cloud server than with an edge server. However, we note that we had very few samples of 5G midband throughput with an edge server, as the UE was connected to either 5G mmWave or LTE-A most of the time in the 5 cities where we deployed an edge server.



(a) Verizon, downlink. (b) Verizon, uplink.

(c) T-Mobile, downlink. (d) T-Mobile, uplink.

(e) AT&T, downlink. (f) AT&T, uplink.

Figure 2.8: Technology-wise throughput breakdown as a function of speed. We cut the downlink plots at 1000 Mbps as the mmWave throughput reaches up to 2.5 Gbps making the rest of the points barely visible.

operators.

Finally, the other two bins (LT-HT and HT-LT) together contribute a very small number of samples in the case of AT&T - Verizon (14% downlink, 10% uplink), as both these operators mainly deploy their HT services in the same locations (cities), but a substantial number of samples in the case of Verizon - T-Mobile and T-Mobile - AT&T, ranging from 32% (Verizon - T-Mobile and T-Mobile - AT&T, uplink) to 48% (T-Mobile - AT&T, downlink). For both operator pairs, in most cases, T-Mobile is the operator using an HT technology, as this is the only operator with a good 5G midband coverage in highways (see Fig. 2.3d). Here, one would expect the operator using an HT technology to always outperform the operator using an LT technology. Interestingly, Figs. 2.7c, 2.7d show that this is not always the case. For example, Verizon and AT&T LT outperform T-Mobile HT in $\sim$20% of the locations in the downlink direction, as T-Mobile's 5G midband throughput can reach up to 760 Mbps but also falls below 2 Mbps 40% of the time (see Fig. 2.5). Similar results are observed for all three operator pairs in the uplink direction, where the performance gap between HT and LT technologies is much smaller compared to the downlink direction.

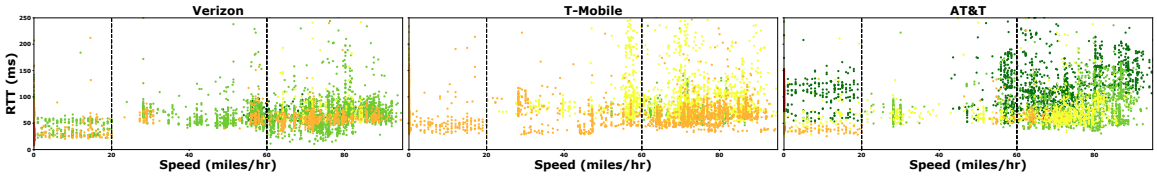


Figure 2.9: Technology-wise RTT breakdown as a function of speed.

2.3.5 Cellular Network Performance Analysis

Given the very wide range of throughput and RTT values under driving (Fig. 2.4b), we examine the impact of different factors on cellular network performance, starting with the vehicle's speed. Fig. 2.8, which plots the throughput (500 ms samples) against the vehicle's speed in both directions for each operator,shows that throughput values can be distinctly divided into three speed regions: (a) low (0-20 mph), (b) mid (20-60 mph), and (c) high (60+ mph).

At low speeds, we observe overall high performance, with all the 5G mmWave points concentrated in this region, as expected. These points are mostly from the cities, where the speed is typically low. On the other hand, the high speed region consists of data mostly from the inter-state highways. This region has the maximum number of points for all the operators as most of our data was collected on highways. Surprisingly, we notice a significant fraction of throughput samples with high values even at such high speeds for Verizon and T-Mobile – several 100s of Mbps in the downlink and several 10s of Mbps in the uplink – mostly due to the deployment of 5G midband and 5G-low (in the case of T-Mobile only) along highways. On the other hand, throughputs are much lower with AT&T, which has much lower 5G midband coverage (Fig. 2.3a). Interestingly, throughputs in the mid-speed region are much lower than in the high-speed region for Verizon and AT&T. Most of the data points in this region are from sub-urban areas in-between cities/towns and inter-state highways. Our data suggest sparser cellular deployments in these areas in comparison to cities and highways, and sparser 5G coverage for Verizon and AT&T, causing the performance to drop considerably for these two carriers. Fig. 2.8 also shows a large number of very-low throughput points in all three regions, suggesting *a weak correlation between the throughput and the vehicle's speed* (see also Table 2.2).

In Fig. 2.9, we plot the RTT against the vehicle's speed in both directions for each operator, using the same three speed bins as in Fig. 2.8. In contrast to throughput, (Fig. 2.8), RTT appears to have a stronger correlation with the vehicle's speed, for two of the three operators (Verizon and T-Mobile); RTTs, regardless of technology, are in general lower in the (0-20 mph) bin and higher in the (60+ mph) bin. On the other hand, for AT&T, the LTE/LTE-A RTTs are in general higher than

Table 2.2: Pearson's correlation coefficient between throughput and a number of KPIs.

	RSRP		MCS		CA		BLER		Speed		HO	
	DL	UL	DL	UL	DL	UL	DL	UL	DL	UL	DL	UL
Verizon	0.06	0.49	0.25	0.40	0.35	0.07	-0.08	-0.04	-0.29	-0.30	-0.02	-0.02
T-Mobile	0.46	0.51	0.34	0.62	0.29	0.05	0.23	0.10	-0.34	-0.10	-0.04	-0.05
AT&T	0.35	0.30	0.23	0.28	0.58	0.29	-0.13	-0.04	-0.37	-0.15	-0.05	-0.05

the 5G RTTs in all three speed bins. Another interesting observation from Fig. 2.9 is the absence of mmWave points for Verizon and AT&T except at very low speeds (close to 0 mph), showing again that operators typically do not upgrade a UE to 5G mmWave in the case of low ICMP traffic. Surprisingly, for AT&T, the RTT of this small number of 5G mmWave samples is higher than the 5G midband RTT.

In the remainder of this section, we focus on throughput. We look at a set of other factors that affect cellular performance and examine if they can explain the large throughput variation regardless of speed in Fig. 2.8. We consider 5 common cellular network KPIs – Primary Cell's RSRP, Primary Cell's MCS, Carrier Aggregation (CA), Primary Cell's Block Error Rate (BLER), and the number of handovers (HO) – and compute the Pearson correlation coefficient of throughput with each of them in Table 2.2. We observe that *(1) none of these KPIs (including speed) has a strong correlation with throughput and (2) the factors that have the highest impact on throughput are different for different operators and even for different traffic directions for the same operator*. Surprisingly, *there is no correlation between throughput and the number of handovers*. We analyze this result in depth in §2.4. In the following, we investigate the root causes of the low correlation of some of the remaining KPIs with throughput.

RSRP. Table 2.2 shows weak-to-medium correlation with throughput for all operator-traffic direction combinations, but no correlation in the case of Verizon downlink. By inspecting our traces, we found that the RSRP for 5G mmWave (which primarily contributes the high downlink throughput samples) is low for most samples in the case of Verizon (-80 to -110 dBm), resulting in almost 0 correlation with throughput, but high in the case of AT&T (-70 to -90 dBm). The reason for this discrepancy lies in the different beamwidths of the phased arrays used by the two operators. In most of the cities, Verizon uses a smaller number of wider beams compared to AT&T, which result in lower gain, and hence, lower RSRP.

MCS. Table 2.2 shows a significantly higher correlation with throughput for T-Mobile uplink compared to the other operator-direction pairs. Here, we note that, because of heavy CA, the primary

cell's MCS is not always a good proxy for the total throughput. For example, if secondary cells have strong link supporting high MCS indices, they can contribute to a high throughput overall even if the primary cell's MCS is low.

CA., Table 2.2 shows weak-to-medium correlation with throughput for all three operators in the downlink and for AT&T uplink, but no correlation in the case of Verizon and T-Mobile uplink. We found that Verizon rarely uses CA in the uplink, which explains the low correlation with throughput. On the other hand, T-Mobile often aggregates 2 carriers in the uplink. However, the use of two carriers does not always manifest to higher throughput, as, in most cases, at least one carrier is LTE (this is true even in the case of 5G midband due to dual connectivity). Since LTE carriers have lower bandwidth than 5G carriers, a larger number of carriers does not necessarily boost the overall throughput.

Overall, we conclude that it is hard to isolate the impact of a single KPI on performance. An in-depth understanding of the impact of multiple KPIs on performance requires a multivariate analysis, which is part of our future work.

2.3.6 Performance Over Longer Time Scales

Up till now, we focused our analysis on short time scales (500 ms). As our mobile app experiments (§2.5) last between 20-180 s each, we now analyze the network performance on the same timescale. This analysis also allows us to compare our data with reports published by commercial bandwidth measurement apps (e.g., Ookla SpeedTest), which measure throughput at similar time scales.

We compute the mean and std. dev. of throughput (as a percentage over the mean) and RTT samples for each 30-second throughput test and each 20-second RTT test and plot their CDFs in Fig. 2.10. The upper rows show that Verizon, T-Mobile, and AT&T, respectively, achieve median DL throughput of 30, 37, 48 Mbps, UL throughput of 13, 14, 10 Mbps, and RTT of 64, 82, 81 ms. Note that the median throughput is higher than that in Fig. 2.4 (which shows the CDF of 500 ms throughput samples), as the throughput of the samples is long-tailed. Nonetheless, the overall performance remains poor in terms of both metrics. Furthermore, the lower row shows that the throughput and RTT experience high fluctuation within 30 s, with median values of 70%, 48%, 52% for DL throughput, 45%, 52%, 44% for UL throughput, and 18%, 29%, 19% for RTT.

To examine if the low throughput and high RTT values are a result of poor high-speed 5G coverage, in Fig. 2.11 we plot the throughput/RTT of each test against the fraction of time during which the UE was connected to 5G mmWave or midband. We observe that only T-Mobile's

Table 2.3: Comparison with Ookla's report for Q3 2022.

	Downlink (Mbps)		Uplink (Mbps)		RTT (ms)	
	Our Data	Speedtest	Our Data	Speedtest	Our Data	Speedtest
Verizon	29.62	58.64	13.18	8.30	63.71	59.00
T-Mobile	37.09	116.14	13.77	10.91	81.68	60.00
AT&T	48.40	57.94	9.80	7.55	80.73	61.00

midband service brings a substantial improvement in throughput and only for the downlink direction (Fig. 2.11a). For the other two operators in the downlink direction and for all three operators in the uplink direction, *the throughput values are similar regardless of the percentage of time the UE was connected to 5G mmWave/midband* (with the exception of a few outliers). The same observations is true for the RTT values (Fig. 2.11c).

Finally, in Table 2.3, we compare the median throughput and RTT values from our dataset against those reported by Ookla SpeedTest [20] during Q3 2022 (which includes the time during which we collected our dataset). Compared to the results in the Ookla report, our results show significantly lower DL throughput, slightly higher UL throughput, and higher RTTs (especially for T-Mobile and AT&T). Assuming that most SpeedTest measurements are performed by static users[2], the significantly lower median DL throughput values and higher RTT values in our dataset demonstrate the performance degradation during driving. Still, this comparison should be taken with a grain of salt, as the numbers in the Ookla report are calculated over the whole country and the app typically selects servers close to the UE's location and uses multiple TCP connections to measure the peak bandwidth. In contrast, as explained in §2.3, most of our measurements are done with remote cloud servers and we used a single TCP connection, as our intent was to measure performance experience by most cloud-based apps.

2.4 Handovers

Handover statistics. We use the data from the throughput tests to quantify the frequency of handovers (HOs) across all cellular technologies. We calculate the total number of HOs (horizontal HOs between cells within the same technology and vertical HOs across technologies) experienced over a 30 s test and normalize it by the distance traveled in that test. Fig. 2.12a shows the results during downlink and uplink tests. Although we observed more than 20 HOs per mile in extreme cases, *in general,*

[2]The Ookla report does not provide any information about mobility.

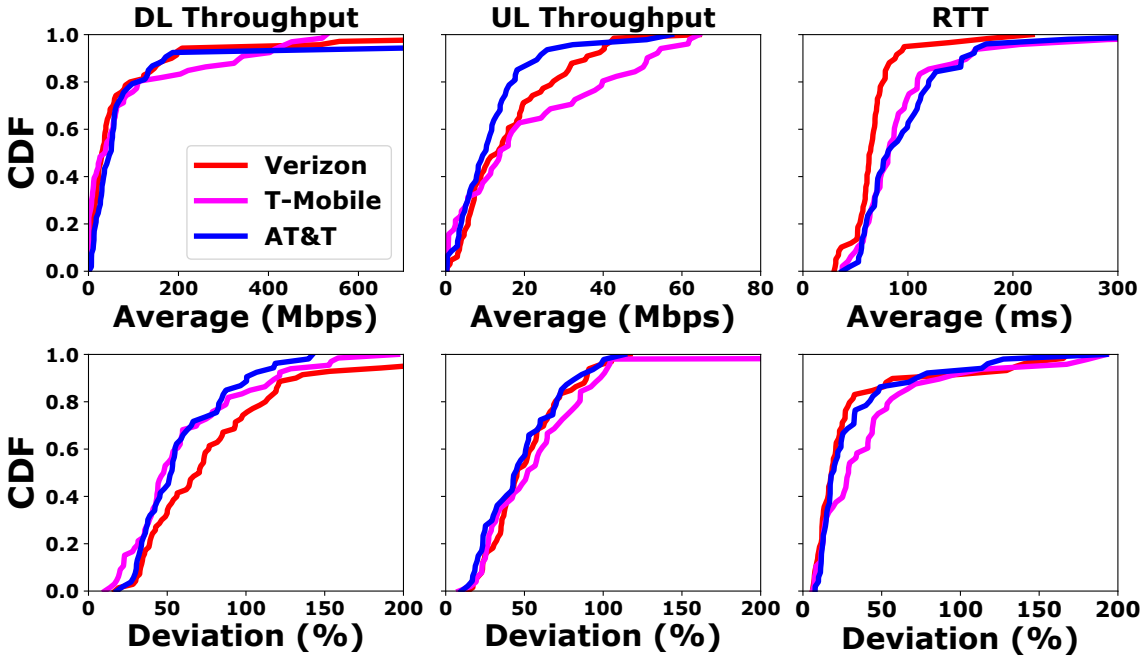
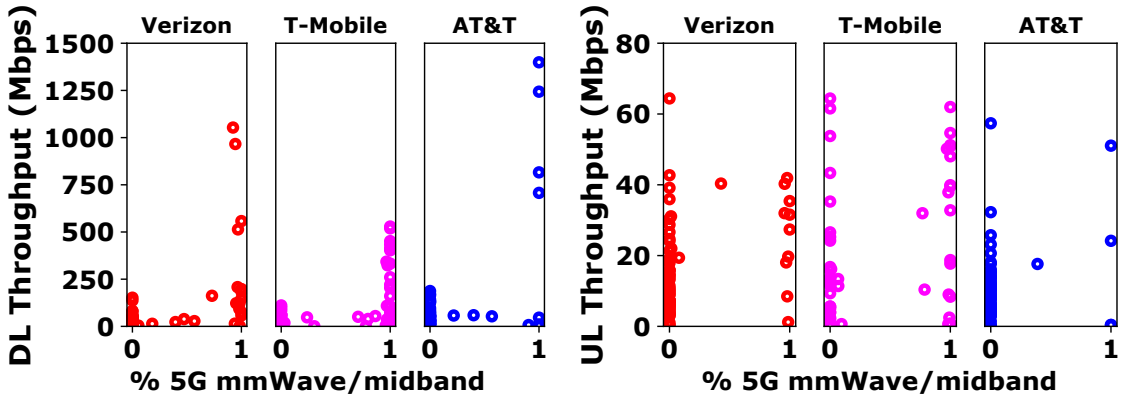


Figure 2.10: CDF of average & standard deviation of individual throughput and RTT tests.

the number of HOs per mile is low. The median (75-th percentile) for Verizon/T-Mobile/AT&T, respectively is 3 (6)/2 (5)/2 (5) in the case of downlink traffic and 2 (5)/2 (6)/1 (3) in the case of uplink traffic. Additionally, the CDF of the HO duration in Fig. 2.12b shows that *most handovers are fast*; the median (75-th percentile) HO duration for Verizon/T-Mobile/AT&T is 53 (73)/76 (107)/58 (74) ms in the case of downlink traffic and 49 (63)/75 (101)/57 (73) ms in the case of uplink traffic.

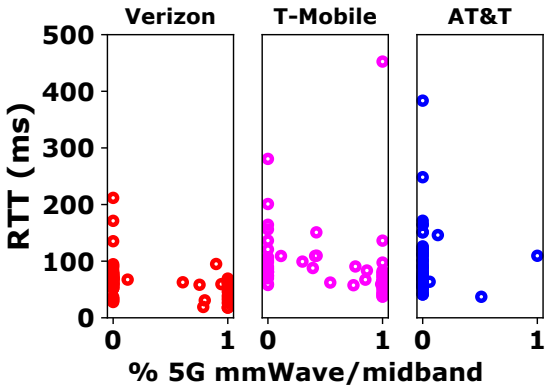
Impact of handovers on throughput. We now analyze the impact of a HO on network throughput. Specifically, we ask two questions: (1) how much does the throughput drop during a HO compared to the average throughput before and after a HO? and (2) how much does the throughput after a HO change compared to the throughput before the HO? Refer to Fig. 2.12c, where the time is measured in multiples of 500 ms (XCAL's throughput logging frequency) and assume that a HO takes place between t_2 and t_3. Then, to answer (1), we calculate $\Delta T_1 = T_3 - (T_2 + T_4)/2$, i.e., the difference between the throughput of the 500 ms interval that experienced one or more HOs and the average throughput over one 500 ms interval before and one 500 ms interval after the HO(s). Similarly, to answer (2), we calculate $\Delta T_2 = (T_4 + T_5)/2 - (T_1 + T_2)/2$, i.e., the difference between the post- and pre-HO throughput, each averaged over 1 s.

Figs. 2.13a-2.13c, 2.13d-2.13f show the CDF of the difference between the throughput during a HO and the average of just before and after the HO, for downlink and uplink, respectively.



(a) DL Throughput.

(b) UL Throughput.



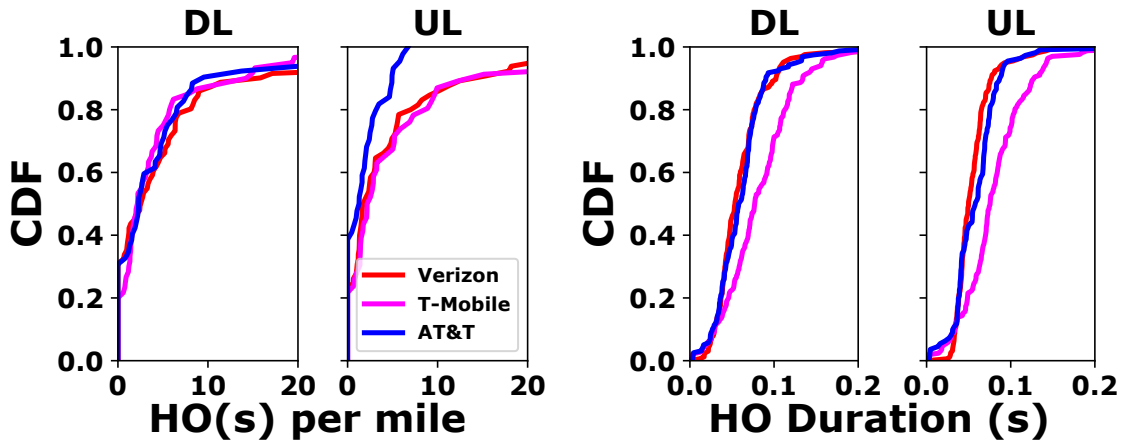
(c) RTT.

Figure 2.11: Performance as a function of the % of time the UE was connected to 5G mmWave/midband.

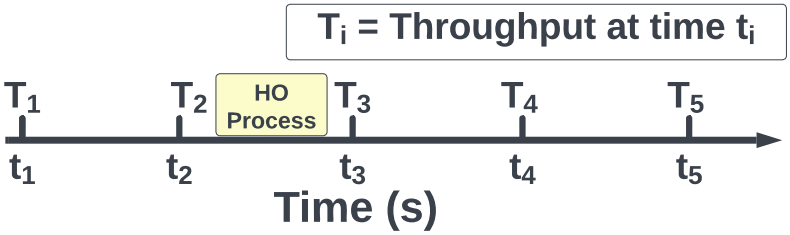
Irrespective of the traffic direction and operator, the values are lower than 0 around 80% of the time, indicating a drop in throughput during the HO. Nonetheless, the throughput drop is in general small, up to 60-80 Mbps in the downlink and up to 20-30 Mbps in the uplink.

Figs. 2.14a-2.14c, 2.14d-2.14f show the CDF of the difference between the post- and pre-HO throughput in the downlink and uplink directions, respectively, for each operator. About 55-60% of the time, the post-HO throughput is higher than the pre-HO throughput, i.e., a HO improves the performance. While the median throughput difference is very low (0.5-2 Mbps), the improvement can be as high as 100 Mbps in the downlink and as high as 20 Mbps in the uplink, which partly counterbalances the throughput drop during a HO.

Interestingly, for all operators and in both directions, the post-HO throughput is lower than the pre-HO throughput for about 25% of the time. To investigate further, we break down in the same



(a) HOs per mile. (b) HO duration.

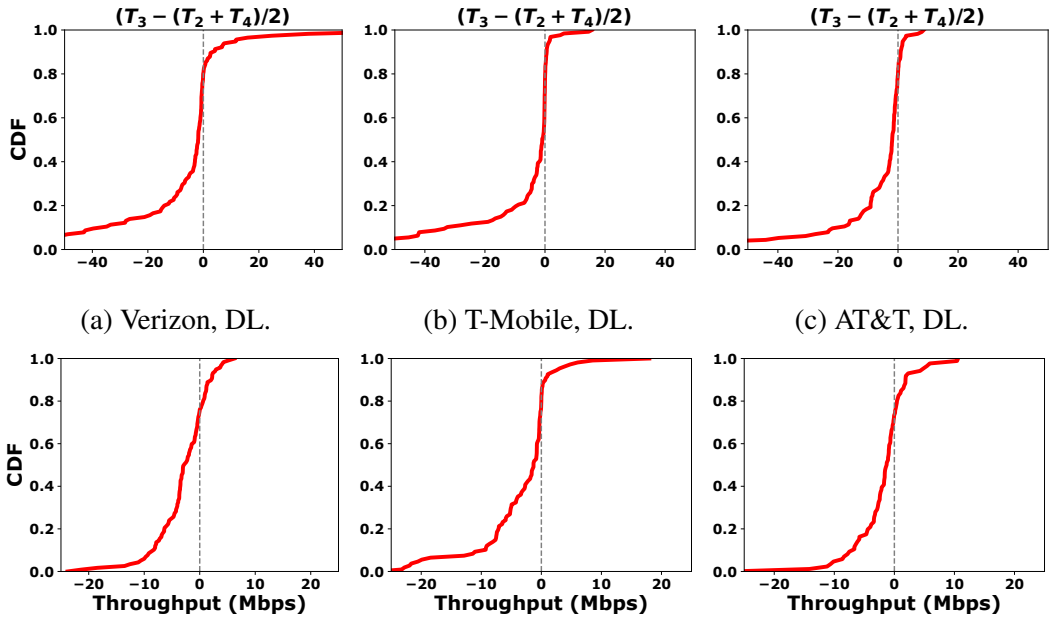


(c) HO process timeline.

Figure 2.12: Handover process and related statistics.

figures the HO types into vertical (4G→5G, 5G→4G) and horizontal (4G→4G, 5G→5G) HOs. We observe that, as expected, among the four types of HO, 5G→4G is the type that mostly results in lower post-HO throughput, while 4G→5G typically improves the throughput. We also observe that horizontal HOs often result in lower post-HO throughput, but the impact is small (the negative part of these two CDFs has lower values than their positive part). A similar observation was reported in [21] for 4G→4G HOs and in [3] for NSA 5G→5G HOs in the mmWave bands.

Together, the small average number of HOs per mile (Fig. 2.12a), the HO duration (Fig. 2.12b), and our analysis of the impact of HOs on throughput (Figs. 2.13, 2.14) explain the low correlation between throughput and HOs in Table 2.2. Assume an average driving speed of 60 mph, then from Fig. 2.12a, there are 1-3 HOs per mile for every 60 s or at most 0.025 HOs for every 500 ms interval in the median case, i.e., most 500 ms intervals do not experience any HO. In addition, the impact of a HO on the overall throughput is small (Figs. 2.13a-2.13c and 2.13d-2.13f), and for about 25% of the cases, the post-HO throughput is actually lower than the pre-HO throughput (Figs. 2.14a-2.14c and 2.14d-2.14f).



(a) Verizon, DL. (b) T-Mobile, DL. (c) AT&T, DL.

(d) Verizon, UL. (e) T-Mobile, UL. (f) AT&T, UL.

Figure 2.13: Impact of handovers on throughput $(T_3 - (T_2 + T_4)/2)$.

2.5 5G Applications

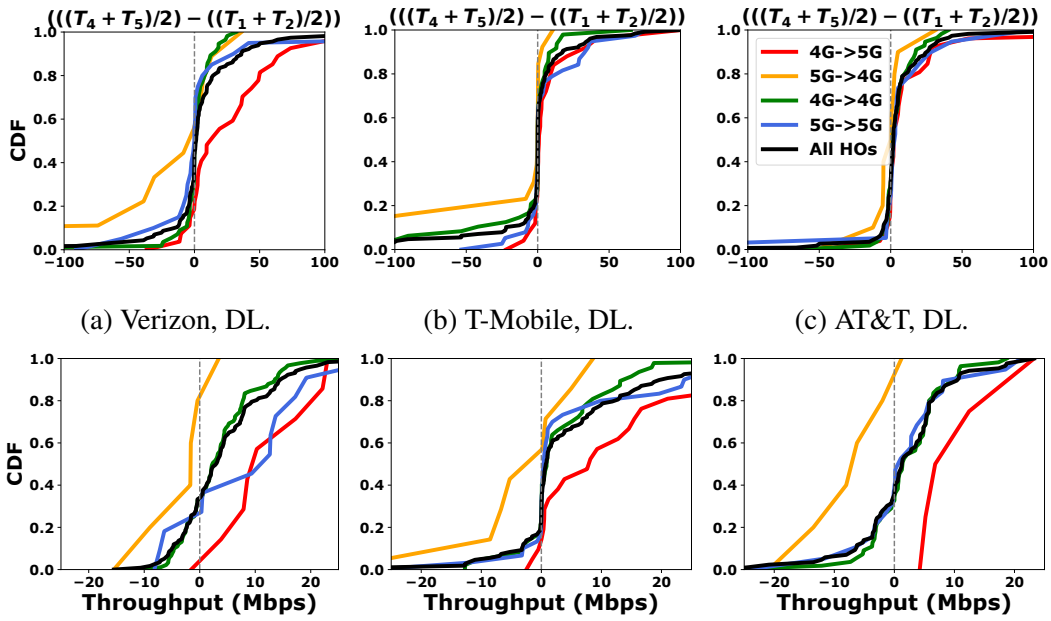
In this section, we analyze the feasibility of today's cellular networks in supporting a set of "5G killer" apps, which are all latency-critical and demand high bandwidth.

2.5.1 Uplink-centric Apps

AR and CAV apps are two representative uplink-centric 5G apps. Various works have shown that offloading the computation to a powerful edge server enables AR to achieve high-quality object detection [22, 11, 23], depth estimation [24, 25] and SLAM [26]. Similarly, CAVs rely on offloading vehicle-captured camera frames or LIDAR point clouds to the edge to enable collaborative vehicle perception [27, 12, 28]. To achieve high QoE, these apps have stringent requirements for high network throughput and low RTT.

2.5.1.1 Methodology

In practice, commercial apps like these still do not exist today. Therefore, we built a canonical edge-assisted AR/CAV benchmark app, consisting of an Android app that offloads pre-recorded frames to an edge GPU server (with Nvidia A100) in a best-effort manner. We list the key



(a) Verizon, DL. (b) T-Mobile, DL. (c) AT&T, DL.

(d) Verizon, UL. (e) T-Mobile, UL. (f) AT&T, UL.

Figure 2.14: Impact of handovers on throughput $((T_4 + T_5)/2 - (T_1 + T_2)/2)$.

parameters in Table 2.4. Both the server and the Android app we wrote can be configured to simulate various application scenarios, by specifying the frame size, frame rate (FPS) of incoming frames, frame compression time (if the app uploads compressed frames), and server inference time. The configuration parameters for AR and CAV are shown in Table 2.4 and are taken from [11, 12].

In each test, we run each app with and without frame compression configured, resulting in four runs each lasting 20 seconds. In each run, we measure the averaged E2E offloading latency, and the number of frames offloaded per second. For the AR app, we additionally estimate the object detection accuracy in mean average precision (mAP).

2.5.1.2 Estimating Object Detection Accuracy from E2E Latency

To understand the relationship between the end-to-end (E2E) offloading latency and object detection accuracy in terms of mAP, we performed an offline study with the phone offloading to an edge-server, where we varied the E2E offloading latency by varying the network bandwidth, and measured the object detection accuracy under different E2E latencies. We used Argoverse [29], a dataset recorded by a RGB camera mounted on a top of the car. We assume the edge server runs Faster R-CNN [30], a popular two-stage object detection model. The AR app runs an off-the-shelf on-device local-tracking algorithm [31], which moves the existing bounding boxes to follow the

Table 2.4: Configurations for the AR & CAV application.

	AR	CAV
Frames per second (FPS)	30	10
Frame size (raw)	450 KB	2000 KB
Frame size (compressed)	50 KB	38 KB
Frame compression time	6.3 ms	34.8 ms
Server inference time (A100)	24.9 ms	44.0 ms
Frame decompression time	1.0 ms	19.1 ms
Duration of a run	20 s	20 s

objects in the frame and is shown to improve edge-assisted object detection performance [27, 12].

Table 2.5 shows the relationship between E2E offloading latency and the object detection accuracy. Since the AR app produces object detection results for each incoming frame based on the latest received result from the server, the object detection accuracy is the same as long as the E2E is within the same bin in the unit of frame times. Since frame compression is lossy and thus results in different object detection accuracy, we report mAP both with and without frame compression.

2.5.1.3 Main Results

Since Verizon benefits from edge servers and achieves the lowest RTT (Fig. 2.4b, Fig. 2.10), we discuss its results in detail and briefly cover the corresponding results for T-Mobile and AT&T later in this section. The findings reported in this section hold true for all three carriers.

Performance of the AR App

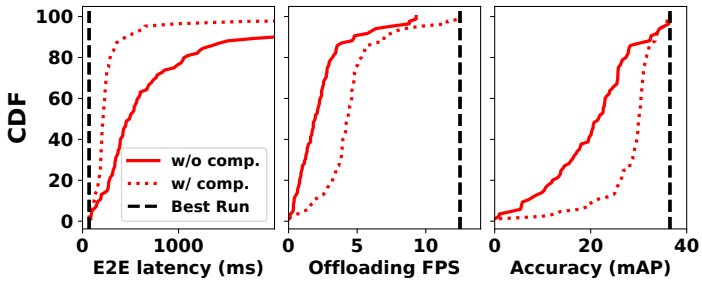
Fig. 2.15 shows the performance of the AR app. We make the following observations. (1) Even in the best static scenario, the app achieves E2E offloading latency of 68 ms, offloads only 12.5 FPS, and achieves an object detection accuracy (mAP) of only 36.5%. (2) *The AR app performance is significantly impacted by driving, achieving much higher E2E latency compared to the best static scenario, marked by a dashed vertical line, and thus much lower offloading frame rate and object detection accuracy.* Even with frame compression, the median E2E latency is 214 ms – 3x that of the best static case. As a result, the app achieves a median offloading frame rate of only 4.35 FPS and mAP of 30.1 – 34.8% and 82.4% of those under the best static scenario, respectively. (2) Fig. 2.15b shows that *5G mmWave/midband improves the worst case performance compared to 4G/5G-low, and the use of an edge server boosts the performance regardless of technology.* However, *the performance overall remains low even when the phone is connected to high-speed 5G 100% of*

Table 2.5: Object detection accuracy (mAP) with the Argoverse dataset and Faster R-CNN model, at each E2E latency bin.

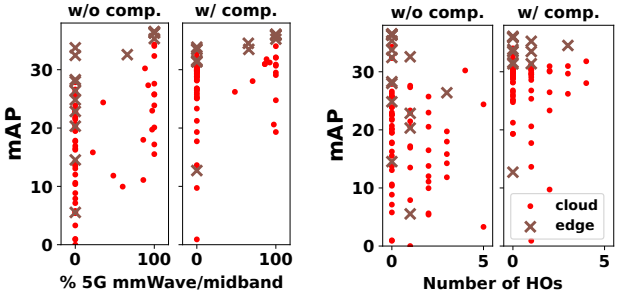
E2E latency (frame times)	mAP w/o compression	mAP w/ compression	E2E latency (frame times)	mAP w/o compression	mAP w/ compression
0-1	38.45	38.45	15-16	20.59	20.35
1-2	37.22	36.14	16-17	20.11	19.69
2-3	36.04	34.75	17-18	19.53	18.95
3-4	34.65	33.12	18-19	18.40	17.61
4-5	33.36	31.82	19-20	18.01	17.85
5-6	32.20	30.50	20-21	17.52	17.00
6-7	31.08	29.53	21-22	16.96	16.55
7-8	28.03	26.99	22-23	16.59	15.97
8-9	27.01	25.73	23-24	15.41	15.16
9-10	25.62	25.21	24-25	15.78	14.94
10-11	25.77	24.35	25-26	15.86	15.37
11-12	23.29	22.44	26-27	14.81	14.71
12-13	22.75	21.56	27-28	14.70	13.77
13-14	22.48	21.64	28-29	14.44	13.62
14-15	21.59	21.16	29-30	14.05	13.70

the time. (3) Fig. 2.15c shows that the app experiences up to 5 handovers within a 20-second run, but *there is no strong correlation between the number of handovers and mAP, suggesting that the impact of handovers on the AR app performance is limited.* We attribute this to the low impact of handovers on second-scale throughput (§2.4) and the AR app's built-in mechanism for mitigating transient throughput drops, local tracking, which effectively reuses the previously server-returned result [25]. (4) Fig. 2.15a and Fig. 2.15b show that compression significantly reduces the E2E offloading latency, enabling more frames to be offloaded and improving the accuracy. This shows that *despite the promised high throughput of 5G, application-level optimizations developed in the pre-5G era such as frame compression are still needed.*

Performance of the CAV App Fig. 2.16 shows the performance of the CAV application over Verizon. We make the following observations. (1) Today's cellular networks fail to achieve an E2E latency of 100 ms needed for the CAV pipeline to achieve accurate view reconstruction [27, 32, 33]. Fig. 2.16a



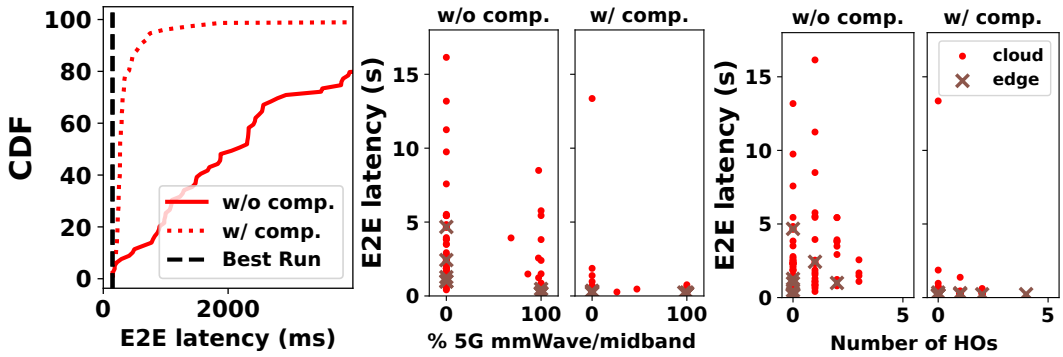
(a) Performance CDFs.



(b) Impact of tech. (c) Impact of HO.

Figure 2.15: Performance of the AR app with Verizon.

shows the median E2E latency during driving is 269 ms if frame compression is used, and the lowest E2E latency recorded throughout our trip is as high as 148 ms. As shown in Table 2.4, the DNN inference time is 44 ms, and assuming the optimal RTT of 15 ms when connected to an edge server via 5G mmWave, to achieve an E2E latency of 100 ms the app needs to transmit a 2000 KB frame in 41 ms, which requires an uplink bandwidth of 390 Mbps – far beyond the capability of today's 5G mmWave. Although point cloud compression significantly reduces the frame size from 2000 KB to 38 KB, it incurs a compression time of 34.8 ms and decompression time of 19.1 ms, which makes it impossible for the app to achieve 100 ms E2E latency.(2) Although frame compression does not reduce the E2E latency to below 100 ms, it still helps by reducing the median E2E latency by 8X, suggesting that *app-level optimizations such as point cloud compression are still needed in the 5G era*. (3) Similar to the AR app, *the use of an edge server improves performance compared to a cloud server regardless of the cellular technology and high-speed 5G only improves the worst case performance compared to 4G/5G-low* (Fig. 2.16b). Also, *there is no obvious correlation between number of handovers and E2E latency* (Fig. 2.16c).

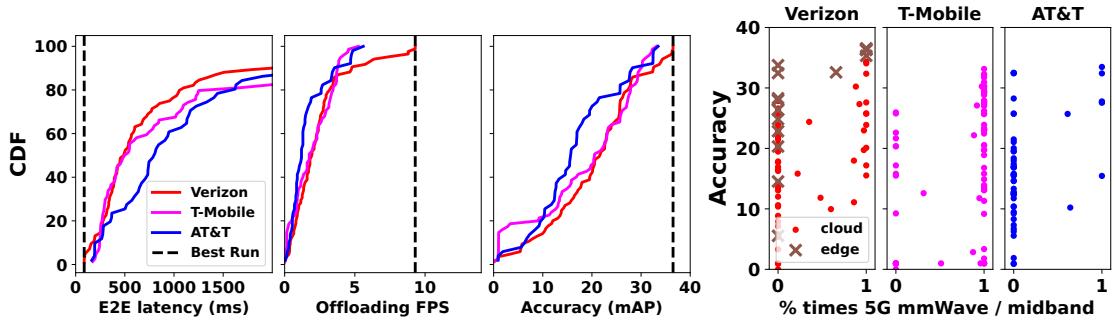


(a) Perf. CDFs. (b) Impact of tech. (c) Impact of HO.

Figure 2.16: Performance of the CAV app with Verizon.

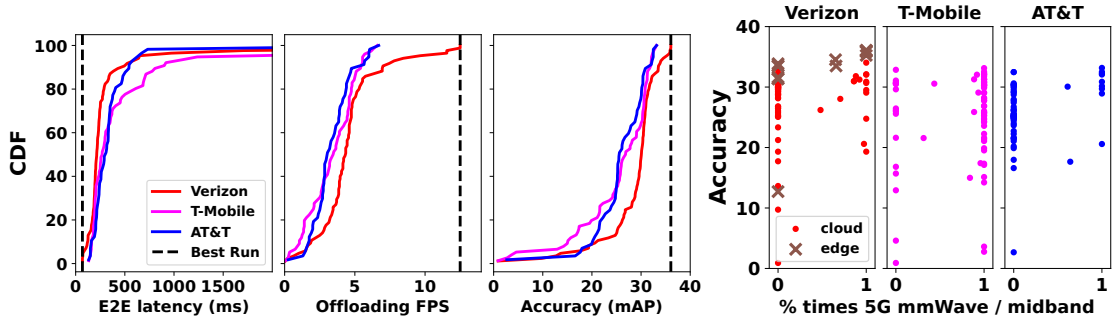
2.5.1.4 Results for all three operators

Figs. 2.17, 2.18, 2.19 show the performance of the AR and CAV apps for all three major U.S. operators. We found that the same observations made for Verizon (§2.5.1.3) also apply to T-Mobile and AT&T. In terms of cross-operator comparison, we make the following observations: (1) For the AR application, Verizon achieves the lowest E2E offloading latency, which results in the highest offloading FPS and object detection mAP across the three operators. This is because Verizon achieves the lowest median RTT of 63.7 ms, in comparison to T-Mobile's 81.7 ms and AT&T's 80.7 ms. AT&T achieves the worst performance without frame compression but slightly better performance than T-Mobile with frame compression. (2) The lead in Verizon is more significant when frame compression is used. Since frame compression reduces the frame size significantly from 450 KB to 50 KB, Verizon's shorter RTT plays a more important role in reducing the E2E latency than the high throughput. (3) For the CAV application, T-Mobile achieves the lowest E2E latency among the three operators, when point cloud compression is not used. This is because uncompressed point clouds have a large size of 2 MB, which benefits from T-Mobile's highest UL throughput compared to Verizon and AT&T (see Fig. 2.10). When point cloud compression is used, there is no significant difference between the performance of the three operators. (4) The impact of cellular technology is limited for both apps. In the case of AR, for AT&T, similar to Verizon (Figs. 2.15b, 2.16b), high-speed 5G improves the worst-case performance, while for T-Mobile, it improves the best-case performance without compression. However, the maximum object detection accuracy remains very low (below 36%) with all three operators regardless of technology. Similarly, in the case of CAV, high-speed 5G improves the worst-case performance for Verizon and AT&T but it has



(a) CDF of app performance. (b) Impact of 5G connectivity.

Figure 2.17: AR without frame compression.



(a) CDF of app performance. (b) Impact of 5G connectivity.

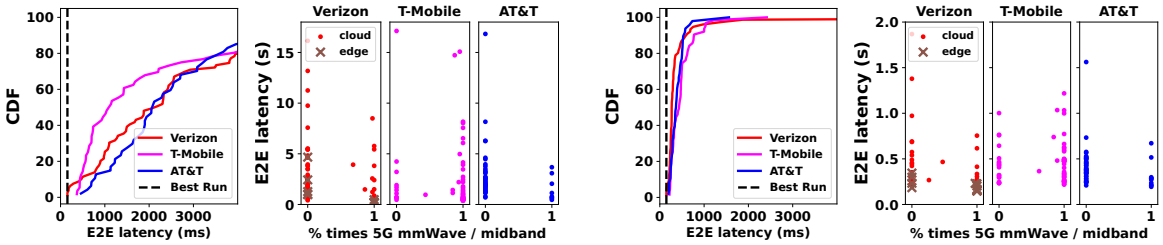
Figure 2.18: AR with frame compression.

no impact for T-Mobile.

2.5.2 360° Video Streaming

Methodology

We evaluate 360° video performance by streaming videos from a media server to a mobile client during driving. We used Puffer, an open source video streaming server [34], and built our own client to receive and display video chunks for 360° videos. We customized Puffer's ABR algorithm to run BBA [35], which only relies on buffer size to choose a video bitrate and skips instances when capacity estimation is not needed. The 360° sample videos are chosen from YouTube and publicly hosted on an AWS instance. Each video is segmented into 2-second chunks and encoded into 4 different quality settings (100, 50, 10, and 5 Mbps) using ffmpeg [36]. Each playback session is run for 3 minutes. During each session, we collect three metrics – avg. QoE, avg. bitrate, and rebuffer time as a % of the duration of a run. We calculate the QoE [37] of a chunk based on the weighted sum of three elements, (1) video bitrate, (2) video bitrate variation between successive chunks, and



(a) App performance. (b) Impact of 5G connectivity. (c) App performance. (d) Impact of 5G connectivity.

Without compression **With compression**

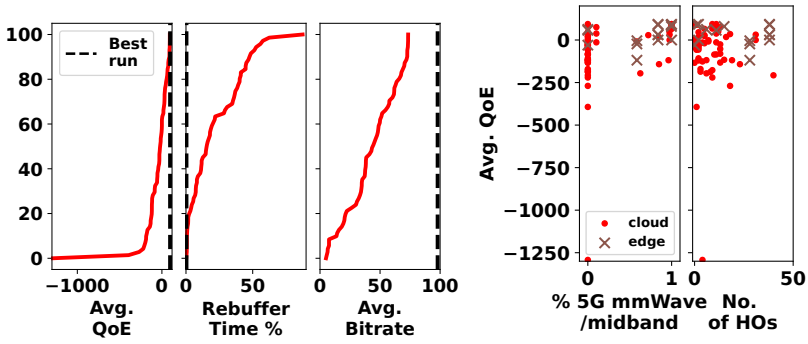
Figure 2.19: CAV performance with and without compression.

(3) rebuffer time: $QoE_k = B_k - \lambda|B_k - B_{k-1}| - \mu T_k$, where B_k is the bitrate of chunk k, T_k is the rebuffer time recorded while downloading chunk k, and λ and μ are weighting parameters. The QoE of a run is the average QoE of all the individual chunks downloaded during that run. We empirically choose λ = 1 and μ = 100 following the guideline in [37].

Main Results Fig. 2.20 shows the results for the Verizon network. We make the following observations. (1) The QoE [37] varies significantly during driving (Fig. 2.20a) with a median value of -53.75, which is significantly lower than the value of the best static run (96.29), suggesting that *high-quality 360° video streaming cannot be supported under driving*. Note that the theoretical best value is 100 assuming no stalls and no bitrate switch. (2) The average QoE during driving is negative for 40% of the runs due to the high rebuffering time, which can be as high as 87% of the total playback time (Fig. 2.20a). (3) Compared to the AR and CAV apps, Fig. 2.20b (left) shows that *the cellular technology has a bigger impact on video streaming QoE*; most of the runs when the UE was connected to high-speed 5G 100% of the time have positive QoE. This is because, compared to the AR and CAV apps, video streaming is not as latency-critical due to buffering, and its performance mostly depends on the network bandwidth, which is typically higher with 5G mmWave/midband. (4) Fig. 2.20b shows that connecting to an edge server generally results in higher QoE. (5) As with AR/CAV, *there is no strong correlation between QoE and the number of handovers*. Some runs with 40+ handovers yielded better QoE than runs with just 5 handovers.

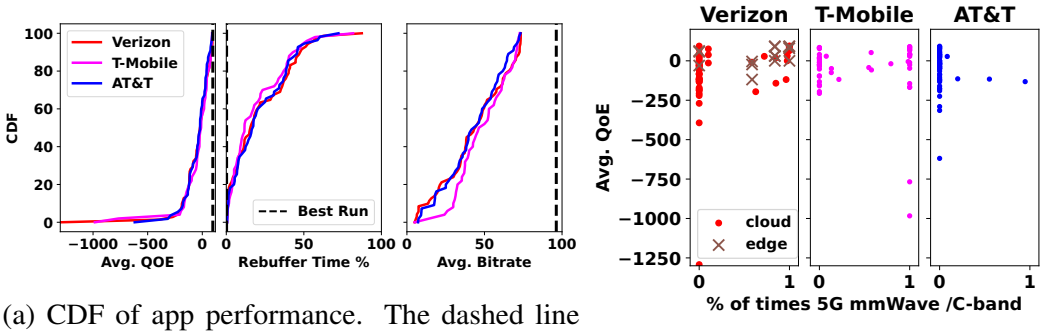
Results for All Three Operators

Fig. 2.21 shows the performance of the 360° video streaming application for all three major U.S. operators. The same observations made for Verizon also apply to T-Mobile and AT&T. In terms of cross-operator comparison, we make the following observations: (1) All operators achieve similar QoE, rebuffering, and average bitrate, with T-Mobile doing slightly better in terms of both rebuffering and average bitrate. (2) The cellular technology appears to have no impact on the performance



(a) Performance CDFs. (b) Tech. & HO impact.

Figure 2.20: 360° video streaming with Verizon.



(a) CDF of app performance. The dashed line represents the best run.

(b) Impact of 5G connectivity.

Figure 2.21: 360° video performance.

with T-Mobile; in fact, the two runs with the worst QoE happened with the phone connected to 5G midband 100% of the time. For AT&T, the phone was only connected to 4G/5G-Low during almost all but 3 runs.

2.5.3 Cloud Gaming

2.5.3.1 Methodology

We deployed an AWS GPU instance to host two popular games: CSGO [38] and Hitman 2 [39] using the popular Steam Remote Play [40] gaming platform. These games are then played on our SGS21 smartphones using the Steam Link [41] Android application with 4K at 60FPS settings. The cloud server streams video frames at 4K resolution; however the frames are downscaled to 2K on the local device to fit SGS21's maximum resolution. We measure cloud gaming performance based on three metrics: send bitrate (Mbps), network latency (ms), and frame drop rate (%). The bit-rate information can be extracted by looking up the bit rate adapter's information from the Steam server's

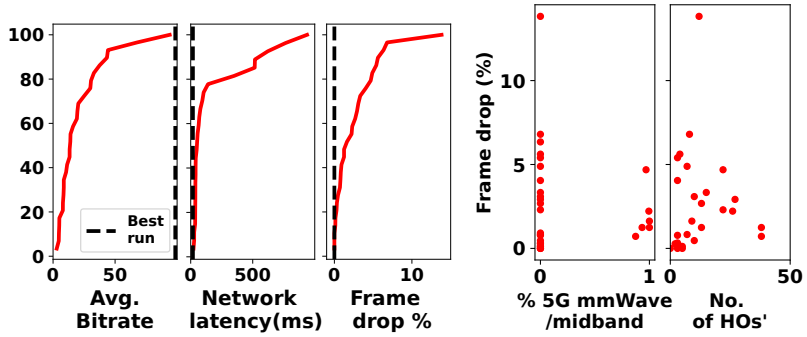
logs. We note that the maximum target value that can be set by the bitrate adapter is 100 Mbps. Additionally, the application server also reports network latency information every time it estimates a significant change in latency values. Finally, the percentage of frames dropped also heavily affects user's experience. To quantify the metric, we collect the number of frame drop events every second and divide it by the most recently set frame rate (FPS).

2.5.3.2 Main Results

Fig. 2.22 shows the results over Verizon. We make the following observations. (1) Fig. 2.22a shows that *the user QoE degrades significantly during driving* compared to the best static run. The sending bitrate varies significantly during driving (Fig. 2.22a) with a median value of only 17.5 Mbps, significantly lower than the value of the best static run (98.5 Mbps). The network latency during driving is always higher than 17 ms (best static run, required to stream at 60 FPS) and higher than 200 ms for 20% of the runs, which results in unacceptable user QoE. On the other hand, the frame drop rate is typically low (median value around 1.6%) but can be as high as 13.2%. In contrast, the best static run has a frame drop rate of 0.5%. (2) These numbers suggest that Steam Remote Play tries to keep the frame drop rate low (by adapting the frame rate) even at a cost of very high latency. (3) Similar to the AR and CAV apps, *5G midband/mmWave improves the worst case QoE* (the frame drop rate is never higher than 5%), but, overall, *does not improve the QoE compared to LTE/5G-low*. In fact, some runs with the UE connected to 5G midband/mmWave 100% of the time had higher frame drop rate than some runs with the UE connected to LTE/5G-low 100% of the time. (4) Similar to all the other apps, *there is no correlation between the QoE and the number of handovers*. **Note:** The work in [3] shows that handovers have a large impact on the QoE of apps such as video streaming and cloud gaming, which contrasts our findings in this section. We note that the authors in [3] drew their conclusions from a small number of experiments (e.g., an 8-min drive in the case of cloud gaming), in contrast to our results which are obtained from a cross-country drive.

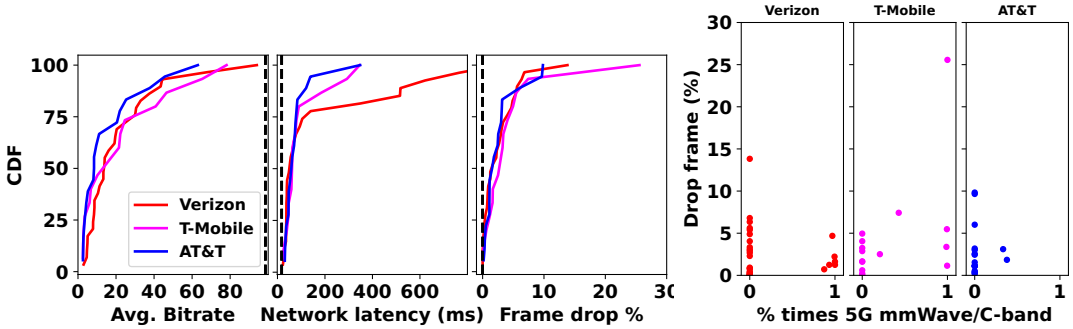
2.5.3.3 Results for All Three Operators

Fig. 2.23a shows the performance of cloud gaming application for all three operators. The same observations made for Verizon also apply to T-Mobile and AT&T. In terms of cross-operator comparison, we make the following observations: (1) Fig. 2.23a shows that T-Mobile achieves the highest median bitrate at 21 Mbps, followed by Verizon (19 Mbps) and AT&T (9 Mbps). (2) In terms of network latency, the median values of all operators are approximately the same (around 50 ms).



(a) Performance CDFs. (b) Tech. & HO impact.

Figure 2.22: Cloud gaming with Verizon.



(a) CDF of app performance. Dashed line = best run. (b) Impact of 5G connectivity.

Figure 2.23: Cloud gaming performance.

However, the app under Verizon sometimes experienced very high network latency of 500-1000 ms, while under the other two operators never exceeded 200 ms. (3) The frame drop rates with Verizon and AT&T are similar whereas T-Mobile sometimes experienced extreme frame drop rate (up to 25%). (4) 5G midband/mmWave has no impact for AT&T and T-Mobile.

2.6 Related Work

2.6.1 Measurements under Limited Mobility

Various works have measured 5G performance in the U.S. [6, 7, 8, 1, 2]. The work in [1] studies 5G network performance, power consumption, and application QoE from a single location to multiple servers across the country, but it does not collect lower-level signaling messages. Other works study 5G performance in one or few cities, e.g. in Minneapolis [6, 7], Chicago [8, 7, 2], Boston [2, 42], Atlanta [7, 42], and Rome [43, 44], and thus the reported numbers do not reflect the

rural or country-wide performance. Furthermore, all these studies are conducted mainly under static and walking scenarios, and although most of them include some driving experiments, they are small in scale, e.g. spanning a few city blocks.

Outside the U.S., the works in [45] and [43, 44] study 5G performance in China and Italy, respectively, but focus only on the midband, and perform their measurements in a single city. The work in [46] studies mobile access bandwidth in China by collaborating with a commercial bandwidth testing app, and the work in [47] conducts a large-scale measurement study of 5G across a whole country (UK) from a mobile operator's perspective, focusing on network usage and temporal evolution. Different from our study, these works do not capture information such as mobility and low-level signaling messages.

Outside academia, several commercial bandwidth testing apps (e.g., Ookla SpeedTest [48], OpenSignal [49], nPerf [50]) provide crowdsourced-based 5G coverage and performance maps and some also publish periodic reports, e.g., [51, 20], comparing the major U.S. operators in terms of 5G coverage, performance, and QoE of a small set of applications (video, gaming, voice). Similar to the works in [46, 47], these measurements do not capture information such as user mobility or 5G band and do not have access to lower layer signaling messages. In §2.3.6, we compare the average driving network performance from our dataset against the average performance reported by OpenSignal for the 3 major U.S. operators.

2.6.2 Network Measurements under Driving

Driving measurements in 5G. The work in [3] performed extensive 5G measurements over the 3 major U.S. carriers while driving, but only focused on handovers. More importantly, the study mostly discusses findings using a single operator or short driving segments as examples, or presents cumulative results over all three operators. In contrast, we provide a detailed comparison of the handover behavior and their impact on network performance and application QoE across all 3 operators. The work in [9] also analyzes the inefficiencies in handover behavior via a 45-hour driving experiment, but only studies a single operator (AT&T).

Driving measurements pre-5G. The work in [52] studied handover policies via driving tests in three U.S. cities, and the works in [53, 54] conducted extensive measurements of cellular performance on high-speed rails (HSRs) in China. However, these measurements were conducted for LTE & 3G networks before 5G was rolled out. Additionally, the network performance on HSRs can be different from driving due to the difference in device speed and infrastructure provisioning.

2.7 Summary and Recommendations

In this study, we collected a first-of-its-kind multi-carrier, multi-technology, multi-band, cross-layer cellular network dataset through a cross-continental US driving trip (5700km+). Using this dataset, we revealed characteristics of today's cellular networks under driving, in terms of coverage, geo-diversity, operator diversity, network performance, and app performance. Our study showed disappointingly low and fragmented 5G coverage and poor network performance, even in areas with full high-speed 5G coverage, which, in turn results in poor user QoE for major "5G killer" apps, compared to static conditions. Our initial analysis, using low-level signaling messages and a number of KPIs, showed that no KPI is highly correlated with throughput, suggesting that further research is required to understand the factors that affect network performance during driving. We make three recommendations based on our results: (1) app developers should continue to explore app-level optimizations (compression, local tracking, buffering, rate adaptation, etc.) from the pre-5G era; (2) smartphone vendors should explore multipath solutions over multiple cellular networks; and (3) network operators and cloud providers should collaborate in deploying more edge services embedded in the operator networks.

The findings above capture a snapshot of operational 5G networks performance and coverage in 2022, three years after its initial rollout. Given how quickly operators expand coverage, adjust carrier aggregation levels, and transition between deployment architectures, a single snapshot cannot reveal whether the gaps we identify here persist, worsen, or close over time. Understanding this requires tracking the evolution of these networks by revisiting the same measurement methodology at a later point in time. We take up this task in the next chapter, where we replicate this study two years later along the same cross-country route.

Chapter 3

Performance of Cellular Networks on the Wheels v.2

In the previous chapter (§2), we discussed the results from a cross-country drive (from Los Angeles to Boston, 5700+ km) in the US, 3 years after the initial 5G rollout (2022), measuring cellular network coverage, network performance, and the QoE of a set of major 5G "killer" apps (cloud-assisted AR, CAVs, 360° video streaming, and mobile cloud gaming) over all three major US operators. The goal was to present a holistic view of operational 5G networks in driving scenarios, capturing the real world network capabilities over a broad geographic footprint. The findings revealed disappointingly low 5G coverage while driving for all three operators – with two of them having less than 25% 5G coverage along the driving route. More critically, our study showed extremely poor network and application performance even in areas with full 5G coverage.

5G deployment has advanced significantly since the 2022 study. Operators are transitioning from Non-Standalone (NSA) to Standalone (SA) 5G, introducing features like higher carrier aggregation (CA) and new frequency bands with diverse duplexing modes – all aimed at improving performance. At the same time, the wireless landscape is evolving with the rise of non-terrestrial networks (NTNs). In particular, Low-Earth-Orbit (LEO) satellite networks are transforming global connectivity by offering internet access in remote or underserved areas where traditional networks fall short. These changes in the wireless ecosystem necessitate revisiting such large-scale measurement studies to reassess the current state of cellular coverage and performance while users are on the move, and to compare it against the state established by the previous chapter back in 2022.

Numerous studies [55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67] have evaluated various

5G aspects after our 2022 study was performed. However, most are limited in geographic scope, focusing on one [56, 61, 62] or a few cities [55, 60, 67, 57, 61, 58, 65]. A few studies use large datasets collected via driving measurement campaigns [59, 64], but each of them focuses on a specific 5G aspect, e.g., CA [59], or multi-carrier access [64], and none of them tracks performance evolution over time. Some works study the evolution of 5G over extended time periods using crowd-sourced datasets from commercial or custom bandwidth measurement apps [63, 66]. While these datasets offer broader geographical and user coverage compared to cross-country drives, they do not capture information such as user mobility or lower layer features, they do not obtain measurements over different networks at the same location and time, and they do not analyze application performance.

To bridge this gap, this chapter replicates our 2022 study presented in the previous chapter (§2), driving along the same route from Los Angeles to Boston, while simultaneously measuring the coverage and performance of all three major US operators. Additionally, we expand our 2022 study on two fronts: (a) by exploring the impact of different lower layer features such as CA, bandwidth configurations, and MIMO on the network performance, and (b) by conducting simultaneous measurements with Starlink and performing a head-to-head comparison with cellular networks. Our key findings are as follows:

• All three operators have significantly expanded their 5G coverage in 2024, with two operators heavily deploying high-speed 5G (5G-mid or 5G-mmWave) and extending it beyond urban centers. However, two out of three operators still offer 5G coverage in less than 50% of the driving route even five years after the initial 5G rollout.

• The effect of improved coverage is clearly visible in the network performance with a dramatic improvement in throughput from 2022 to 2024 for all three operators. On the other hand, the RTT remained roughly unchanged for two operators and improved by less than 15% in the median case for the third operator, suggesting that latency remains a more complex challenge than throughput.

• All three operators significantly expanded their 5G-mid downlink (DL) CA levels and aggregate bandwidth, with T-Mobile and AT&T also increasing their uplink (UL) bandwidth. The expansion is more limited in the 5G-low bands, where only T-Mobile introduced CA in 2024. In 5G-mmWave, we saw increased CA levels with Verizon in the UL direction, enabled by the capabilities of newer UE models, but, surprisingly, a drop in CA levels in the DL direction. Furthermore, we saw a sharp decline in MIMO capabilities, with all three operators significantly reducing the number of 4x4 MIMO base stations (gNBs) in 2024.

• The total number of handovers (HOs) per mile increased substantially in 2024 compared to 2022. On the other hand, with the exception of 5G→LTE HOs in T-Mobile's 5G SA deployments, the HO

duration either remained the same or reduced in 2024.

• Our head-to-head comparison of cellular networks against Starlink shows that Starlink's network availability along the driving route was 98% (vs. 100% for the cellular networks). Its RTT is the lowest among the four networks and its median DL throughput is commensurate with that of cellular networks. On the other hand, its peak DL throughput and its UL throughput are significantly lower than that of cellular networks.

• Our replication of the mobile app study from the previous chapter for two 5G "killer" apps using a different methodology confirmed the main finding of the previous chapter: cellular networks were not able to support those apps in 2022. The improved 5G-mid coverage and network performance in 2024 results in a large QoE improvement for these two apps, however, fallbacks to 5G-low/LTE still hurt the performance. Importantly, application-level optimizations developed in the pre-5G era, such as frame compression and local tracking, remain relevant even in 2024, despite the drastic improvements in cellular performance. Despite its lower RTT compared to cellular networks, Starlink offers the lowest QoE for both apps due to its lower throughput.

We make our dataset available to the research community to facilitate additional replicability studies [68]. For brevity, throughout this chapter we refer to the study presented in the previous chapter (§2) as the *previous study*.

3.1 Methodology

We adopt the same methodology as the previous study with a few key modifications. Our measurement campaign was conducted along the same cross-country route on November 1–8, 2024. Our testbed consisted of four smartphones, each connected to a different network operator (Verizon, AT&T, T-Mobile, Starlink) to *concurrently* measure the performance over all the available networks. Three of the phones were connected to XCAL Solo [14] devices to collect lower-layer cellular KPIs. For Starlink-specific data collection, grpcurl [69] was utilized to fetch and save terminal information from gRPC APIs exposed on the router's local network. Similarly to the previous study, we deployed AWS EC2 cloud servers in California and Ohio, and AWS EC2 Wavelength edge servers in Los Angeles, Las Vegas, Denver, Chicago, and Boston for tests done with Verizon in those cities.

A major difference from the previous study lies in the user equipment (UE). While the previous study used Samsung Galaxy S21 Ultra devices (featuring the Snapdragon X65 modem), we used Samsung Galaxy S24 Ultra phones equipped with the newer Snapdragon X75 modem. S24 supports CA of up to 4 component carriers (CCs) in 5G-mid/5G-low in the DL direction and in

5G-mmWave in the UL direction, compared to 2 CCs in S21. Our goal is to capture the evolution of cellular network coverage and performance driven not only by infrastructure advancements but also by improvements in end-user devices. For our LEO satellite measurements, we used a Gen 3 standard Starlink terminal with a dish router with the unlimited "roam" plan. The terminal is tightly mounted on the roof of the vehicle. The UE connects to the Starlink router via an 802.11ax 6e link. The complete testbed is shown in Fig. 3.1.

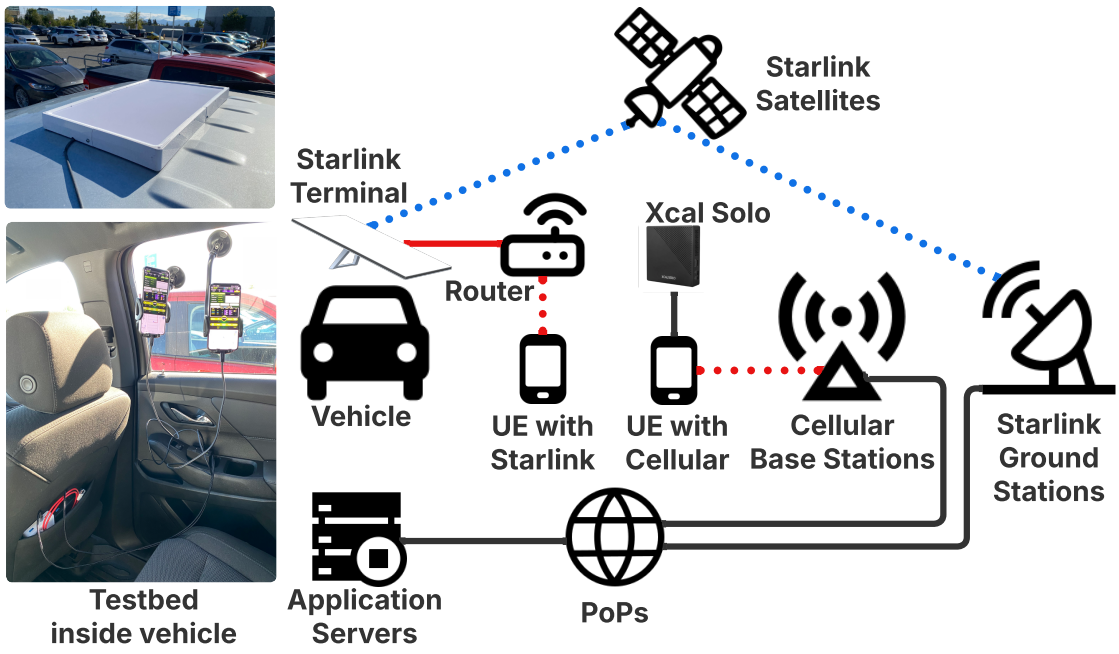
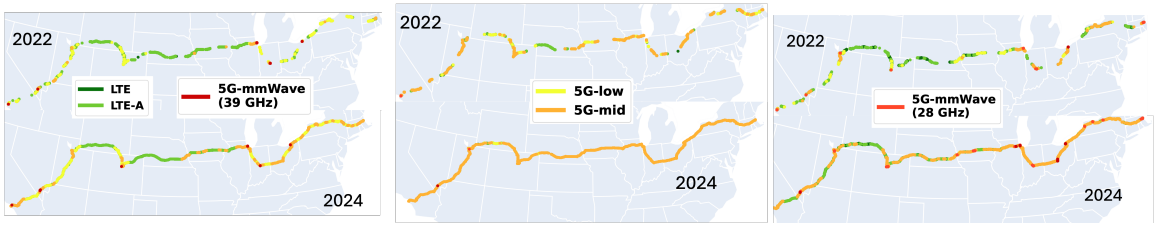


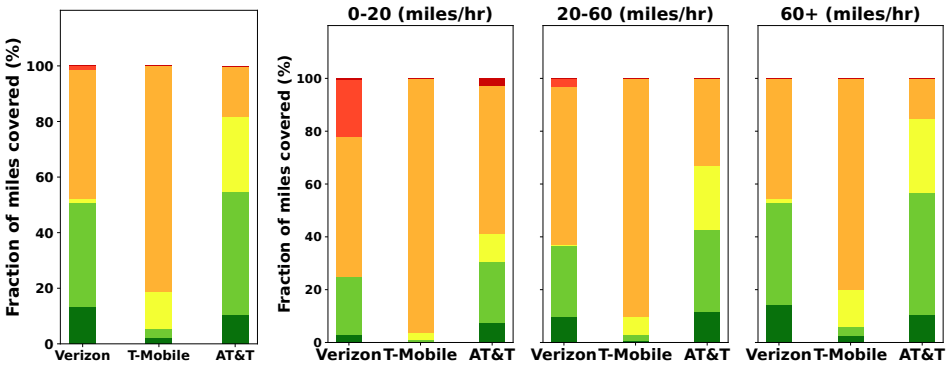
Figure 3.1: Overview of our measurement testbed.

A second difference from the previous study lies in the type of measurements we conducted. The previous study executed a suite of tests in a round-robin fashion: a 30-35 s DL/UL TCP throughput test using nuttcp, followed by a 20 s RTT test using ICMP ping, and 6 min of application tests per round. This limited the number of collected throughput and RTT samples, as most of the time in each round was spent on application tests. Since backlogged throughput tests and RTT tests are better suited for revealing peak network capabilities, our methodology prioritizes collecting more throughput and RTT samples. Thus, we increased the duration of DL/UL tests to 2 min and the duration of RTT tests to 30 s, and unlike the previous study, we did not measure the performance of "5G killer apps" in real time. Instead, we evaluate two of these representative applications – AR (UL-oriented) and 360° streaming (DL-oriented) – via trace-based emulation using the throughput and RTT data from the publicly available dataset in the previous study and from our dataset. The

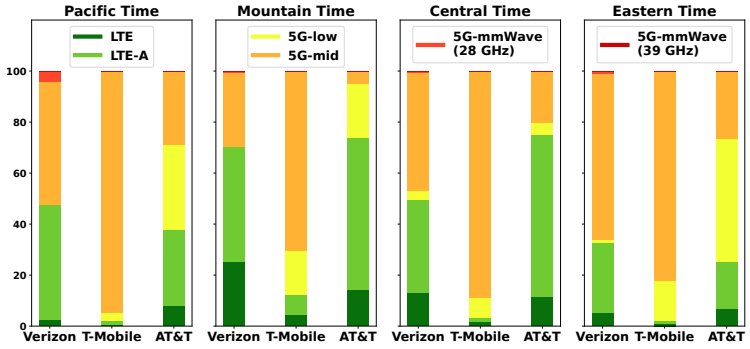


(a) AT&T. (b) T-Mobile. (c) Verizon.

Figure 3.2: 2022 vs. 2024: Driving route and cellular network coverage.



(a) Overall. (b) By vehicle speed.



(c) By timezone.

Figure 3.3: Cellular technology breakdown per operator in 2024.

emulation methodology is described later in detail in §3.7.

3.2 Coverage

We begin by comparing the cellular network coverage of the three major US operators between 2022 and 2024. Following the definition used in the previous study, we define coverage for a given cellular technology as the fraction of miles driven under that technology. Fig. 3.2 presents

a bird's-eye view of the cellular network coverage along our driving route, segmented by different technologies, showing a substantial increase in 5G coverage in 2024 compared to 2022 for all three operators. The total 5G coverage has increased from 22%/68%/18% in 2022 (Fig. 2.3a in the previous study) to 49%/94%/45% in 2024 (Fig. 3.3a) for Verizon/T-Mobile/AT&T, and the high-speed 5G coverage (5G-mid and 5G-mmWave) has increased from 16%/37%/2% to 48%/81%/18%.

Although all three operators demonstrate substantial improvement in terms of 5G coverage in 2024, Fig. 3.3a shows that Verizon and T-Mobile primarily rely on high-speed 5G technologies. In contrast, AT&T continues to depend significantly on the low-speed, but robust 5G low-band spectrum. Nevertheless, five years after the initial 5G rollout, Verizon and AT&T still rely on LTE along considerable portions (49% & 45%) of the driving route.

We also found that T-Mobile is the only operator with an extensive country-wide 5G SA deployment in 2024 (60.5% vs. only 2.5% in 2022). Verizon only recently started deploying SA services; we found a very small percentage (0.003%) of 5G SA in the 2024 dataset. Finally, for AT&T, we found no 5G SA in any of the two datasets.

We further analyze the coverage based on vehicle speed in Figure 3.3b. We observe a notable decline in 5G coverage when moving from low to high-speed bins for Verizon and AT&T, aligning with observations in the previous study. Different from the previous study, T-Mobile still maintains an impressive 5G coverage ($\geq$90%) on highways in 2024. Additionally, while high-speed 5G technologies decreased notably under higher vehicle speeds for all three operators in 2022, our results highlight a shift in deployment strategies in 2024. Specifically, for Verizon and T-Mobile, high-speed 5G coverage remains dominant across all speed bins, accounting for $\geq$98% and $\geq$90% of their total 5G coverage respectively. In contrast, AT&T continues to rely predominantly on low-band 5G, especially in high-speed regions (highways).

Similar to the observations in the previous study, only AT&T and Verizon have non-negligible 5G mmWave coverage in 2024. However, there is a notable change for Verizon between the two years. In 2022, both operators' 5G-mmWave was restricted to low-speed areas, typically downtowns of large cities. In contrast, in 2024, we observe roughly 2.4% 5G-mmWave coverage for Verizon in the mid-speed bin (20–60 mph). Further analysis revealed that we were able to connect to 5G mmWave from freeways in metropolitan regions such as Los Angeles and Chicago. Interestingly, Verizon's mmWave coverage in the low-speed bin has declined from 35% in 2022 to 24% in 2024, revealing a shift towards increased reliance on 5G-mid band deployments over mmWave.

Finally, we break down the coverage by timezone in Fig. 3.3c. Similarly to 2022, operators continue to adopt diverse deployment strategies in different regions in 2024. For Verizon and AT&T,

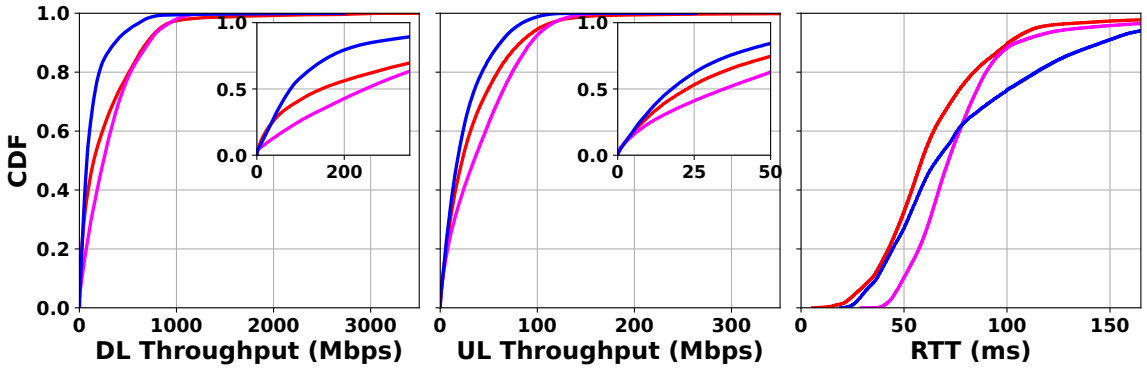


Figure 3.4: Overall throughput and RTT of all three operators during driving in 2024.

5G coverage is noticeably higher in the Pacific and Eastern timezones, likely driven by higher population densities and greater demand in those areas. T-Mobile, in contrast, maintains relatively uniform 5G coverage across all timezones. When focusing on high-speed 5G coverage, we observe notable regional disparities. For T-Mobile and AT&T, the Pacific timezone has the highest high-speed 5G coverage (28% and 94%, respectively), while for Verizon, the highest high-speed 5G coverage is found in the Eastern timezone (66%). Notably, the Mountain timezone still remains the most underserved region in terms of high-speed 5G coverage in both years for all three operators.

In contrast to 2022, the 2024 deployment landscape shows notable shifts across timezones. Verizon, which had lower overall 5G coverage in the Pacific timezone compared to the Central timezone in 2022, reversed that trend in 2024. For T-Mobile, while the Pacific timezone continues to lead in high-speed 5G coverage, in 2024 we see a significant boost in high-speed 5G across all other timezones as well.

Summary. 5G coverage has improved significantly in 2024, but two of the three operators still deliver 5G along less than 50% of the route. Verizon and T-Mobile have shifted toward broader deployment of high-speed 5G technologies, extending beyond urban centers into mid- and high-mobility areas, whereas AT&T continues to rely heavily on 5G low-band, particularly on highways. 5G-mmWave deployment remains sparse; Verizon shows some signs of extending it beyond city areas, but, at the same time, it replaces it with 5G-mid in urban areas. The Pacific and Eastern timezones lead in both overall and high-speed 5G availability, reflecting operator prioritization in denser, high-demand regions.

Table 3.1: Cellular performance metrics (25th/50th/75th percentiles) across technologies: 2022 vs. 2024.

(a) DL throughput (Mbps).

	Verizon				T-Mobile				AT&T			
	LTE	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**
2022	2/11/32	3/15/38	31/101/207	401/1035/1508	1/4/21	1/3/23	1/21/234	-	7/22/50	22/50/109	9/46/72	4/740/1334
2024	10/29/71	42/72/128	164/336/566	185/763/1632	1/6/26	34/72/149	143/293/502	-	24/55/101	30/64/118	81/218/409	87/680/1395

(b) Uplink throughput (Mbps).

	Verizon				T-Mobile				AT&T			
	LTE	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**
2022	3/8/17	5/13/25	7/18/34	38/42/51	2/6/15	1/4/21	1/9/42	-	3/7/12	10/17/29	7/18/33	0/0/0
2024	4/11/22	1/7/51	15/37/65	27/51/109	0/1/4	5/13/34	16/41/72	-	4/10/18	12/27/47	11/29/53	7/48/105

(c) Round-trip time (ms).

	Verizon				T-Mobile				AT&T			
	LTE	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**	**LTE**	**5G-low**	**5G-mid**	**5G-mmWave**
2022	52/64/73	60/69/88	51/57/65	18/24/33	63/70/82	71/84/111	47/62/78	-	61/91/122	56/62/74	36/46/63	88/112/129
2024	54/65/84	58/94/109	41/53/72	16/26/49	77/100/151	68/79/94	58/69/83	-	66/96/129	44/56/75	36/47/59	83/96/121

3.3 Network Performance

Our results in §3.2 showed a substantial 5G coverage growth for all three operators from 2022 to 2024. To understand whether this increased coverage translates into better user experience, we now perform a head-to-head comparison of the network performance across the two years.

3.3.1 Overall Performance While Driving

Fig. 3.4 presents the throughput and RTT CDFs for the three operators during driving. We observe that *all three operators exhibit a multi-fold improvement in throughput in 2024 in both traffic directions*. The median DL throughput increased from 16/6/34 Mbps in 2022 to 148/248/77 Mbps in 2024 for Verizon, T-Mobile, and AT&T, respectively. Similarly, the median UL throughput increased from 9/6/8 Mbps to 23/35/18 Mbps. The peak performance also improved, most notably for Verizon – the DL (UL) throughput increased from $\sim$2.2 Gbps ($\sim$100 Mbps) in 2022 to over 3.5 Gbps ($\sim$350 Mbps) in 2024.

Interestingly, the ranking among operators has changed drastically in 2024 compared to 2022. In the DL direction, AT&T outperformed the other two operators 80% of the time in 2022 (Fig. 2.4b in the previous study), despite its low 5G coverage. In contrast, in 2024, T-Mobile consistently achieves the best performance, followed by Verizon. In fact, the relative gain in median throughput from 2022 to 2024 is the lowest in case of AT&T (2x) in comparison to Verizon and T-Mobile (9x and 41x, respectively). In the UL direction, T-Mobile lagged behind the other two

operators 55% of the time in 2022, but it is clearly the best performing operator in 2024, with a median improvement of 5.8x compared to 2.25x for AT&T and 2.5x for Verizon.

A comparison between Fig. 3.4(right) and Fig. 2.4b in the previous study reveals that *the latency did not follow the throughput trend*. For Verizon and T-Mobile, the 25th and 50th RTT percentiles remain largely the same between 2022 and 2024 (Verizon: 48/60 ms in 2022 vs. 46/59 ms in 2024, T-Mobile: 57/71 ms in 2022 vs. 60/72 ms in 2024). T-Mobile saw a modest improvement from 93 ms to 86 ms at the 75th percentile, but Verizon's 75th percentile increased from 71 ms to 78 ms. AT&T is the only operator that reduced its latency in 2024, with its 25th/50th/75th percentiles dropping from 57/77/111 ms to 48/66/103 ms. Nonetheless, Verizon still provides the lowest latency in 2024, partly due to the use of edge servers, although the gap with T-Mobile reduces drastically, especially above the 80th percentile; and AT&T's latency remains the highest among the three operators about 40% of the time.

3.3.2 Performance Breakdown By Cellular Technology

Table 3.1 breaks down the 2022 and 2024 throughput and RTT metrics by cellular technology. We make the following observations.

(1) 5G-mmWave still provides the best DL/UL performance at the 50th and 75th percentiles in 2024. In particular, AT&T's mmWave UL performance has improved drastically (the previous study reported 0 UL throughput 90% of the time). However, at the 25th percentile, T-Mobile's and Verizon's 5G-mid throughput is higher than AT&T's mmWave throughput in the DL direction; and all three operators' 5G-mid throughput is higher than AT&T's mmWave throughput in the UL direction. Additionally, Verizon's DL 5G mmWave throughput has substantially decreased in 2024 compared to 2022, even though it still remains higher than the throughput of all other technologies. Recall that Verizon extended its 5G mmWave service to freeways outside urban centers in 2024. These newer deployment zones may be less optimized or more prone to mobility-related impairments. Another possible reason for the performance drop is increased congestion in the mmWave bands, as more users upgraded to mmWave-capable devices in 2024 (see §3.4.1.1).

(2) The large improvements in the overall throughput in 2024 (Fig. 3.4) mostly stem from the boost in the 5G-mid performance with improvements ranging from 3x-14x in the median case. This is true particularly for T-Mobile and Verizon, whose 5G coverage in 2024 is predominantly based on 5G-mid deployments (Fig. 3.3). In contrast, while AT&T also shows substantial improvements in 5G-mid throughput in 2024, its 5G coverage is still largely dominated by the low-band spectrum,

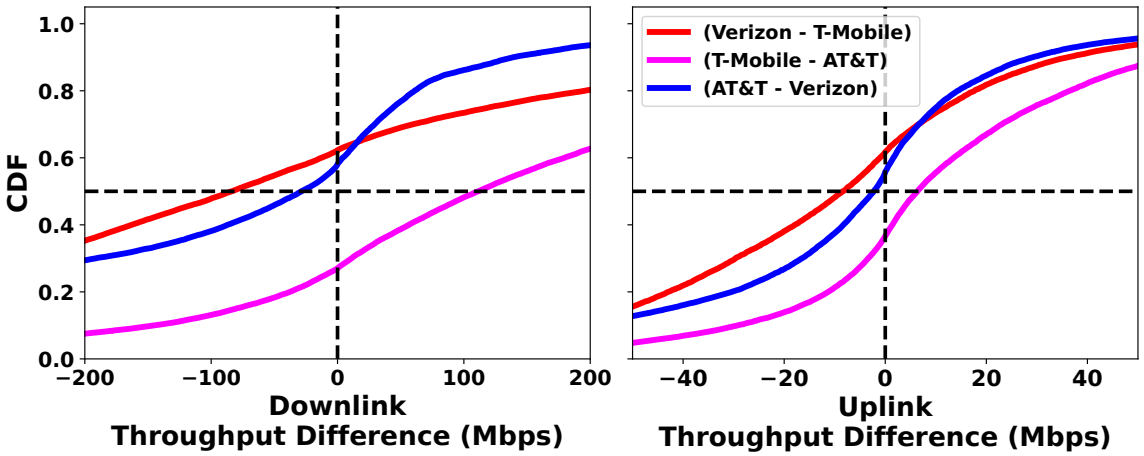


Figure 3.5: CDF of throughput difference for each pair of operators in 2024.

which provides much lower throughput than 5G-mid, as shown in Table 3.1. This explains AT&T's overall lower throughput in 2024 compared to the other two operators.

(3) The latency shows different trends compared to throughput. For Verizon and T-Mobile, the latency has increased for most technologies in the median case from 2022 to 2024. Specifically, Verizon's median RTT rose by 2% for LTE, 36% for 5G-low, and 8% for 5G-mmWave, while 5G-mid was the only exception, improving from 57 ms to 53 ms. T-Mobile shows a similar pattern, with the median RTT increasing by 43% for LTE and 11% for 5G-mid, while 5G-low saw a modest 6% improvement. In contrast, AT&T shows a more favorable trend: RTT dropped for all 5G technologies but slightly increased for LTE.

Summary. We observe a drastic improvement in cellular network performance under driving from 2022 to 2024, driven largely by the maturation of 5G mid-band deployments. T-Mobile and Verizon show the largest performance boost due to their aggressive mid-band rollouts, while AT&T, with heavier reliance on 5G low-band, sees more modest gains. On the other hand, latency remains a complex challenge, shaped by multiple factors, including backend architecture and scheduling policies, which likely explains the modest or even negative improvements observed from 2022 to 2024 for two of the three operators.

3.3.3 Operator Diversity in Cellular Network Performance

The previous study explored operator diversity in cellular network performance at a given location and time and found that the performance at a given location and time can be highly diverse across operators in both directions. We repeat this analysis for our 2024 dataset (Fig. 3.5).

While operator diversity still exists in 2024, we observe a fundamental difference compared to 2022. In 2022 (Fig. 2.7a in the previous study), the throughput difference CDFs for all operator pairs in both traffic directions were symmetric around 0, indicating roughly equal probability for one operator to outperform the other at a given location and time. In contrast, in 2024, the CDFs are no longer symmetric around 0, especially in the DL direction. T-Mobile, due to its extensive 5G-mid coverage in 2024, outperforms Verizon/AT&T 65%/75% of the time in the DL direction and 60%/65% of the time in the UL direction.

We also observe that the spread of the CDFs widens in 2024 compared to 2022 – especially in the DL direction. The steeper slopes in 2022 reflect smaller, more consistent differences, whereas the flatter CDF curves in 2024 capture the growing divergence in operator capabilities. Nevertheless, the core insight from the previous study still holds – leveraging operator diversity through multi-carrier access continues to offer strong potential for improving overall performance.

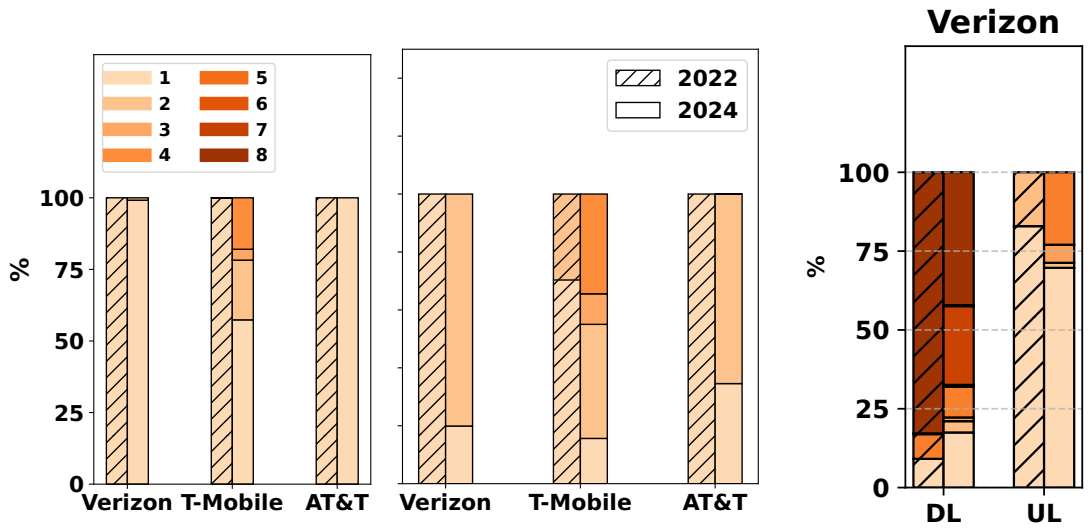
3.4 5G Features

In the previous section, we showed that cellular network performance – particularly for 5G – has improved markedly from 2022 to 2024. To better understand the underlying factors driving this performance boost, this section takes a closer look at two factors that theoretically boost 5G throughput – CA and MIMO – and examines their evolution over the past two years. Although we did not examine these 5G-specific features in the previous study, we extracted the relevant information from the 2022 dataset to enable a deeper exploration of these metrics and a comparison with the present study.

3.4.1 CA and Aggregate Cell Bandwidth

One of the key innovations 5G introduced to boost performance is the usage of high bandwidth channels ($\geq$ 50 MHz and going up to 100 MHz) and high CA, which aggregates multiple channels (Component carriers – CCs) leading to aggregate bandwidths of up to 800 MHz (in 5G-mmWave). Hence, it is imperative to understand the evolution of CA adoption and bandwidth configurations from 2022 to 2024.

Figs. 3.6, 3.7 compare the number of DL/UL aggregated CCs and the corresponding distribution of aggregate cell bandwidth for 5G-low, 5G-mid, and 5G-high (mmWave) in 2022 vs. 2024. Since CA is only performed under heavy traffic, we only use the data collected during



(a) DL, 5G-low. (b) DL, 5G-mid. (c) 5G-high.

Figure 3.6: CA comparison.

download/upload tests for this comparison, and plot the fraction of miles driven under a particular CA/bandwidth configuration over the total miles driven under 5G. We omit the mmWave-related analysis for AT&T and T-Mobile due to the very low 5G mmWave coverage across the driving route for these operators. Note that in 5G-mmWave all CCs belong to the same band and each CC typically provides 100 MHz of bandwidth,[1] i.e., there is a direct mapping between the number of aggregated CCs and the total bandwidth. Hence, we only show the number of CCs for 5G-mmWave. In contrast, in 5G-low and 5G-mid bands, different CCs can belong to different bands[2] and support different bandwidths from 15 MHz to 100 MHz, i.e., a higher CA level does not necessarily lead to higher bandwidth.

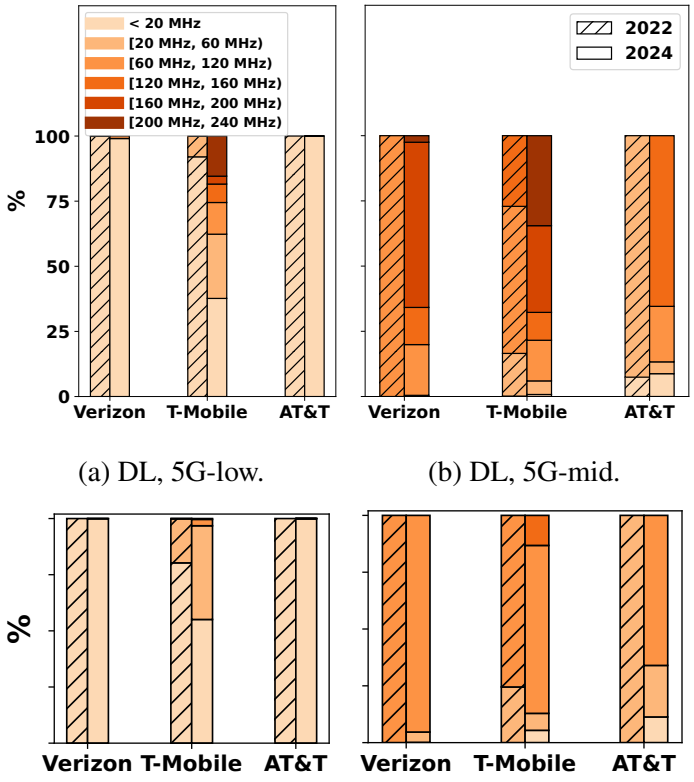
3.4.1.1 Downlink

Overall, we observe a mixed trend for the CA level and the aggregate cell bandwidth across different bands in Figs. 3.6a-3.6c and Figs. 3.7a, 3.7b.

5G-low. Figs. 3.6a, 3.7a show that, for Verizon and AT&T, there's practically no change in the CA level or aggregate bandwidth over the two years. Both operators use a single CC with bandwidth ranging from 10-20 MHz. On the other hand, with T-Mobile, we see more than one CC about 40% of the time in 2024 (vs. no CA in 2022), and 3–4 CCs nearly 20% of the time. This increase in CA

[1] except for occasional 50 MHz primary CCs

[2] Our classification into 5G-low, 5G-mid is based on the primary CC.



(a) DL, 5G-low. (b) DL, 5G-mid.

(c) UL, 5G-low. (d) UL, 5G-mid.

Figure 3.7: Sum BW comparison.

is also reflected in the bandwidth distribution; while 2022, similar to the other two operators, was mainly dominated by 10-20 MHz connections, the aggregate bandwidth exceeds 20 MHz 60% of the time in 2024, with a non-negligible share exceeding 160 MHz. As seen in Table 3.1, all three operators improved their median 5G-low DL performance in 2024 compared to 2022, but the gain was substantial for T-Mobile compared to the other two operators (nearly 24x vs. 5x for Verizon and 28% for AT&T).

5G-mid. Unlike 5G-low, we observe a clear increase in CA usage from 2022 to 2024 for all 3 operators in 5G-mid (Figs. 3.6b, 3.7b). In 2022, Verizon and AT&T relied on single-carrier configurations while T-Mobile used 2 CCs about 30% of the time. In contrast, in 2024, 2-CC aggregation became the dominant setup for Verizon and AT&T, while T-Mobile stands out as the only operator using 3 and 4 CCs.

We note here that these CA patterns are also influenced by the UE model. Recall that the previous study used Samsung S21 devices, which do not support more than 2-CC aggregation. In the case of T-Mobile, it is possible that the operator also allowed for aggregation of 3 and 4 CCs back in

2022, but this is not reflected in the 2022 dataset due to the UE model used by the previous study. On the other hand, both Verizon and AT&T continue to use only 2-CC aggregation in 2024, failing to fully utilize the capabilities of newer UE models.

The higher CA in 2024 leads to a clear increase in the aggregate bandwidth for all three operators compared to 2022. Verizon, which operated exclusively in the 60–120 MHz range in 2022, expands to 120+ MHz over 80% of the time and 160+ MHz over 60% of the time in 2024. T-Mobile exhibits an even more aggressive bandwidth expansion. Most of its connections were within the 60–120 MHz range in 2022, but over 79% of its connections use 120+ MHz bandwidth in 2024, with a significant share (around 35%) reaching 200–240 MHz. In contrast, AT&T, while it also increased its aggregate bandwidth in 2024, is capped out at 120–160 MHz. This aligns with the lower 5G-mid DL throughput observed for AT&T in Table 3.1 compared to the other two operators, even though it uses the same CA configuration (2-CC) as Verizon. Interestingly, Table 3.1 shows that Verizon achieves slightly higher 5G-mid DL throughput than T-Mobile at the 50th and 75th percentile,[3] despite the latter's higher bandwidth configurations, suggesting that a few extra 10s of MHz do not necessarily lead to an additional throughput boost.

5G-mmWave. Surprisingly, for 5G-mmWave connections with Verizon, the number of aggregated CCs declined noticeably in 2024 (Fig. 3.6c). The fraction of 8 CCs dropped by nearly half in 2024 (40% in 2024 vs. 82% in 2022), with 6-CC, 4-CC, and 1-CC configurations becoming more prevalent. This in turn explains the significant 5G-mmWave throughput drop at the 25th and 50th percentile in 2024 compared to 2022 in Table 3.1. We conjecture that the reduced levels of CA in 2024 are (partly) due to increased congestion in the 5G mmWave bands, as more users have upgraded their devices to advanced models that support 5G mmWave and/or higher CA levels. In contrast, the 75th percentile is higher in 2024, possibly due to improvements in S24's modem and signal processing capabilities. A similar result w.r.t. the performance of the two UE models was recently reported in [65].

3.4.1.2 Uplink

Figs. 3.6c, 3.7c, 3.7d show that the CA levels and aggregate bandwidth are much lower in the UL direction.

5G-low, 5G-mid. In 5G-low and 5G-mid, we found that only T-Mobile performed CA, aggregating 2 CCs less than 5% of the time for 5G-low and around 25% of the time for 5G-mid in 2024. In

[3]The same holds true even at the 99-th percentile.

contrast, Verizon and AT&T did not perform UL CA in any of the two years. Hence, we omit the CA breakdown, and only show the aggregate bandwidth breakdown in Figs. 3.7c, 3.7d.

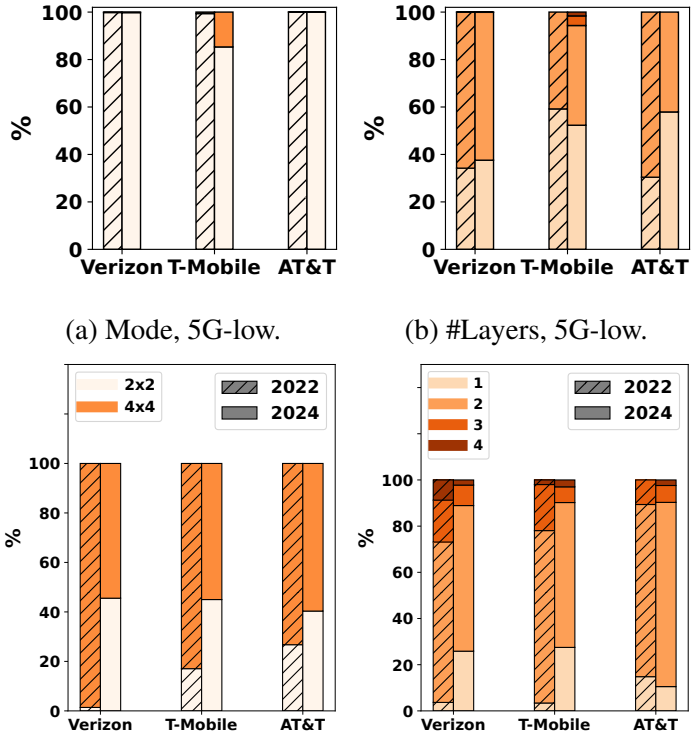
Figs. 3.7c, 3.7d show limited changes for the UL bandwidth between 2022 and 2024. For Verizon, there is practically no shift in aggregate bandwidth in either band. 5G-low (5G-mid) cells provided 10-20 MHz (60-120 MHz) bandwidth in both years. AT&T also had no change in 5G-low bandwidth (10-20 MHz in both years) but expanded its 5G-mid bandwidth. It used exclusively 20-60 MHz cells in 2022, but wider 60–120 MHz cells 60% of the time in 2024. T-Mobile showed the most aggressive expansion of UL bandwidth – using cells with 20-60 MHz aggregate bandwidth (40% of the time) and even 60-120 MHz ($\leq$5% of the time) in the low bands in 2024, and introducing cells of 120–160 MHz (about 10%) in the mid bands, the highest among all three operators.

The higher 5G-low aggregate bandwidth in 2024 for T-Mobile led to an increase in its UL throughput compared to 2022 (Table 3.1). However, AT&T maintained the highest UL performance in 5G-low in both years, in spite of lower bandwidth. In 5G-mid, one can attribute the throughput increase for T-Mobile and AT&T in 2024 vs. 2022 to the higher aggregate bandwidth used in 2024; but a substantial increase in throughput is also observed for Verizon in Table 3.1. Overall, our results suggest that, for low cell bandwidths, the UE's hardware capabilities probably play a more important role than the bandwidth itself.

5G-mmWave. In contrast to the DL direction, where the aggregated mmWave bandwidth for Verizon actually decreased in 2024, we observe 3-CC and (primarily) 4-CC CA in the UL direction (400 MHz aggregate bandwidth) for about 25% of the driving distance (Fig. 3.6c). Recall that the S21 phone used by the previous study supports at most 2-CC CA in the UL direction. That being said, most UL transmissions still use a single mmWave CC in 2024. As a result, Table 3.1 shows only a small increase in the median 5G mmWave UL throughput (42 Mbps vs. 51 Mbps), but a substantial increase (more than 2x) at the 75th percentile.

3.4.1.3 Summary.

Over the past two years, all three operators expanded aggressively their 5G-mid DL CA levels and aggregate bandwidth. T-Mobile and AT&T also expanded their UL bandwidth, although AT&T does not perform UL CA. In contrast, in the low bands, only T-Mobile introduced CA in 2024 and increased its cell bandwidth. In the mmWave bands, with Verizon, we observed a non-negligible amount of 3- and 4-CC aggregation in the UL direction in 2024, which were not supported by the S21 devices used by the previous study in 2022, but, surprisingly, a significant drop in the CA levels



(a) Mode, 5G-low. (b) #Layers, 5G-low.

(c) Mode, 5G-mid. (d) #Layers, 5G-mid.

Figure 3.8: MIMO comparison (Downlink).

in the DL direction. We also found that small bandwidth upgrades (by a few tens of MHz) do not always lead to throughput improvements.

3.4.2 MIMO

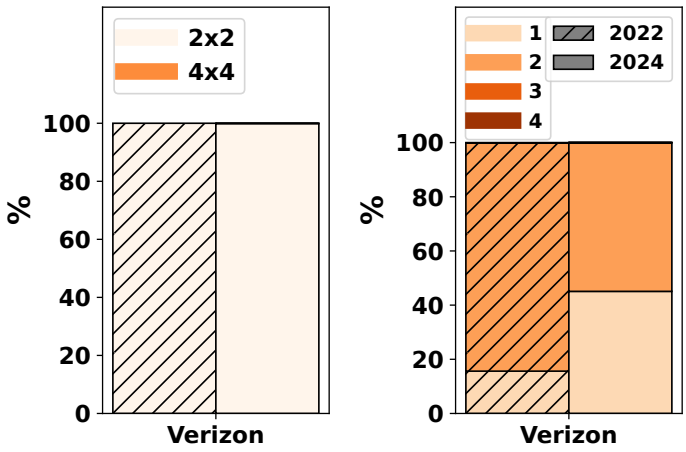
MIMO is another well-established technique for boosting wireless capacity by transmitting multiple data streams in parallel over multiple antennas. In this section, we analyze how 5G MIMO capabilities have evolved across all three operators. In our analysis, we consider both the MIMO mode, which reflects the number of Tx/Rx antennas on the gNB, and the number of MIMO layers, which indicates how many of the antennas were actually used to transmit parallel data streams during data download/upload tests.

3.4.2.1 Downlink.

Figs. 3.8a, 3.8c compare the distribution of DL MIMO modes in 5G-low and 5G-mid, respectively, in 2022 vs 2024. We observe that all encountered gNBs (across all operators and bands)

supported MIMO (2x2 or 4x4) in both years. The same is true for 5G-mmWave (Fig. 3.9a). However, 5G-low and 5G-mmWave gNBs support exclusively 2x2 MIMO except for a small fraction (15%) of T-Mobile 5G-low in 2024. In contrast, in 5G-mid, 4x4 MIMO was the dominant mode across all three operators in 2022 – 99/83/73% for Verizon/T-Mobile/AT&T – but, surprisingly, it reduced to around 55-60% for all three operators in 2024.

To understand this shift in the MIMO mode trend from 2022 to 2024, we look at the corresponding MIMO layers in Figs. 3.8b, 3.8d. Fig. 3.8d shows that in 5G-mid the combined fraction of 3 and 4 layers is very small (10-25%) for all three operators. The dominant number of MIMO layers in both years is 2, indicating that, even when higher MIMO capabilities are available, the physical channel often cannot support a high number of layers. Similar trends are observed for 5G-low in Fig. 3.8b. The fraction of 2 layers across all three operators was only 41-70% in 2022 and further reduced to 42-62% in 2024, in spite of the fact that all gNBs supported 2x2 MIMO. Additionally, for T-Mobile, the fraction of 3/4 layers is negligible in 2024 (6%) although the fraction of 4x4 MIMO mode was 15%. Similarly, in 5G-mmWave (Fig. 3.9b), the fraction of 2 layers reduced from more than 80% in 2022 to about 55% in 2024.



(a) Mode, 5G-high. (b) #Layers, 5G-high.

Figure 3.9: 5G mmWave MIMO comparison, 2022 vs. 2024 (Downlink).

A similar result for 5G-mid, based on a small-scale measurement campaign in Minneapolis and Chicago in 2022, was reported by the authors of [55], who concluded that increasing the MIMO order in future cellular generations might not be the best way to boost throughput. Additionally, equipping gNBs with multiple antennas has negative implications on energy efficiency. Our results suggest that operators have learned a lesson over the past two years and either reverted to 2x2 MIMO configurations (by disabling some antennas) or deployed more 2x2 gNBs in 2024.

Table 3.2: 5G-mmWave uplink MIMO comparison.

	MIMO Mode		**MIMO Layers**	
	1x1	**2x2**	**1**	**2**
2022	4%	96%	82%	18%
2024	23%	77%	79%	21%

3.4.2.2 Uplink.

In the UL direction, no operator supported MIMO in 5G-low and 5G-mid. Hence, we limit our discussion to 5G mmWave with Verizon in Table 3.2, which shows a similar trend as for the DL direction. Although the fraction of 2x2 MIMO is high in both years (96%/77% in 2022/2024), the fraction of 2 layers is very low (only 18%/21% in 2022/2024), again indicating that the channel cannot support MIMO in most cases. Additionally, the fraction of 2x2 MIMO reduced substantially in 2024 compared to 2022 (from 96% to 77%).

3.4.2.3 Summary.

Our results show that the physical channel often does not enable a high number of MIMO layers (3/4 in 5G-mid, 2 in 5G-low and 5G-mmWave) even when the infrastructure provides support. As a result, all three operators decreased drastically the number of 5G-mid gNBs supporting 4x4 MIMO and Verizon decreased the number of 5G-mmWave supporting 2x2 UL MIMO in 2024. Overall, increasing the cell bandwidth appears a much more promising way to further boost throughput in future cellular generations compared to increasing the MIMO order.

3.5 Handovers

Handovers (HOs) are essential mobility procedures in cellular networks that ensure seamless connectivity as users move. In this section, we examine how HO behavior, specifically in terms of frequency and duration, has evolved from 2022 to 2024 across different technologies and operators. While the previous study limited its analysis to HOs during throughput tests, we expand the scope to the entire 2022 and 2024 datasets to gain a more comprehensive understanding of this evolution.

HO frequency. Table 3.3 shows the number of HOs per mile for all three operators for 2022 and 2024. The results include the total numbers as well as a breakdown by HO type.

We observe a significant increase in the total number of HOs per mile across all three operators, from 1.2/1.72/1.46 HOs per mile in 2022 to 3.86/2.69/2.86 in 2024, for Verizon/T-

Table 3.3: HOs per mile along the driving route.

Verizon					
	Total	**LTE→LTE**	**5G→5G**	**5G→LTE**	**LTE→5G**
2022	1.2	0.74	0.18	0.09	0.19
2024	3.86	1.65	1.5	0.17	0.53
T-Mobile					
	Total	**LTE→LTE**	**5G→5G**	**5G→LTE**	**LTE→5G**
2022	1.72	0.58	0.4	0.17	0.59
2024	2.69	0.67	1.61	0.08	0.32
AT&T					
	Total	**LTE→LTE**	**5G→5G**	**5G→LTE**	**LTE→5G**
2022	1.46	0.77	0.24	0.15	0.3
2024	2.86	1.51	0.62	0.16	0.58

Mobile/AT&T. This is expected primarily due to the increased 5G coverage for all three operators and the smaller size of 5G-mid and 5G mmWave cells, which leads to more frequent HOs. In fact, the number of 5G→5G HOs shows the largest increase for all three operators, by a factor of 2.6x (for AT&T) up to 8.3x (for Verizon).

Interestingly, for Verizon and AT&T, Table 3.3 shows a noticeable disparity in the number of 5G→5G HOs per mile in 2024 despite both operators having a similar overall 5G coverage fraction of around 50% (Fig. 3.3). This difference is a result of contrasting deployment strategies. Verizon's 5G is primarily based on mid-band deployments, whereas AT&T relies heavily on low-band 5G (Fig. 3.3). Since low-band 5G operates at much lower frequencies ($\leq$ 1 GHz) than mid-band ($\geq$ 2.5 GHz), it offers significantly larger coverage per base station, resulting in fewer HOs.

The number of LTE→LTE HOs per mile have also increased for all three operators between 2022 and 2024, despite the reduced LTE coverage compared to 2022. While the rise is modest for T-Mobile, the numbers are almost double for Verizon and AT&T in 2024 compared to 2022. Recall that these two operators primarily deploy 5G in NSA mode, where LTE serves as the control plane. As a result, even a 5G→5G HO often requires an additional HO of the anchor LTE cell, inflating the LTE→LTE HO count in 2024. This inflation is much smaller for T-Mobile, where NSA accounts only for 40% of the operator's 5G deployment in 2024.

Table 3.3 reveals distinct trends for Inter-RAT HOs, LTE→5G and 5G→LTE. For 5G→LTE HOs, T-Mobile sees a decline in the number of HOs per mile from 0.17 to 0.08 between 2022 and 2024, a direct consequence of its 5G coverage expanding from 68% to 94%, resulting in fewer

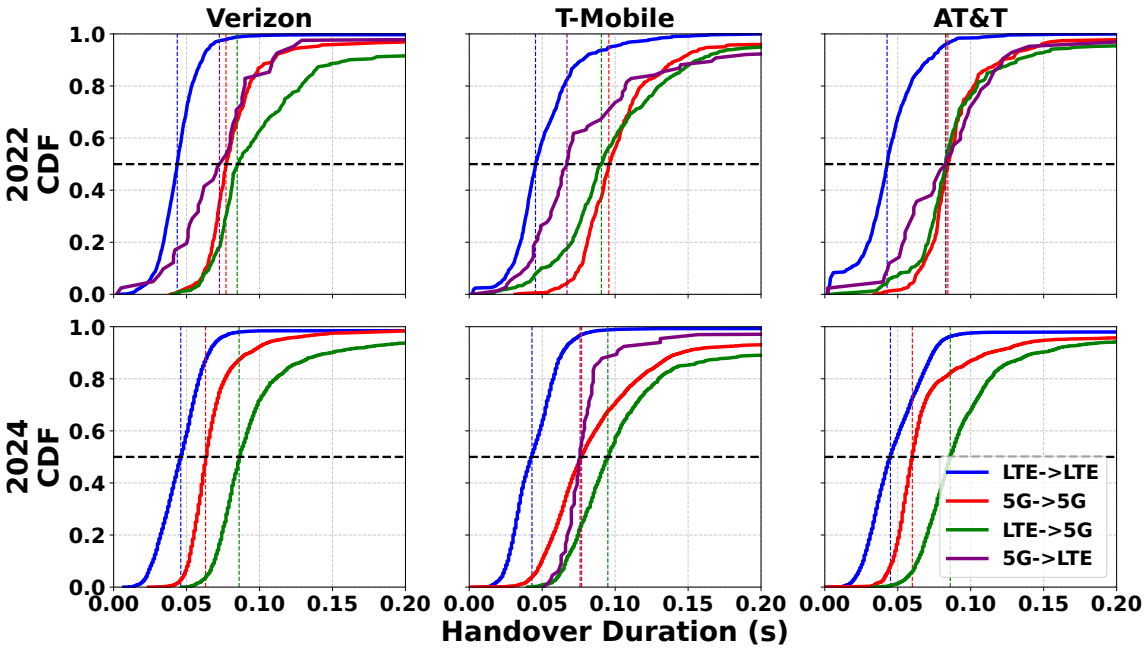


Figure 3.10: HO duration.

fallbacks to LTE. In contrast, Verizon and AT&T experience an increase in 5G→LTE HOs per mile, despite improved 5G coverage in 2024. This is because their 5G footprint, although larger than in 2022, still covers only around 50% of the route, increasing the likelihood of dropping back to LTE as users move in and out of 5G zones. The increase is much smaller for AT&T, due to the operator's heavy reliance on 5G-low.

Interestingly, the number of LTE→5G HOs is consistently higher than the number of 5G→LTE HOs for all three operators in both years. This asymmetry stems from the device behavior under idle conditions – to conserve energy, devices often stay on LTE in the absence of active traffic. When a throughput test begins, the device upgrades to 5G, triggering an LTE→5G HO. However, once in 5G, the network may not immediately fall back to LTE, particularly during the same test session – resulting in fewer 5G→LTE transitions relative to LTE→5G. Similar to 5G→LTE HOs, T-Mobile sees a decline in the number of LTE→5G HOs in 2024, due to its extremely low LTE coverage, while the other two operators see an increase due to their balanced LTE/5G coverage.

HO duration. We define the HO duration as the time from the moment the network sends the RRC Reconfiguration message initiating a HO till the moment the random access procedure (PRACH) is completed. Fig. 3.10 compares the HO duration for different HO types across the two years.

For intra-RAT LTE→LTE HOs, the median HO duration remains similar in both years for all three operators. In contrast, for the intra-RAT 5G→5G, we notice a substantial drop in the median

HO duration – from 77/96/84 ms in 2022 to 63/77/60 ms in 2024 for Verizon/T-Mobile/AT&T. As operators rolled out more 5G gNBs, the likelihood of encountering neighbor cells with better signal strength increased from 2022 to 2024, leading to fewer failures during random channel access, which reduces the overall 5G→5G HO duration.

In the case of inter-RAT HOs, for LTE→5G HOs, the median HO duration remains unchanged across the two years for all three operators. On the other hand, for 5G→LTE HOs, we see a different picture. In 2022, 5G→LTE HOs either were triggered by A2 events[4] or occurred due to 5G failures. In contrast, in 2024, we found that the only event-triggered 5G→LTE HOs were those in T-Mobile's 5G SA deployments (triggered by B2 events); all other 5G→LTE HOs occurred due to 5G failures. Since we do not compute HO duration in failure scenarios, the 2024 5G→LTE durations are unavailable for Verizon and AT&T. For T-Mobile, we observe a more nuanced trend: although the 2022 HO durations are lower than 2024 up to the 60th percentile, the tail behavior is noticeably worse. The 2022 durations cross the 100 ms mark at the 70th percentile, whereas in 2024, approximately 85% of the HOs complete within 100 ms, indicating tighter control and reduced long-tail delays.

Summary. The total number of HOs per mile increased substantially in 2024 compared to 2022 due to (i) the increased 5G coverage and the smaller size of 5G-mid and 5G mmWave cells, which leads to more frequent 5G→5G HOs and (ii) the extensive use of 5G NSA, which often triggers an additional LTE→LTE HO along with a 5G→5G HO. On the other hand, with the exception of 5G→LTE HOs in T-Mobile's 5G SA deployments, the HO duration either remained the same (for LTE→LTE HOs and LTE→5G HOs) or reduced (for 5G→5G) in 2024, likely due to improved 5G cell density, which reduces the probability for RACH failures.

3.6 Cellular vs. LEO Satellite Networks

Up to this point, our analysis has focused on comparing the coverage, performance and key 5G metrics of cellular networks between 2022 and 2024. With LEO satellite networks gaining momentum and expected to play a pivotal role in the next-generation of wireless networks, it is imperative to understand how they compare to existing terrestrial deployments. In this section, we present a head-to-head comparison between cellular networks and Starlink's LEO satellite network using the dataset collected in 2024. Note that the previous study did not consider Starlink in their evaluation.

[4]A brief description of the HO triggering events is given in Table 5.1 in §5.1.

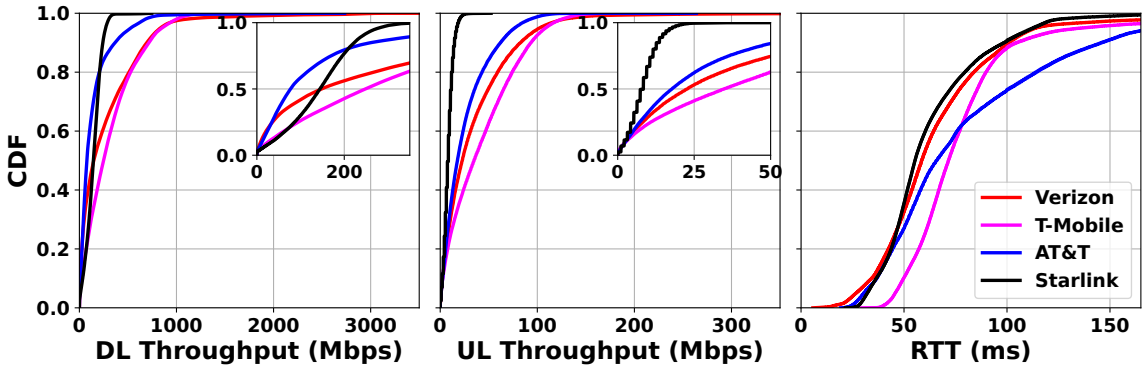


Figure 3.11: Throughput and RTT comparison of Starlink and cellular networks.

3.6.1 Network Availability

Before comparing the performance of cellular networks against that of the Starlink network, it is important to understand the availability of the two networks. We analyze the outage rate of the two networks, defined as the fraction of outage duration relative to the total measurement duration. For cellular networks, outages are recorded as ”No Service” events logged by XCAL, while for Starlink, the Starlink grpc API directly reports outage intervals.

The three cellular networks did not exhibit any ”No Service” events throughout the entire driving route, resulting in a 0% outage rate. In contrast, Starlink experienced outages for approximately 2.03% of the route. To understand where these outages occur, we further categorize them by area type: urban, suburban, and rural. These labels were manually annotated during data collection. Interestingly, nearly 70% of Starlink outages occurred in urban areas – especially dense downtown regions – while the remaining 30% was split evenly between suburban and rural zones. This pattern is consistent with prior work [70], and reflects the challenges posed by urban obstructions like tall buildings, which limit the line-of-sight required for stable satellite connectivity.

3.6.2 Network Performance

Fig. 3.11 compares DL/UL throughput and RTT between Starlink and the three cellular operators.

Downlink performance. Fig. 3.11 (left) shows that Starlink outperforms AT&T (the operator with the lowest DL throughput) around 80% of the time. Its throughput at the 25-th/50-th/75-th percentile is 34/144/196 Mbps vs. 12/77/170 Mbps for AT&T. However, AT&T outperforms Starlink 20% of the time, with a peak throughput of 2.7 Gbps compared to only 745 Mbps for Starlink.

Starlink outperforms Verizon around 50% of the time with both operators achieving similar median throughput. However, for the upper 50-th percentile, Verizon achieves much better performance, for example 428 Mbps vs. 196 Mbps at the 75th percentile. Finally, Starlink performs similar to T-Mobile at the 25th percentile, but beyond that, T-Mobile outperforms Starlink by a margin even higher than that of Verizon.

Uplink performance. We notice a completely opposite behavior in terms of UL performance in Fig. 3.11 (middle); Starlink achieves the worst performance among the four networks, with a median value of only 8 Mbps and a peak value of 52 Mbps vs. 18-35 Mbps and 216-531 Mbps, respectively. Much like the cellular networks, Starlink also faces a significant asymmetry in terms of DL and UL performance.

Similar to previous studies [70], we also found that Starlink provides the best performance in both directions in rural areas and the worst performance in urban areas, due to obstructions, while cellular networks provide the best performance in urban areas due to high availability of high-speed 5G (see Table 3.4).

RTT. Interestingly, Starlink achieves overall the best RTT among the 4 networks (Fig. 3.11(right)). Its 25th/50th/75th percentiles are 46/55/73 ms vs. 46-60/59-72/78-103 ms for the three cellular operators. This result is similar to the result in [71], based on an 860 km driving measurement campaign on the East Coast but very different from the result in [70], based on another driving campaign spanning 5 states in 2023, where Starlink's RTT was higher (by 8-10 ms in the median case) than Verizon's and T-Mobile's RTT. While the Starlink RTT values reported in [70] are similar to the values in our dataset, the cellular RTTs with Verizon and T-Mobile are lower than the RTTs in our dataset.

Summary. Starlink provides almost (but not) 100% availability and better RTT compared to cellular networks. Its DL performance is commensurate with that of cellular networks in the median case (100+ Mbps) and significantly better compared to 2/3 cellular networks at the 25-th percentile, making it a solid competitor for mobile Internet access in rural areas, where two cellular operators have limited high-speed 5G coverage. However, its peak throughput is bounded at 745 Mbps while 5G-mid and 5G mmWave exceed 1 Gbps, and its UL throughput is much lower than that of cellular networks.

Table 3.4: Starlink and cellular performance by area type.

(a) Urban

	Verizon	T-Mobile	AT&T	Starlink
DL Throughput	104/285/632	136/324/602	46/180/404	41/117/175
UL Throughput	18/42/75	35/59/84	15/32/57	2/5/8
RTT	26/35/52	50/59/67	33/41/56	44/57/81

(b) Suburban

	Verizon	T-Mobile	AT&T	Starlink
DL Throughput	25/143/465	77/228/442	33/70/197	79/134/184
UL Throughput	9/24/54	10/34/66	8/18/35	4/7/10
RTT	47/65/81	59/69/86	49/76/125	47/57/75

(c) Rural

	Verizon	T-Mobile	AT&T	Starlink
DL Throughput	37/132/386	99/245/450	34/75/143	95/152/203
UL Throughput	7/20/44	10/31/62	6/16/33	5/8/12
RTT	49/59/78	64/76/88	53/67/99	46/55/72

3.7 Application Performance

The previous study analyzed the performance of four representative latency-critical and bandwidth-intensive "5G killer apps" – cloud-assisted AR, CAVs, 360° video streaming, and mobile cloud gaming – and showed that cellular networks were not able to support those apps. In this section, we repeat the same analysis with two of these apps, AR (UL-oriented) and 360° video streaming (DL-oriented) Instead of using real-experiments, we opted for trace-driven emulation, using throughput traces from the 2022 publicly available dataset and our dataset, to enable reproducibility.

Emulation methodology. We connect the UE to a server over 802.11ax, and emulate the UL (for AR) and DL (for video streaming) bandwidth and latency using Linux tc. Since throughput and RTT traces were collected separately, we merge them as follows. We use fixed-length throughput traces of 40 s with samples recorded every 500 ms. To assign RTTs to the throughput traces, we extract the minimum RTT observed from ping tests for each operator-technology-timezone combination, and assign it to the throughput trace based on the cellular technology of each throughput sample.

For Verizon, we further distinguish between traces collected via cloud servers and those using edge servers, assigning separate RTT values accordingly. We assume this minimum RTT represents the propagation delay between the UE and the server; every other delay component (e.g., RAN delay, queuing delay, etc.) is reflected in the throughput trace.

3.7.1 Cloud-Assisted AR

We used the same canonical cloud-assisted AR benchmark Android app as in the previous study, which offloads pre-recorded camera frames (at a frame rate of 30 FPS) to a cloud/edge GPU server (Nvidia A100) that performs object detection in a best-effort manner. We also used the same camera dataset [29] and the same methodology as the previous study to compute the object detection accuracy (mAP metric [72]).

Figs 3.12a-3.12d plot CDFs of the average E2E offloading latency, the number of offloaded frames per s, and the mAP with raw and compressed (H264) frames, for each operator in 2022 and 2024 (the 2024 results also include Starlink). With a frame rate of 30 FPS, the target E2E latency is 33 ms. The offline accuracy is 38.45%. Similar to [73], we consider the offloading performance to be satisfactory if the mAP score of the application with offloading is within 90% of the offline accuracy (34.6%). We make the following observations.

(1) Although the results for 2022 are quantitatively different from those in the previous study, due to the different methodology, the main conclusion still holds: *cellular networks were not able to support cloud-assisted AR under driving in 2022*. Even with compression, the median E2E latency in 2022 is 191-217 ms – 5.8x-6.6x that of the target value. As a result, the app achieves a median offloading frame rate of only 4.3-4.8 FPS and mAP of 31.6-32.5% (82-85% of the offline mAP). The performance is much worse without compression, with a median E2E latency of 510-610 ms, median offloading frame rate of only 1.6-1.9 FPS, and median mAP of 19.2-21.3% (50-55% of the offline accuracy).

(2) The performance improves drastically in 2024, mainly due to the higher bandwidth provided by all three cellular operators. The median E2E latency, offloading frame rate, and mAP with raw/compressed frames is 243-380 ms/137-171 ms, 2.5-3.8 FPS/5.3-6.5 FPS, and 24.8-30.1%/33.3-34.5% (64-78%/87-90% of the offline accuracy), respectively. The app achieves satisfactory accuracy about 46% of the time with Verizon, 27% with AT&T, and 19% with T-Mobile.

(3) Figs. 3.14a, 3.14b, which plot the accuracy vs. the percentage of high-speed 5G for each run with a cloud/edge server, confirm two more findings from the previous study: *high-speed 5G improves*

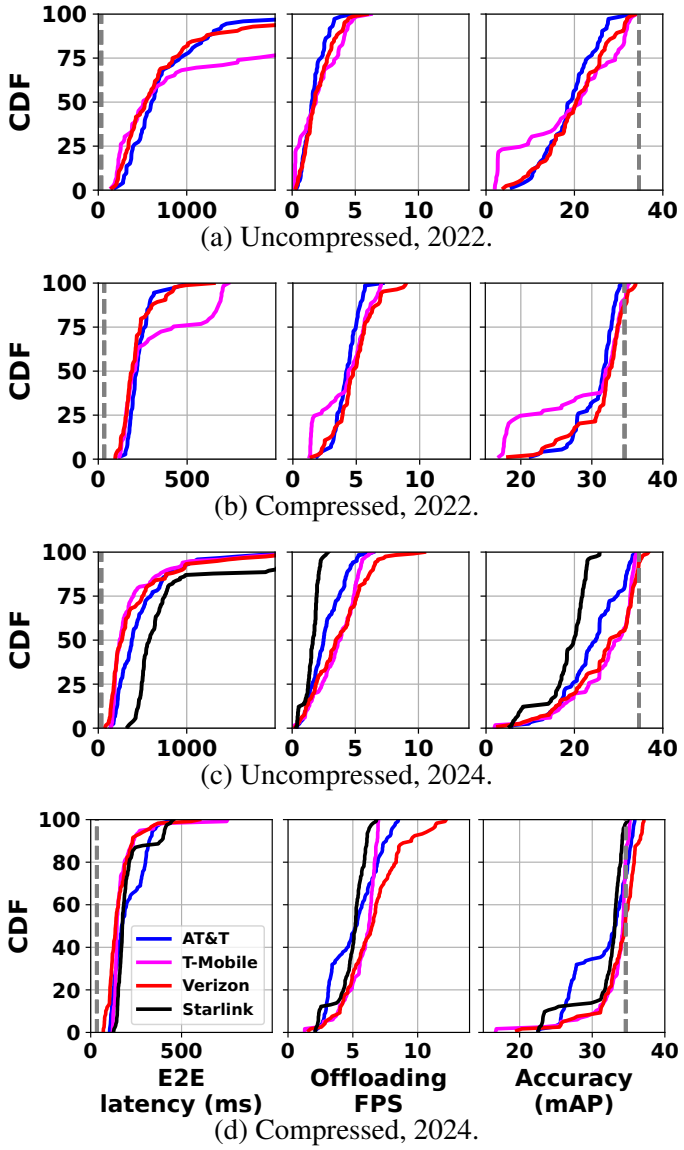


Figure 3.12: 2022 vs. 2024: AR performance.

the worst-case performance compared to 4G/5G-low, and *the use of an edge server boosts the performance regardless of the cellular technology*. Note that in 2024, all the runs with an edge server experience almost 100% high-speed 5G coverage, as the operator significantly expanded its 5G mmWave/midband coverage in large urban centers (where the AWS Wavelength servers are located). **(4)** In spite of its lower latency compared to cellular operators (Fig. 3.11(right)), Starlink provides the worst AR performance due to its much lower UL throughput (Fig. 3.11(middle)). The gap is particularly pronounced with raw frames, where the median E2E latency, offloading frame rate, and mAP with Starlink are 576 ms, 1.7 FPS, and 19.9% vs. 380 ms, 2.5 FPS, and 24.8% with the

worst-performing operator. The gap closes substantially with compressed frames that reduce the bandwidth requirements significantly (174 ms vs. 171 ms, 5.2 FPS vs. 5.3 FPS, 33.1% vs. 33.3% mAP). However, Starlink provides satisfactory accuracy only 2.4% of the time.

(5) Our results show that *application-level optimizations developed in the pre-5G era, such as frame compression and local tracking [22, 74], remain relevant even in 2024, in spite of the drastic improvements in cellular performance over the past two years.* Compression, which was shown to significantly improve performance in the previous study, helps the app achieve satisfactory accuracy 25-50% of the time with the 2024 traces. This result is achieved in spite of the app offloading at most 12.5 FPS (Fig. 3.12d(middle)), thanks to local tracking, which helps mask the long E2E latency.

3.7.2 360° video streaming

We used the 360° video streaming system from the previous study and the same YouTube videos for our evaluation. The system uses Puffer, an open source video streaming server [34], with BBA [35] as the ABR algorithm, and a custom mobile client. We encoded each video into 5 different quality settings (5, 10, 50, 100, 250 Mbps), adding one extra setting (250 Mbps) compared to the previous study, since the cellular network bandwidth increased drastically in 2024. Each playback session runs for 40 s. We used the same video chunk QoE metric [37] as the previous study with λ = 1 and μ = 250, following the guideline in [37], which results in a theoretical maximum QoE value of 250 (assuming no rebuffering and no bitrate change). All other settings are the same as in the previous study. Fig. 3.13 plots CDFs of the average QoE, average bitrate, and rebuffering time per session for all operators in 2022 and 2024. We make the following observations.

(1) Similar to the AR results, our results in this section for 2022 confirm the main conclusion of the previous study: *cellular networks were not able to support 360° video streaming under driving in 2022*. The QoE is negative 75-90% of the time with all three operators, with a median value between -152 and -96 and a maximum value of only 62-170. Although the average bitrate exceeds 150 Mbps for a few sessions with T-Mobile and Verizon, the median bitrate is only about 40 Mbps for all 3 operators. Similarly, the rebuffering time is very high, with median values of 22-45%. In particular, for T-Mobile, 25% of the sessions experience a rebuffering time of 100%. Note that the QoE and the rebuffering time are much worse in our trace-driven emulation compared to the results reported in the previous study, even though the bitrate is similar, due to the additional encoded bitrate of 250 Mbps we used in our evaluation. While this bitrate can occasionally be supported, the highly fluctuating network conditions under driving result in increased rebuffering time when the player requests a

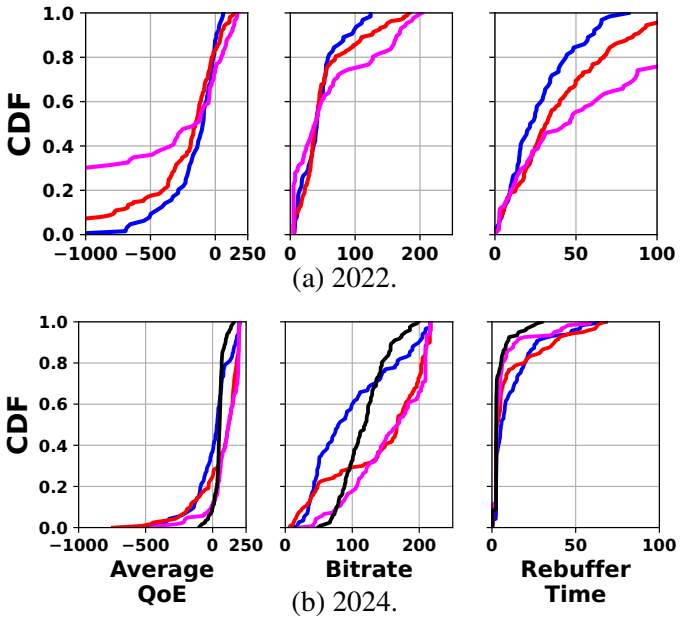


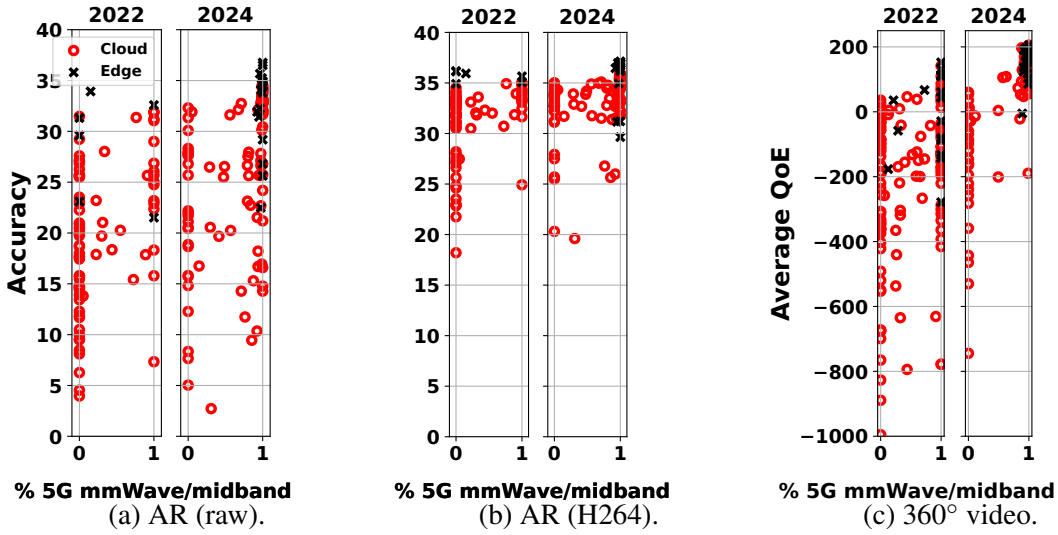
Figure 3.13: 2022 vs. 2024: 360° video performance.

chunk at 250 Mbps. Interestingly, AT&T provides the lowest bitrates, but also the lowest rebuffering time. This is because AT&T's DL throughput was higher than 250 Mbps only with 5G-mmWave, and hence the player rarely requested chunks at that bitrate.

(2) The performance improves drastically in 2024. The rebuffering time is less than 5% for 48-71% of the time, with a median value of only 2.9-5.6%, and the median bitrate increases to 79-168 Mbps. As a result, the QoE is positive 50-90% of the time, with a median value of 32-121. The improvement is more pronounced for T-Mobile and Verizon, the two operators with the largest 5G-mid coverage in 2024.

(3) Fig. 3.14c shows that the combination of high-speed 5G and an edge server improves the worst-case QoE in 2022 but almost always results in very good to excellent QoE in 2024. However, the use of edge servers is less critical compared to the AR use case, as video streaming can tolerate latencies due to buffering. In fact, T-Mobile achieves better performance than Verizon in 2024 (Fig. 3.13b) due to its extensive fast 5G-mid coverage in spite of the absence of edge servers. We conclude that *5G-mid performance in 2024 is sufficient to provide high QoE for 360° video streaming under driving*, but coverage holes and fallbacks to 5G-low/LTE still hurt the performance.

(4) Starlink provides the lowest rebuffering time among the four operators but also lower bitrates than two cellular operators (Verizon and T-Mobile). As a result, Starlink provides the highest worst-case, but the lowest best-case QoE. This result is a direct reflection of Starlink's DL throughput relative to



(a) AR (raw). (b) AR (H264). (c) 360° video.

Figure 3.14: Impact of high-speed 5G and edge servers.

the throughput of the cellular operators (Fig. 3.11(left)).

3.8 Related Work

The related work on 5G (and pre-5G) measurement studies before the previous study was covered in the previous study and the work after the previous study is discussed in the introduction. Here, we briefly summarize related work on LEO satellite networks. Since the launch of Starlink, several works have measured various aspects of its performance [75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 71, 86, 70]. With the exception of [71, 70], all these works perform measurements from *fixed* locations. Furthermore, only a handful of works [71, 86, 70] have performed a head-to-head comparison between Starlink and cellular networks. Among them, [71, 70] use a measurement methodology similar to ours, conducting measurements during long drives on the East Coast and in the Midwest, respectively. Our cross-country measurement campaign confirms several of the findings in [71, 70] with respect to outage, performance, and impact of area type. Interestingly, our RTT results (§3.6.2) agree with those in [71], which were obtained in the same year as our study, but are different from those in [70], which were obtained one year earlier.

3.9 Summary

The previous study offered the first comprehensive, large-scale characterization of operational 5G networks in driving scenarios, assessing coverage, performance, and mobile application QoE across the US. To assess how 5G has evolved at its midlife, we conducted a replicability study in 2024 along the same coast-to-coast route.

Our results reveal substantial improvements in both coverage and throughput across all three major US operators, largely attributable to aggressive 5G-mid deployments, but modest or no improvement in latency. This progress in 5G-mid deployment has led to a marked enhancement in the QoE for two "5G killer" applications analyzed in the previous study. However, coverage holes and fallbacks to 5G-low or LTE still hurt the performance, and application-level optimizations developed in the pre-5G era are still needed, even in areas with excellent 5G-mid or 5G-mmWave coverage.

To broaden our analysis, we compared these terrestrial network improvements with Starlink's LEO satellite network. Our results show that, while non-terrestrial systems now offer competitive median downlink speeds and RTTs, they continue to fall short in peak downlink speeds and overall uplink performance.

Together, this chapter and the previous one provide a broad characterization of operational cellular networks in the US, covering coverage, network performance, and mobile application QoE, and tracking how they evolve over a two-year period. Among our findings, the results reveal an ongoing transition from Non-Standalone (NSA) to Standalone (SA) 5G deployments – a shift with significant implications for network performance and mobility management that remains largely unexplored. We focus on this transition in the next chapter.

Chapter 4

A First Large-Scale Study of Operational 5G Standalone Networks

To accelerate the deployment of 5G, most operators initially adopted the Non-Standalone (NSA) architecture, which leverages the existing 4G LTE core for control plane signaling while introducing 5G New Radio (NR) as a data connectivity layer. NSA has facilitated the rapid rollout of 5G, delivering significant improvements in throughput over LTE [6, 7, 2, 57, 58, 59, 66, 45], including in our own earlier measurements (§2). However, NSA's reliance on LTE introduces additional signaling overhead and limits the full potential of 5G, particularly in achieving ultra-low latency and maximizing network efficiency.

In contrast, the Standalone (SA) architecture represents a fully independent 5G network, operating without reliance on the 4G core and enabling advanced features such as network slicing, enhanced security protocols, and lower latency [87, 88]. While SA deployments began in the US in as early as 2020, they gained significant traction only in late 2022 [89]. Despite this progress, SA adoption remains in its early stages compared to NSA, with T-Mobile being the only US network operator offering 5G SA service in 2024. Understanding how commercial SA networks compare to NSA networks in terms of coverage, performance, mobility management, and network characteristics is essential as operators transition towards a fully SA 5G future.

Since the 5G rollout in 2019, a large number of studies [6, 7, 1, 2, 90, 55, 56, 8, 91, 67, 57, 45, 44, 46, 47, 60, 58, 62, 61, 92, 3, 59, 66, 65, 93] have measured various aspects of 5G performance including coverage, throughput, latency, carrier aggregation (CA), resource allocation, handovers, beamforming, and application QoE. However, all these studies either focus exclusively on 5G NSA

networks [6, 7, 1, 2, 90, 55, 56, 8, 91, 67, 57, 45, 44, 46, 47, 60, 58, 62, 66, 65] or inspect 5G SA networks in extremely limited scope [3, 61, 92, 59, 93], and hence do not provide a complete picture of 5G SA coverage and performance today. In particular, a number of important questions remain unanswered: *(i) What is the current coverage of 5G SA deployments compared to the existing NSA deployments? (ii) How do SA deployments differ from NSA deployments in terms of throughput, latency, mobility management, and other network characteristics? (iii) How has SA performance evolved over time, and does it address the initial gaps identified during its early deployment?*

To answer these questions, we conduct the *first large-scale* measurement study of T-Mobile's[1] 5G SA and NSA deployments in the US via two large-scale driving measurement campaigns (Boston to Chicago, Chicago to Boston) conducted 18 months apart (May 2023 and November 2024). Our study spans multiple frequency bands, both in the low-band (n71) and mid-band (n25, n41), and different duplexing modes (FDD and TDD). Unlike prior studies, which focused on short-term or limited-scale measurements, our dataset includes over 3200 km of drive tests and in-field network measurements across urban, suburban, and highway environments. The collected dataset covers key aspects of 5G SA and NSA network behavior, including downlink (DL) and uplink (UL) performance, latency, CA, mobility management, and transmit power control. Our analysis provides a multi-faceted comparison of SA and NSA deployments, highlighting their evolution and trade-offs. The key findings of our study are summarized as follows:

• 5G SA emerges as the leading deployment mode in 2024 with a drastically improved coverage over the past 18 months, from 22.3% in May 2023 to 61.9% in Nov. 2024. Our results show an initial stronger focus on urban deployment; however, by Nov. 2024, SA coverage has expanded significantly even outside large urban centers. Overall, our results highlight the operator's strategic focus on driving a substantial transition from LTE-dependent NSA to fully standalone 5G networks.

• SA has maintained higher DL throughput in both years despite NSA's improvements in 2024 due to refinements in LTE-5G dual connectivity. Further, SA demonstrates significant UL throughput improvements from 2023 to 2024, driven by the focused CA efforts and the capabilities of new smartphone models. In contrast, NSA UL throughput remained largely unchanged from 2023 to 2024 and NSA's peak performance across both years and both traffic directions appears to have plateaued. 5G SA also provides comparable or better RTT performance in comparison to NSA in both years. Overall, these results highlight the growing maturity of SA networks to provide better performance, as well as the operator's focus on gradually phasing out NSA in favor of standalone 5G for long-term

[1] T-Mobile is the only major US operator with non-negligible 5G SA coverage.

scalability.

• 5G SA supports higher CA levels than 5G NSA, resulting in a higher aggregate bandwidth and improved network performance. Additionally, the introduction of new frequency bands and duplexing modes in 5G SA, combined with the advanced capabilities of newer UE models, enables a large number of diverse CA configurations with intra-/inter-band CA and FDD/TDD combinations of up to 4 carriers in 2024 (vs. only 2 in 2023). The large diversity of available configurations offers the operator a variety of choices for its SA deployments that trade off performance, latency, and robustness, while reflecting the increasing complexity and maturity of SA deployments as 5G networks continue to evolve. In contrast, our results show that the operator has not advanced NSA's CA capabilities. The NSA mode aggregates at most 2 carriers in both years, in spite of the fact that newer UE models support higher aggregation levels.

• Surprisingly, we found that the 5G→5G handover duration is higher in 5G SA compared to 5G NSA due to rigorous signaling and advanced security mechanisms employed in the current 5G SA deployments. Nonetheless, the handover duration for both deployment modes and for intra-frequency SA handovers in particular has reduced significantly from 2023 to 2024.

• Our analysis shows that NSA uses higher control channel (PUCCH) transmit (Tx) power, largely attributed to NSA's deployments in non-urban areas, where greater pathloss necessitates higher Tx power to maintain effective communication with base stations. However, SA exhibits higher data channel (PUSCH) Tx power around 60% of time, even though it always achieves lower pathloss than NSA. Further breakdown of PUSCH power based on different traffic types reveals the operator's policies to elevate Tx power in the presence of UL-heavy traffic. This policy, combined with SA's high UL CA helps SA achieve superior UL performance in comparison to NSA.

Contributions. In summary, this study makes the following contributions: (1) We collect a large first-of-its-kind 5G SA and NSA dataset via two cross-country driving measurement campaigns, conducted 18 months apart, and spanning 3,200+ km. (2) Leveraging this dataset, we conduct the first large-scale, detailed characterization of T-Mobile's 5G SA network, examining coverage, network performance, mobility management, and other critical network features. (3) Our study offers a head-to-head comparison of 5G SA and NSA networks across two years, shedding light on the evolution and maturity of SA deployments. (4) Our dataset and scripts are publicly available [94].

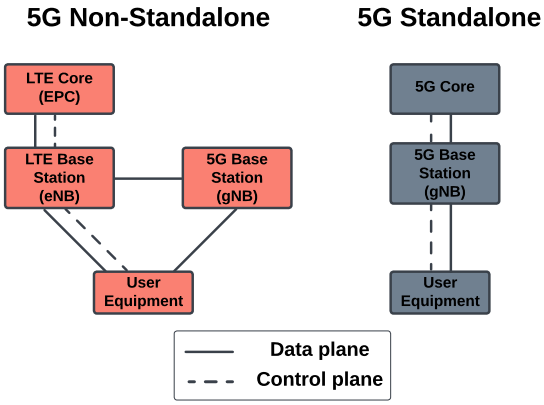


Figure 4.1: 5G NSA vs. SA architecture.

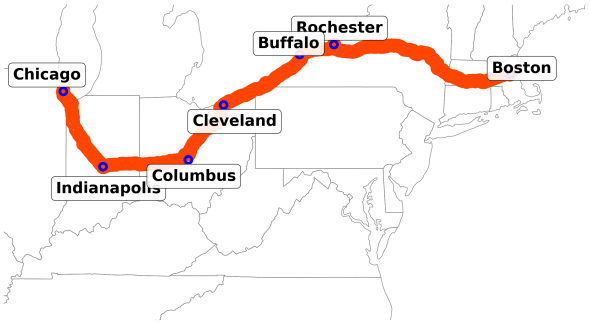


Figure 4.2: Measurement Route.

4.1 5G SA and NSA architectures

Fig. 4.1 shows the fundamental architectural differences between the 5G NSA and SA modes. In 5G NSA, the 5G base station (gNB) is primarily responsible for data transmission, while the control plane operations are controlled by the 4G LTE core (EPC) via the LTE base station (eNB). The LTE base station can be used for data transmission in addition to the 5G base station in dual connectivity mode (EN-DC). In contrast, 5G SA operates independently, with its own 5G Core (5GC), allowing both control and data planes to function solely through the 5G infrastructure. This eliminates the dependency on LTE, enabling advanced 5G features such as reduced latency, enhanced security protocols, and network slicing.

4.2 Methodology

Measurement Route. We conducted two measurement campaigns, 18 months apart (May 2023 and November 2024), spanning over 3200 km of driving across the US. The route of the measurement campaigns is shown in Fig. 4.2. In May 2023 (05/13/2023 - 05/16/2023), we conducted a driving trip from Boston to Chicago, while in November 2024 (11/06/2024 - 11/08/2024), we drove from Chicago to Boston. We covered major cities along the route, including Rochester, Buffalo, Cleveland, Columbus, and Indianapolis. All measurement data reported in this study were collected on interstate highways, in suburban areas, and within city limits. Additional measurements were conducted within several major cities encountered along the routes. Our dataset includes more than 0.5 TB of data and 2580 minutes of measurements.

5G Carriers, User Equipment (UE), and Servers. We conducted measurements with all 3 major

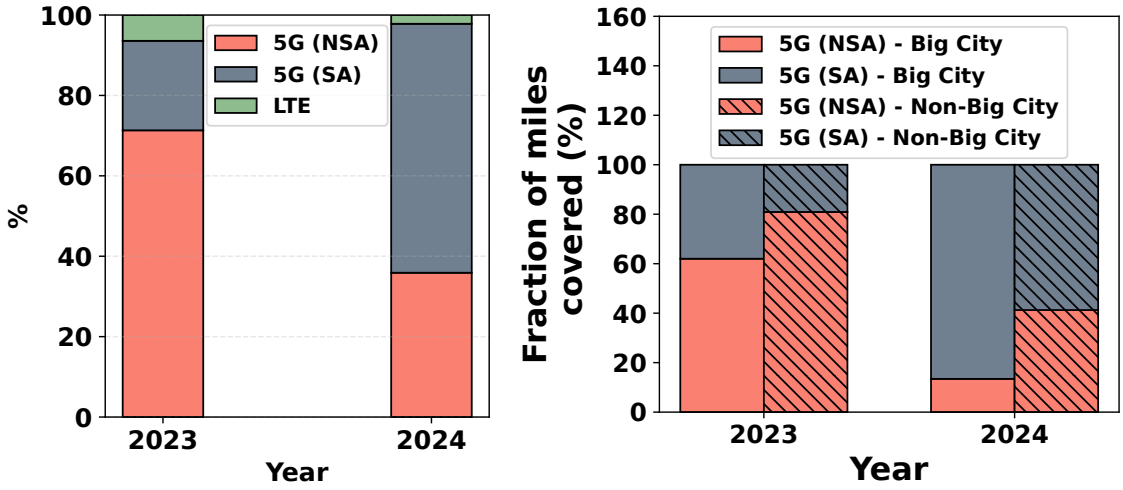
US operators – T-Mobile, Verizon, and AT&T. However, AT&T provided no SA coverage while for Verizon the SA coverage was 0% and $\leq 1\%$ in 2023 and 2024, respectively. Hence, we only consider T-Mobile in this study. T-Mobile has a nationwide deployment of 5G low- and mid-band, and a very limited deployment of high-band (mmWave).

We used two smartphones – Samsung Galaxy S21 Ultra and Samsung Galaxy S24 Ultra – as our user equipment (UE) for data collection. In our 2023 drive, we used the Samsung Galaxy S21 Ultra model, which was still considered a state-of-the-art smartphone at the time of our measurements and was used in other 5G studies conducted around the same time (e.g., [58, 57, 70, 95]). In contrast, in our 2024 drive, we used the Samsung Galaxy S24 Ultra model, released in the U.S. in January 2024. Our goal was to capture the evolution of cellular performance driven by both infrastructure and end-user device advancements. The S24 Ultra features a more advanced modem (Qualcomm Snapdragon X75 vs. Qualcomm Snapdragon X65 in the S21 Ultra). This upgrade enables the S24 Ultra to support higher CA in mid/low bands, with up to 4 component carriers (CCs) in the DL direction and up to 2 CCs in the UL direction, compared to the S21 Ultra's 2-CC/1-CC CA capabilities, and take full advantage of 5G SA's advanced CA features, as we show in §4.4.2. For our server-side infrastructure, we deployed an AWS EC2 server in Virginia.

Experimental Methodology. We conducted backlogged DL/UL TCP throughput tests using nuttcp [17] and measured round-trip time (RTT) using ICMP ping in a round-robin sequence throughout each drive. Application-layer throughput and ping-RTT samples were logged at 500 ms and 200 ms intervals, respectively. Since commercial off-the-shelf smartphones do not grant access to low-level KPIs, such as 5G mode (SA vs. NSA), handover details, frequency bands, or CA, we employed Accuver XCAL Solo [14] devices to extract those features. These devices access the Qualcomm diagnostic interface of the smartphone, allowing us to capture signaling messages between the UE and the base station and to gather low-level KPIs essential for a comprehensive analysis of network coverage and performance.

4.3 Coverage

We begin by taking a closer look at T-Mobile's 5G SA and NSA coverage in the US. We define coverage for a particular technology as the fraction of miles driven under that technology over the total miles driven during our trips. Fig. 4.3a shows that in both years 5G (SA and NSA combined) was the prevalent technology during our driving route (93.6% of the total miles driven in 2023 and 97.8% in 2024). In 2023, 5G NSA was the leading 5G deployment mode with a coverage



(a) Overall SA vs. NSA coverage. (b) Location-wise difference: SA vs. NSA.

Figure 4.3: 5G Standalone and Non-Standalone coverage analysis.

of 71.3% of the total miles driven while SA coverage accounted only for 22.3%. In contrast, in 2024, 5G SA became the dominant deployment mode with a coverage of 61.9% vs. 35.9% for NSA. This significant shift reflects T-Mobile's focused and aggressive efforts to transition toward a fully standalone 5G infrastructure.

Next, we explore whether the SA and NSA deployment strategies vary based on location type. We categorize test locations as either "Big City" (within a 5-mile radius of the downtown of a city with a population over 300K) or "Non-Big City" (outside a 10-mile radius from the downtown of any big city)[2] and present the results in Fig. 4.3b. In big cities, SA coverage increased significantly from 35% in 2023 to 90% in 2024, highlighting the prioritization of SA deployments in urban, high user density regions, where there is a higher need for high-bandwidth and low-latency networks. Interestingly, even outside major urban centers, we observe non-negligible SA coverage both in 2023 and 2024: 19% and 58%, respectively. These results show a slower growth in SA coverage outside urban centers compared to the big cities, suggesting that T-Mobile first concentrated their efforts on SA deployments in urban areas before expanding into less populated regions. Nevertheless, we observe a clear decline in NSA coverage for both area types in 2024 indicating the operator's strategic move towards phasing out LTE-dependent 5G NSA networks in favor of a fully standalone 5G infrastructure.

[2]Ideally, we would like to use the gNB location for our classification, but we only have the UE location, which can result in some misclassifications, when the UE is outside the 5-mile radius of a large city but is connected to a gNB inside the 5-mile radius. To avoid such misclassifications, we omit all the locations between a 5-mile and 10-mile radius.

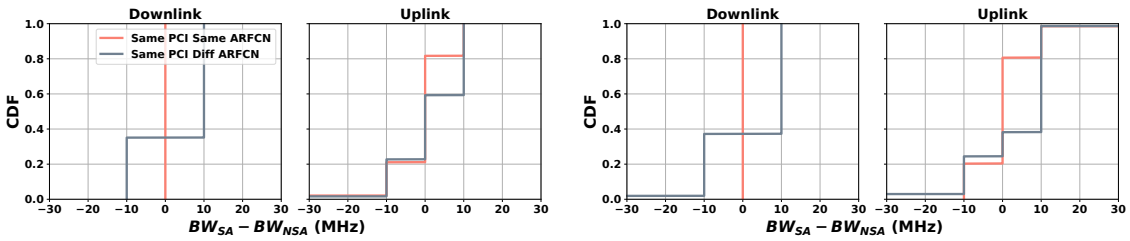
Table 4.1: T-Mobile NSA and SA head-to-head analysis.

(a) Coverage

	Chicago		Boston	
	LTE	5G	LTE	5G
NSA	5%	95%	8%	92%
SA	17%	83%	2%	98%

(b) Cell id (PCI) and channel number (ARFCN) Distribution

	Same PCI Same ARFCN	Same PCI Different ARFCN	Different PCI Same ARFCN	Different PCI Different ARFCN
Chicago	44%	40%	6%	10%
Boston	43%	38%	8%	11%



(a) Chicago. (b) Boston.

Figure 4.4: Difference between NSA and SA PCell bandwidth in Chicago and Boston.

We point out that during our driving tests, we had no control over the cellular technology to which the UE was connected. It is possible that a given location offers both NSA and SA coverage, and the operator's policies determine which of the two 5G technologies the UE will connect to at a given time. As such, the low NSA coverage in 2024 in Figs. 4.3a, 4.3b does not necessarily mean that T-Mobile has removed 5G NSA gNBs or replaced them with SA gNBs; it could also suggest that the operator heavily prioritized their SA service in areas where both modes were available.

To study this aspect in more detail, we conducted two smaller *controlled* driving experiments: one in Chicago, West Lafayette, IN, and the highway connecting the two cities, which was also part of our main drives in 2023 and 2024 (referred to as "Chicago"), and a second one in the city of Boston. In these experiments, we used two S24 phones side-by-side, and locked one of them to NSA mode and the other one to SA mode using the operator service code *#2263#. When the locked mode was unavailable, the UE fell back to LTE. The coverage results are summarized in Table 4.1a.

Table 4.1a shows very high coverage for both 5G deployment modes, confirming that T-Mobile often deploys both modes at a given location. In the Chicago drive, NSA coverage was 95% vs. 83% for SA. Interestingly, in the Boston drive, SA coverage is slightly higher than NSA coverage (98% vs. 92%), suggesting the existence of a small number of 5G gNBs offering only SA service.

Resource sharing between SA and NSA. For those locations where both 5G modes were available, we further look at the serving cell (Physical Cell ID/PCI) and the channel number (ARFCN) used by each mode in Table 4.1b. Table 4.1b shows that the prevalent deployment mode (84% in Chicago,

81% in Boston) in both datasets is to use the same serving cell for both modes. In contrast, cells offering only one mode appear less frequently – in 16% of the locations in Chicago and 19% of the locations in Boston.

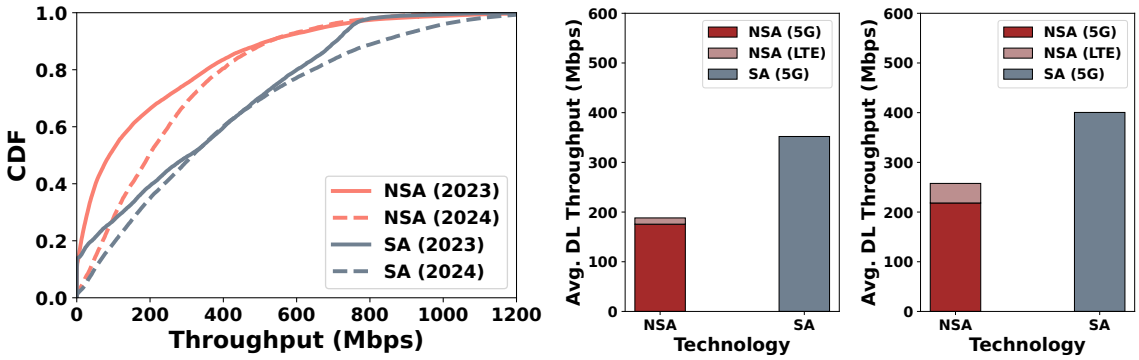
For the locations where both phones connect to the same serving cell, the operator has two options to support both modes. It can either use the same channel for both modes and share the resources (e.g., via RAN slicing) or use a dedicated channel for each mode, offering performance isolation at the cost of higher spectrum usage. Interestingly, Table 4.1b shows a balance between the two options (44% vs. 40% in Chicago, 43% vs. 38% in Boston), with a slight preference towards the first option, which reduces spectrum usage at the cost of (potentially) reduced performance. In contrast, for the locations where each mode is served by a different cell, we observe a slight preference towards different channels for each mode (10% vs. 6% in Chicago, 11% vs. 8% in Boston) potentially to reduce inter-cell interference.

For the cases where both modes are served by the same primary cell, we further plot the difference in the primary cell's bandwidth used by each mode ($BW_{SA} - BW_{NSA}$) in Figs. 4.4a, 4.4b. When the same channel is used by both modes, the operator allocates the full channel bandwidth to both modes ($BW_{SA} - BW_{NSA} = 0$) in the DL direction (Figs. 4.4a(left), 4.4b(left)). In contrast, in the UL direction (Figs. 4.4a(right), 4.4b(right)), the operator allocates the full bandwidth to both modes 60% of the time, but splits the bandwidth *unequally* 40% of the time, favoring either of the two modes. When the two modes are served by different channels, SA is allocated a wider channel most of the time.

Overall, T-Mobile's 5G SA coverage has shown significant growth from 2023 to 2024, surpassing NSA as the leading 5G deployment mode. Although a higher prioritization of deployment is observed in large urban centers, the SA coverage has grown significantly even in smaller cities and rural areas. Our smaller focused studies suggest that the most common deployment mode is to upgrade NSA gNBs to also offer SA services (using either the same or a different channel) instead of replacing NSA gNBs with SA gNBs or deploying additional SA cells.

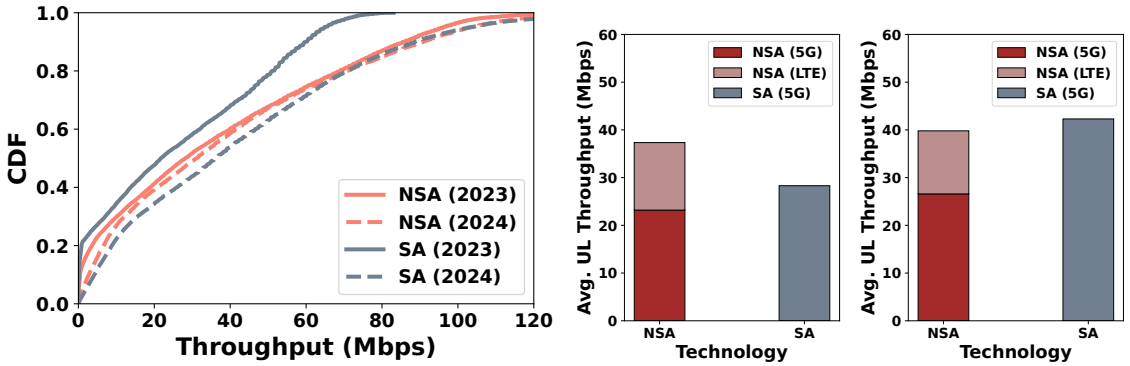
4.4 Network Performance

In the previous section, our results showed high 5G SA coverage in 2024. In this section, we explore whether the high 5G SA coverage actually translates to better performance compared to 5G NSA.



(a) Overall. (b) Breakdown (2023) (c) Breakdown (2024).

Figure 4.5: NSA vs. SA: DL Throughput.

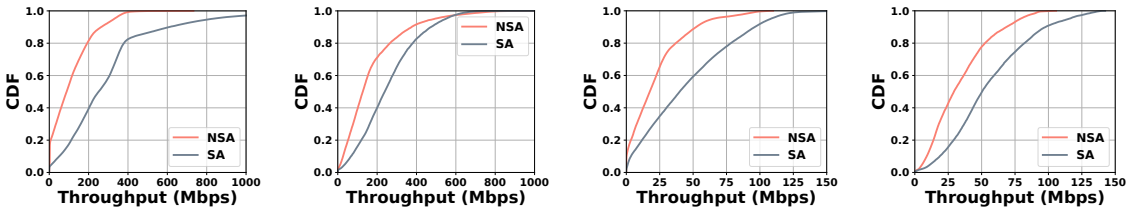


(a) Overall. (b) Breakdown (2023) (c) Breakdown (2024).

Figure 4.6: NSA vs. SA: UL Throughput.

4.4.1 Throughput

Fig. 4.5a shows the CDFs of DL throughput by 5G NSA and 5G SA in 2023 and 2024. For 5G SA, the median throughput remains the same in both years, but we observe an improvement in peak throughput (90th percentile) and worst-case throughput (20th percentile). For 5G NSA, there is a noticeable throughput improvement in 2024, especially in the lower and median percentiles, but the peak throughput remained unchanged. The improvement in 5G NSA throughput in 2024 is attributed to the refinements in LTE-5G dual connectivity (DC), as seen in Figs. 4.5b and 4.5c. Both the 5G and LTE (DC) average throughput have increased significantly to provide a combined overall higher throughput in 2024. Despite these improvements, 5G SA outperforms 5G NSA in terms of DL throughput in both years. Although the gap has narrowed in 2024 (from 218 Mbps to 119 Mbps in the median case), 5G SA still provides substantial gain over 5G NSA in terms of DL throughput.



(a) Downlink (Chicago). (b) Downlink (Boston). (c) Uplink (Chicago). (d) Uplink (Boston).

Figure 4.7: NSA and SA head to head performance comparison in Chicago and Boston.

Next, we analyze the UL performance of 5G SA and NSA. Fig. 4.6a shows the CDFs of 5G NSA and SA UL throughput for 2023 and 2024. We observe a completely different trend here compared to the DL performance. The UL performance of 5G NSA has remained largely unchanged from 2023 to 2024, whereas there is a significant improvement in UL performance of 5G SA over the same duration. The 25-th/50-th/75-th percentile of 5G SA UL throughput increased from 3/22/47 Mbps in 2023 to 12/36/64 Mbps in 2024. In fact, the 5G SA throughput was lower than the 5G NSA throughput in 2023, but higher in 2024.

Similar to DL throughput, we again break the 5G NSA UL throughput into 5G and LTE (DC) in Figs. 4.6b and 4.6c. We observe that in 2023, the throughput contribution from the 5G cell in NSA was actually lower than 5G SA throughput, but the LTE (DC) component helped 5G NSA achieve an overall higher throughput compared to 5G SA. In 2024, we observe a small increase in the contribution of the 5G component and a small reduction in the contribution of the LTE component, but the 5G SA UL throughput slightly surpasses the combined 5G and LTE NSA throughput. Again, this finding is contrasting to the DL performance (Figs. 4.5b and 4.5c), where we observed that both the 5G and LTE throughput for NSA increased from 2023 to 2024.

We note that the results in Figs. 4.5a, 4.6a are obtained by measurements at different locations, as the UE was connected only to one mode at a given location, and hence, they may be affected by various confounding factors (for example, the area type). For a more fair, head-to-head comparison, we again use our smaller datasets from the Chicago and Boston drives, collected with two UEs locked in NSA and SA mode, respectively. The results in Figs. 4.7a-4.7d confirm the results from our large-scale measurement campaign in 2024, showing a significant throughput gain in the median case for SA over NSA in both the DL (163/112 Mbps in Chicago/Boston vs. 119 Mbps in Fig. 4.5a) and UL direction (22/21 Mbps in Chicago/Boston vs. 5 Mbps in Fig. 4.6a).

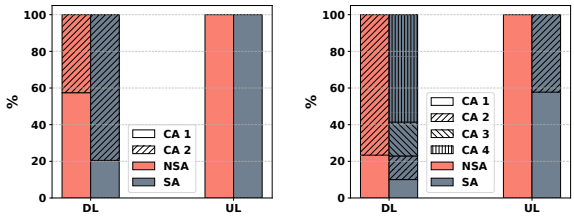
Overall, we observe an increasing trend in 5G SA performance in both traffic directions, with SA consistently outperforming NSA in most scenarios. While NSA shows a noticeable improvement in DL performance from 2023 to 2024, its UL performance remained largely unchanged.

Interestingly, the peak throughput values for NSA appear to have plateaued, indicating a possible shift in the operator's deployment priorities.

4.4.2 CA in 5G SA vs. NSA and its Impact on Performance

5G SA is expected to support higher CA [96, 97, 98, 99], aggregating carriers from multiple frequency bands (mid and low bands) and different duplexing modes (FDD and TDD) in both traffic directions. High CA increases the aggregate bandwidth and has the potential to improve the overall network performance. In this section, we take a deeper look at the diversity of CA among 5G SA and NSA deployments, the different band combinations that exist, and the overall impact on network performance. Since CA is only performed under heavy traffic, we only use the data collected during download/upload tests for this comparison.

Overall CA Distribution. Fig. 4.8 shows the CA distribution for 5G NSA and 5G SA for 2023 and 2024 in both traffic directions.



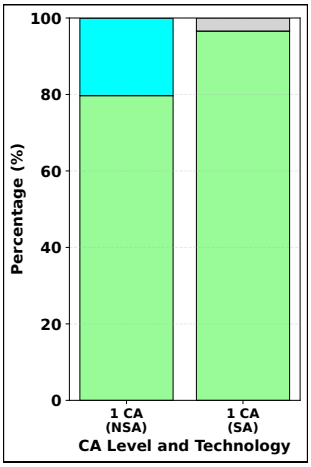
(a) 2023 (b) 2024

Figure 4.8: CA distribution.

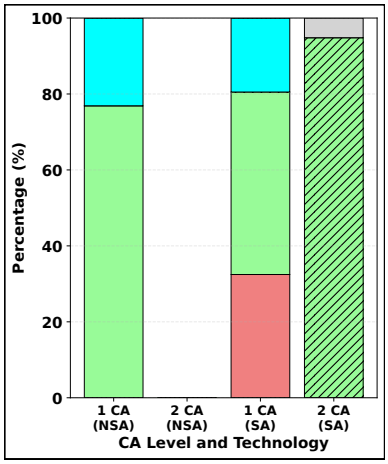
In the DL direction, in 2023 both NSA and SA relied exclusively on 1-CC and 2-CC configurations, but the percentage of 2-CC configurations for SA is much higher than for NSA (80% vs. 42%). In 2024, 2-CC CA becomes the primary configuration for NSA (78%), while we observe the introduction of 3-CC and 4-CC configurations for SA (20% and 60%, respectively). We note here that these CA patterns are also influenced by the UE model. Recall that our measurements in 2023 were done with a S21 Ultra smartphone, which supported a maximum of 2 CCs. Hence, it is possible that the operator also allowed for CA of 3 and 4 CCs in 2023, but this is not reflected in our 2023 dataset due to the UE model. On the other hand, in 2024, even with the use of a better phone supporting higher CA, our results show that T-Mobile refrained from configuring higher CA for 5G NSA.

In the UL direction, we see lower levels of CA overall for both deployment modes in both years. In 2023, we observe no CA for either mode, which again could be due to the limitation of the S21 UE model, which did not support UL CA. In 2024, similar to the DL direction, even though the S24 model supports 2-CC aggregation, we observe 2-CC configurations only for SA (41% of the time), while 5G NSA still supports only 1 CC.

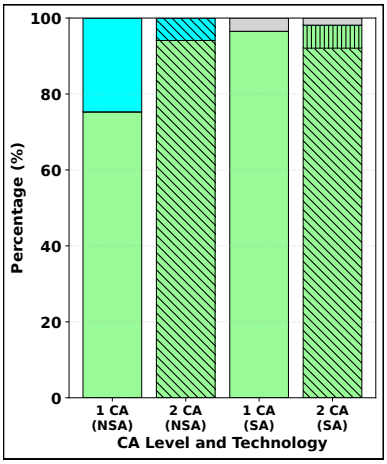
Band Combinations in CA. Fig. 4.9 shows the distribution of the different frequency bands



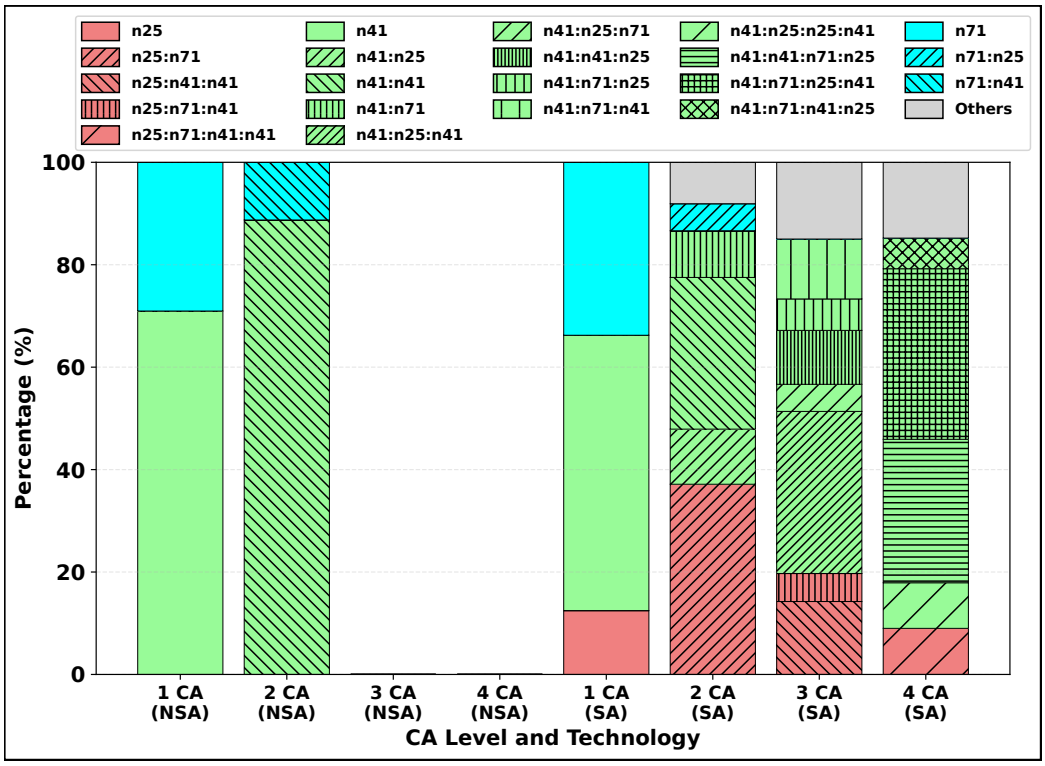
(a) UL (2023).



(b) UL (2024).



(c) DL (2023).



(d) DL (2024).

Figure 4.9: 5G NSA vs. SA: CA and frequency bands distribution.

aggregated for 5G NSA and SA in 2023 and 2024 in both traffic directions. Any band or band combination that appears less than 5% of the time is included in the ”others” category.

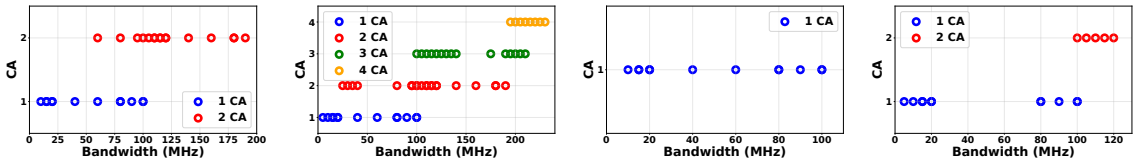
Figs. 4.9c and 4.9d show the band combinations for DL CA for 2023 and 2024. For NSA, the frequency band distribution has not changed drastically from 2023 to 2024. In 1-CC

configurations, n41, which is the high-capacity TDD 5G-mid band, dominates as the primary band in both years, with a smaller percentage (25% in 2023, 29% in 2024) contributed by n71, which is a low frequency FDD band. With 2-CC aggregation, we again see a dominance of two n41 CCs aggregated in both years, and a small percentage (less than 6%) of 5G-low band (n71 FDD) and 5G-mid band (n41 TDD) aggregation in 2023, which slightly increases (11%) in 2024.

On the other hand, we observe several interesting trends in 5G SA. While in 2023 most of the 1-CC SA deployments were using the 5G-mid TDD n41 TDD band, in 2024, we observe around 46% of the time the use of 5G-low and 5G-mid FDD bands, including the newly available FDD 5G-mid band n25. For 2-CC configurations, in 2023, similar to NSA, we mainly observe the combination of two n41 CCs or n41 combined with n71 (with the difference that the primary cell here always uses the high-capacity n41 band). In contrast, in 2024, we observe a large diversity, with several combinations of n41, n71, and n25. Further, in 2024, for 3-CC and 4-CC configurations, we observe a significantly broader variety of band combinations, with a notable share of 'others,' indicating the presence of numerous less-common combinations that, while smaller in number, contribute to the overall diversity. These configurations show the growing complexity and maturity of SA deployments that leverage higher CA to maximize spectrum utilization and ultimately lead to better performance, as observed in §4.4.1.

The availability of a large number of TDD and FDD combinations in 2024, particularly under 3-CC and 4-CC configurations, offers the operator a variety of choices for its SA deployments that trade off performance, latency, and robustness. The combination of a TDD band (n41, with a total bandwidth of 194 MHz) with an FDD band (n71 or n25, with a total bandwidth of 35 MHz and 65 MHz, respectively) can potentially reduce throughput compared to aggregating two wide n41 channels, but offers extended coverage and improved robustness [59]. As both FDD bands operate at lower frequencies – 600 MHz (n71) and 1.9 GHz (n25) vs. 2.5 GHz (n41), they have a higher range. As a result, the UE can continue using the lower-frequency FDD 5G band when it moves outside of the boundaries of the TDD cell, rather than downgrading to LTE. An additional benefit of the FDD bands is reduced latency, as DL and UL transmissions can take place simultaneously.

Next, we look at the distribution of frequency bands in the case of UL CA. Fig. 4.9a and 4.9b show the frequency band breakdown for different CAs in 2023 and 2024. Recall from Fig. 4.8 that NSA always uses 1 CC for UL communication. We observe similar band distributions for NSA in 2023 and 2024 with n41 being the dominant band and n71 used for a smaller fraction of time, similar to the DL direction. For SA, the trend for 1-CC configurations in the UL direction is also similar to the DL direction. The n41 band is used almost exclusively in 2023, while in 2024, low-



(a) NSA (DL). (b) SA (DL). (c) NSA (UL). (d) SA (UL).

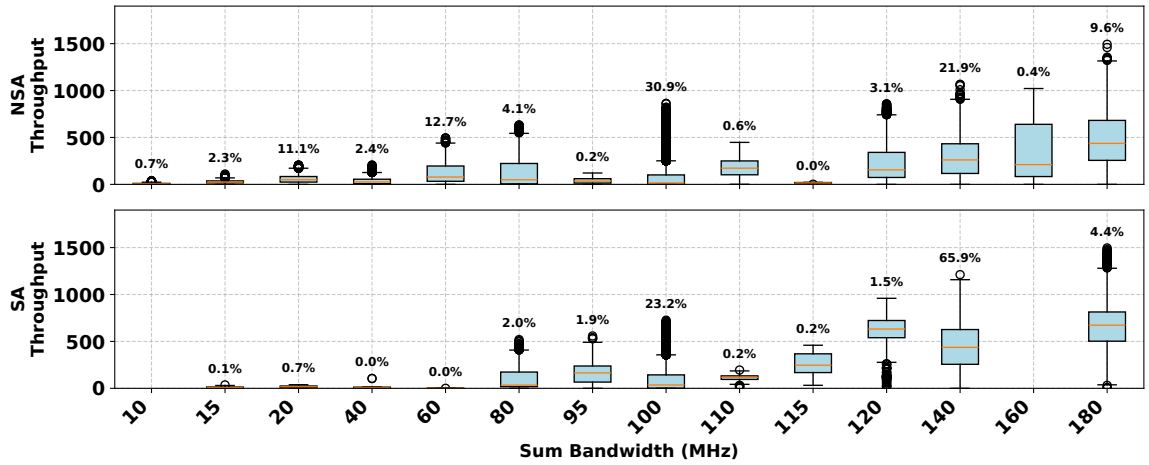
Figure 4.10: CA vs. aggregate bandwidth.

and mid-band FDD n71 and n25 bands are often used, but n25 is given higher preference over n71, unlike in the DL direction. Interestingly, we observe almost no diversity in 2-CC configurations; the combination of TDD mid band n41 and FDD mid band n25, which balances performance and robustness, is used 95% of the time.

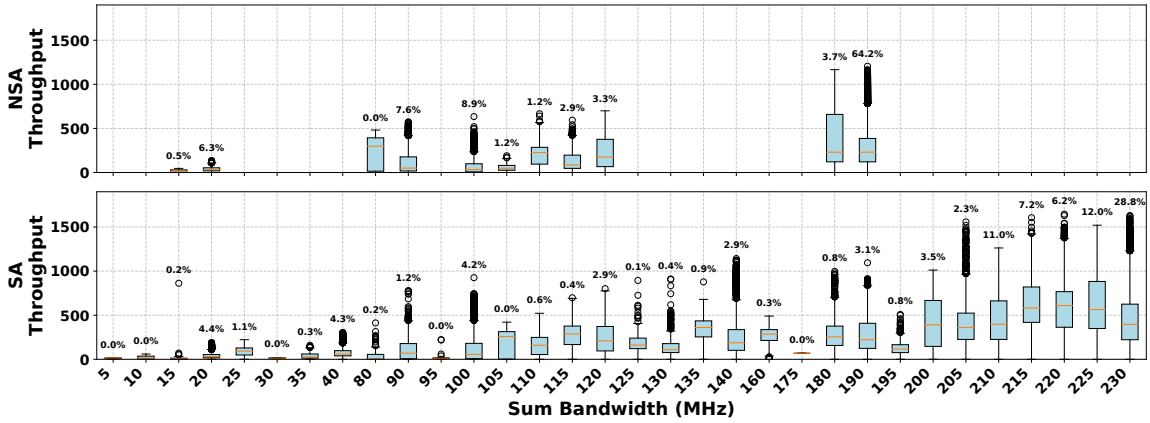
We note that, given the large number of available bands and channel bandwidth configurations (10-100 MHz), a higher CA does not necessarily lead to a higher aggregate bandwidth. This is clearly visible in Figs. 4.10a, 4.10b, which show the different CA and aggregate bandwidth combinations for NSA and SA, respectively, in the DL direction. In the case of NSA, we observe 2-CC configurations with aggregate bandwidths of 60, 80, and 90 MHz vs. 1-CC configurations over a wide channel of 100 MHz. Similarly, in the case of SA, we observe 2-CC configurations with lower aggregate bandwidth than 1-CC configurations, 3-CC configurations with lower aggregate bandwidth than 2-CC configurations, and 4-CC configurations with lower aggregate bandwidth than 3-CC configurations. In contrast, in the UL direction (Figs. 4.10c, 4.10d), 2-CC configurations always lead to higher bandwidth than 1-CC configurations.

Throughput and bandwidth correlation. In the previous section, we explored the diversity of band combinations in 5G SA and NSA and showed that a higher CA does not always lead to a higher aggregate bandwidth. In this section, we examine the distribution of the aggregate bandwidths resulting from using those diverse band combinations and investigate how the aggregate bandwidth correlates with throughput. Understanding this relationship will provide insights into the performance advantages enabled by wider bandwidths and more advanced CA configurations. Since the LTE component of the total NSA throughput is not related to the 5G bandwidth, we use the 5G MAC throughput in this section instead of the application layer throughput.

Figs. 4.11a, 4.11b plot the DL throughput vs. the aggregate bandwidth for NSA and SA in 2023 and 2024, respectively. The percentages above the boxplots show the fraction of samples of a given bandwidth in our dataset. The first observation we make from these figures is that the relationship between throughput and aggregate bandwidth is not strictly monotonic. The throughput of a cell is affected by many factors in addition to the cell bandwidth (e.g., MIMO, cell load, channel



(a) 2023.



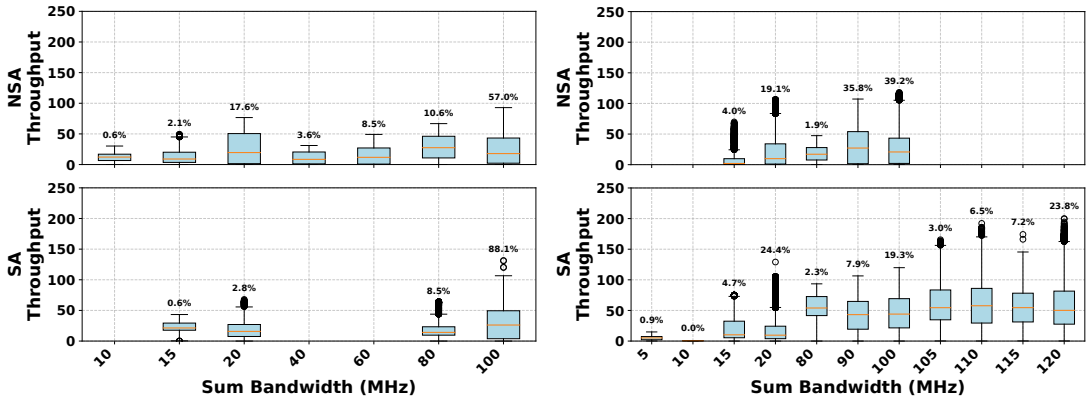
(b) 2024.

Figure 4.11: NSA vs. SA: DL throughput vs. aggregate DL bandwidth in 2023 and 2024.

quality), which are outside our control. However, despite the lack of a strict monotonicity, we do observe that very low bandwidths ($\leq$ 60 MHz) generally yield very low throughput and high bandwidths ($\geq$120 MHz in 2023, $\geq$200 MHz in 2024) generally yield the highest throughput.

In 2023 (Fig. 4.11a), NSA exhibits a more balanced distribution across low, medium, and high aggregate bandwidths, while SA demonstrates a strong tendency towards higher bandwidths, with a substantial number of samples (95.2% vs. 66.5% for NSA) achieving bandwidths of 100 MHz or more. Consequently, the higher bandwidth configurations in SA allow it to achieve better overall throughput compared to NSA, as shown previously in Fig. 4.5a.

In 2024 (Fig. 4.11b), we observe a large fraction (71%) of high-bandwidth configurations



(a) 2023. (b) 2024.

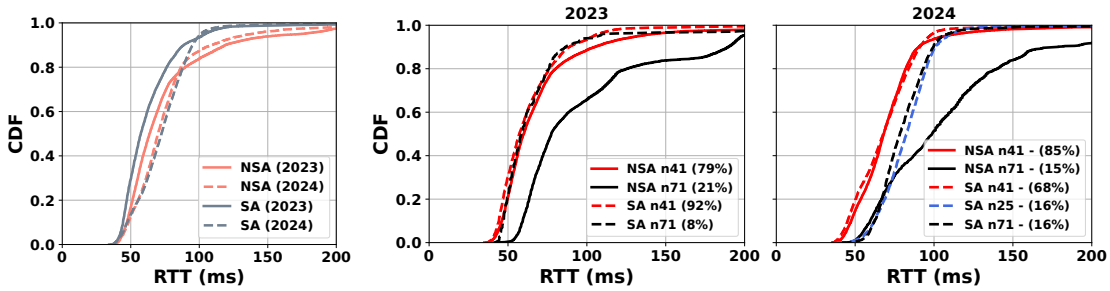
Figure 4.12: NSA vs. SA: UL throughput vs. aggregate UL bandwidth in 2023 and 2024.

(200+ MHz) for SA, contributed by aggregating 3 and 4 CCs. On the other hand, since NSA aggregates a maximum of 2 CCs, the maximum aggregate bandwidth it achieves is limited to 190 MHz, which is the most commonly used aggregate bandwidth (64.2% of the samples vs. 0 samples in 2023). The large fraction of samples at 190 MHz bandwidth with a median throughput of 231 Mbps leads to a substantial improvement in NSA's overall throughput in 2024, as shown previously in Fig. 4.5a. On the other hand, the high-bandwidth configurations for SA do not improve its overall median throughput compared to 2023; in fact, the median throughput at the highest bandwidth of 230 MHz is much lower than the median throughput at lower bandwidths (215-225 MHz). However, they do lead to a substantial increase in its peak throughput, above the 90th percentile (Fig. 4.5a).

Next, we perform a similar analysis with UL bandwidth and throughput in Figs. 4.12a, 4.12b. Similar to the DL case, higher bandwidths generally lead to higher throughput, although the throughput-bandwidth relationship again is not strictly monotonic.

Fig. 4.12a shows that in 2023, both deployment modes had a high percentage of 100 MHz aggregate bandwidth samples, which is the highest available bandwidth for a single CC (recall that in 2023 both SA and NSA used only 1-CC configurations in the UL direction). However, 5G NSA had a non-negligible distribution of low bandwidths ($\leq$ 40 MHz) as well compared to SA, while SA used almost exclusively (96.6% of the samples) cells with 80 MHz or 100 MHz bandwidth, resulting in overall higher 5G throughput (Fig. 4.6b).

In 2024 (Fig. 4.12b), we see a very different picture for both NSA and SA. NSA now uses only 5 bandwidth configurations (vs. 7 in 2023). The fraction of 100 MHz samples has dropped significantly, from 57% to 39.2%, but we also observe a large fraction (35.8%) of 90 MHz samples (absent in 2023), which actually yield slightly higher throughput than the 100 MHz samples.



(a) Overall RTT. (b) RTT breakdown by frequency bands.

Figure 4.13: SA vs. NSA RTT.

In contrast, for SA, the introduction of 2-CC CA in 2024 leads to a large number of bandwidth configurations (9 vs. only 4 in 2023) from 5 MHz up to 120 MHz. The large percentage of samples above 100 MHz (40.5%) due to CA boost SA's throughput significantly in 2024 compared to 2023 (Fig. 4.6a).

In summary, our analysis shows the growing maturity of 5G SA deployments, leveraging higher CA and bandwidth configurations and diverse band combinations, which, combined with the advanced capabilities of newer UE models, enable SA to achieve higher performance compared to NSA. In contrast, the CA configurations for NSA remained largely static across both years (2-CC in the DL direction, 1-CC in the UL direction), even with more capable UE models, marking a clear plateauing of the peak achievable throughput. Overall, SA deployments show a clear trajectory towards higher bandwidth utilization and enhanced performance, underscoring their critical role in the ongoing evolution of 5G infrastructure.

4.4.3 RTT

Fig. 4.13a shows the CDF of RTT over NSA and SA in 2023 and 2024. In both years, the SA RTT is either similar to or shorter than the NSA RTT. This is expected, as in 5G SA, there is no reliance on the underlying LTE core for control plane interactions, which inflates the overall end-to-end latency for NSA. However, we observe that, regardless of the deployment mode, the RTT has increased in 2024 compared to 2023. The increase is higher for SA, while for NSA, the RTT actually has decreased slightly above the 80th percentile. We conjecture that with T-Mobile's aggressive deployment of 5G across the country, more users have been using 5G services, causing congestion and eventually inflating the RTT. A similar observation was reported earlier in this thesis (§3) for the years 2022 and 2024.

We further break down the RTT samples based on the 5G frequency bands in Fig. 4.13b. In 2023, the NSA RTT was much higher than the SA RTT in the n71 band, while the gap was much smaller in the n41 band. NSA's inflated RTT over the n71 band and the largest percentage of n71 samples in NSA compared to SA (21% vs. 8%) resulted in an inflation of the overall NSA RTT in Fig. 4.13a. In 2024, the RTT over the n71 and n41 bands increased substantially for both deployment modes. However, in the case of SA, we also see the appearance of the new n25 band, which operates in mid-band with FDD, and has the highest RTT among the three SA bands. The introduction of the n25 band and the much smaller percentage of n41 samples in SA in 2024 compared to 2023 (68% vs. 92%) contribute to the overall inflation of RTT for SA in 2024 and close the gap with NSA.

The high RTT for the n25 band appears counter-intuitive, as one would expect the RTT to be lower in an FDD system with dedicated DL/UL channels compared to a slot-based TDD system, where the latency should be higher because of asymmetric DL/UL slot distribution prevalent in today's cellular networks. Upon further inspection, we found that n25 uses a smaller subcarrier spacing (SCS) of 15 KHz compared to 30 KHz in the n41 band. The subcarrier spacing is the frequency gap between individual subcarriers in an OFDM signal. A larger SCS means that the frequency separation between adjacent subcarriers in the OFDM wave form is larger. This reduces the duration of each OFDM symbol and enables faster data transmission, which reduces the end-to-end RTT.

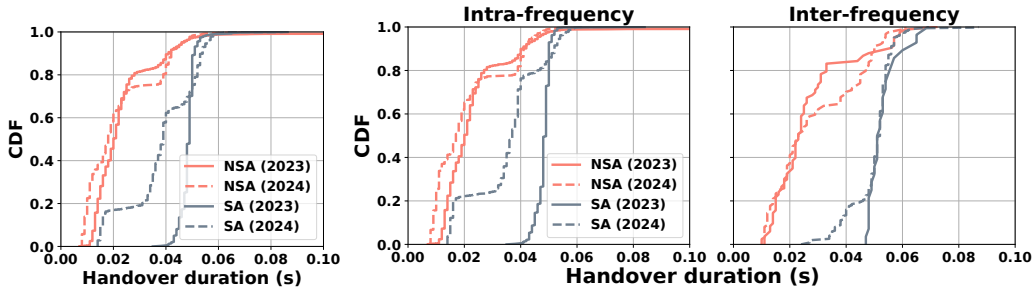
In general, SA consistently achieved either comparable or lower RTT than NSA, benefiting from the absence of the LTE anchor, which inflates the latency due to added control plane signaling. However, the inclusion of the new FDD band n25 in 5G SA in 2024 inflated the SA RTT and reduced the gap with NSA.

4.5 Handover Duration in 5G SA and NSA

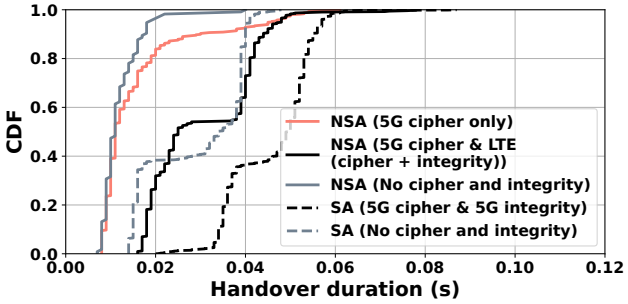
Mobility management in cellular networks, especially the handover (HO) process, plays a key role in providing seamless and ubiquitous Internet access. NSA relies on the 4G LTE core for signaling and control-plane interactions while SA operates independently with its own 5G core. In this section, we focus on comparing the 5G→5G HO durations in the two deployment modes. [3]

Overall HO duration analysis. We define the HO duration as the time from the moment the UE receives an RRC Reconfiguration message from the network that instructs the UE to start the HO

[3] We also examined the HO failures for both modes. We found that the HO failure rate is very low – 1-2% for NSA and <1% for SA – in both years. Hence, we do not further analyze HO failure rate in this study.



(a) Overall. (b) Inter-/intra-frequency HO duration.



(c) Impact of security keys exchanged.

Figure 4.14: SA vs. NSA 5G→5G handover duration.

process until the moment the UE sends an RRC Reconfiguration Complete message to the network indicating a successful HO. Fig. 4.14a shows the 5G→5G HO duration for SA and NSA in both years. We make two observations. First, the HO duration for both technologies has decreased from 2023 to 2024. The drop is more pronounced in case of 5G SA, where the median HO duration decreased from 49 ms to 39 ms in 2024. Second, surprisingly, the NSA 5G→5G HO duration is always lower than the SA counterpart.

The second observation is counter-intuitive, as one would expect the NSA HO latency to be higher because of the involvement of an additional LTE cell over which the 5G cell is anchored. However, we found that 5G→5G HOs in T-Mobile's NSA deployment are done via a Secondary Node (SN) modification, which can be performed without the involvement of the LTE anchor [100, 101], thus significantly reducing the HO signaling. On the other hand, in the case of SA, the HO happens always between two 5G master nodes and involves a substantial amount of signaling between the two cells as well as the 5G core network. Higher signaling exchanges naturally lead to higher delay in the HO execution process.

Intra-frequency vs. inter-frequency HOs. We further break down the HOs into intra- and inter-frequency HOs in Fig. 4.14b. We observe that switching to a different frequency cell incurs higher

latency for both NSA and SA in both years. This is expected, as a higher amount of signaling is required in the case of inter-frequency HOs, where the UEs need to adjust the radio carrier frequency for communication.

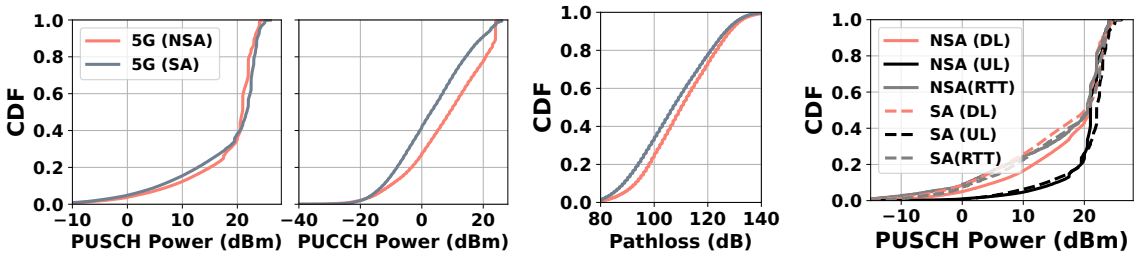
The reduction in HO duration for both intra- and inter-frequency HOs in 2024 compared to 2023 is lower for NSA than for SA. For SA, especially in intra-frequency HOs, we observe a drastic improvement in 25-th/50-th/75-th percentile HO duration from 2023 to 2024 highlighting the maturity and optimization of SA deployments. Additionally, the minimum inter-frequency HO duration for SA in 2024 decreased to 24 ms (from 47 ms in 2023) and around 25% of the samples are considerably lower in 2024 compared to 2023. However, across both HO types and years, the HO duration for NSA is still lower than for SA.

Impact of security key exchanges. 5G SA networks introduced enhanced security mechanisms [102, 87, 88] compared to existing 5G NSA and LTE networks by incorporating stricter security measures, particularly in the context of key exchanges. The key exchange process includes both ciphering and integrity protection that ensures confidentiality and authenticity of signaling messages and user data. These security mechanisms play a crucial role in determining the HO duration. We check the RRC Reconfiguration message sent by the network to initiate a HO for the type of security keys that are exchanged and break down the HO duration based on the type in Fig. 4.14c.

We observed 3 and 2 distinct scenarios w.r.t. key exchanges in NSA and SA, respectively. In NSA, the three scenarios are: (a) 5G cipher NEA2 (New Encryption Algorithm 2) used only for user plane data, as integrity protection for user plane data in dual connectivity is not supported [102], (b) 5G cipher NEA2 used for user plane data and LTE EEA2 (EPS Encryption Algorithm 2) cipher and LTE EIA2 (EPS Integrity Algorithm 2) integrity protection for signaling layers, and (c) No security keys exchanged. For SA, we have: (a) 5G cipher NEA2 and integrity protection NIA2 for both signaling and user plane data, and (b) No security keys exchanged.

In Fig. 4.14c, the HO duration for the cases where both the cipher and integrity protection keys are exchanged is the highest for both modes. In this scenario, the duration for NSA is comparatively shorter than for SA, probably because the 5G user plane does not have integrity protection due to dual connectivity in NSA. In contrast, in SA, both signaling and user plane data have ciphering and integrity protection key exchanged during the HO. Moreover, NSA relies on LTE's EEA2 and EIA2 for ciphering and integrity protection, whereas in SA, more advanced NEA2 and NIA2 keys are exchanged with stringent security measures leading to higher HO duration.

When only 5G cipher is exchanged without any LTE cipher or integrity protection (for NSA only), the HO duration reduces significantly. This is expected, as in this scenario, there is no



(a) PUSCH and PUCCH power. (b) Pathloss. (c) PUSCH by traffic type.

Figure 4.15: Overall PUSCH power and PUCCH power for SA and NSA; Pathloss in SA and NSA; PUSCH power breakdown by traffic type.

active involvement of the LTE anchor, and the dearth of signaling between the 5G cell and the LTE anchor cell reduces the latency considerably.

Finally, we take a look at those cases where no security-related keys were exchanged in SA and NSA. For SA, the duration drops to a minimum duration of as low as 14 ms and most samples are less than 40 ms. This is a significant reduction w.r.t. the case where both the keys are exchanged in SA. For NSA, the HO duration when no keys are exchanged, is mostly similar to the duration when only 5G cipher key is exchanged for user plane data, with the gap between the two cases becoming substantial only after the 80th percentile.

Overall, there are multiple factors that affect the HO duration in SA and NSA. NSA benefits from reduced signaling overhead in cases where the LTE anchor is not involved in the 5G→5G HO, which minimizes the HO delay. In contrast, SA introduces more signaling requirements and enhanced security mechanisms, such as exchange of NEA2 and NIA2 keys for signaling and user plane data. These rigorous checks ensure robust security but inherently increase latency. Additionally, the type of HO – intra- vs. inter-frequency – also impacts the HO duration. These findings underscore the trade-offs between reduced signaling complexity and latency in NSA versus robust security and independence of SA, highlighting the distinct priorities and challenges in mobility management for the two architectures.

4.6 Tx Power in SA and NSA

UL transmit (Tx) power is a critical metric in cellular networks, reflecting the strength of the signal used by the UE to communicate with the base station. UL Power Control (ULPC) in cellular networks [103, 104] is a mechanism used to adjust the Tx power of mobile devices in the

UL direction. ULPC optimizes the quality and efficiency of the communication while minimizing interference to other users. It continuously adjusts the transmit power levels based on feedback from the base station, environmental conditions, and network requirements to maintain an optimal balance between signal quality and power consumption. In this section, we compare the PUSCH (Physical UL Shared Channel) power and PUCCH (Physical UL Control Channel) power for 5G NSA and SA deployments.

Fig. 4.15a shows the PUSCH power and PUCCH power for SA and NSA. We observe that the PUSCH power is comparable for the two deployment modes. The SA PUSCH power is slightly higher than the NSA PUSCH power 60% of the time but lower 40% of the time. On the other hand, the control channel PUCCH power for NSA is always higher compared to SA and the gap is much more pronounced compared to the PUSCH power. The median PUCCH power for NSA is 9 dBm, compared to only 3 dBm for SA.

To understand these differences, we plot the pathloss of NSA and SA in Fig. 4.15b. We observe that the pathloss for NSA is always higher than for SA, with the NSA median being 110 dB and the SA median being 107 dB. This explains why the PUCCH power for NSA is always higher than for SA. Since NSA has a higher percentage of deployments in non-urban areas, i.e., highways, where the base stations are typically further away from the UEs, the UEs use higher PUCCH power to compensate for higher path loss due to longer UE-base station distance.

However, the higher pathloss for NSA does not explain why in Fig. 4.15a around 60% of the time the PUSCH power is lower for NSA than for SA. To explore further, we break down the PUSCH power based on the traffic type: backlogged DL traffic, backlogged UL traffic, ICMP-ping RTT tests in Fig. 4.15c. We observe that the PUSCH power is slightly higher for SA compared to NSA in the case of UL backlogged traffic, whereas for the other two types of traffic, where there is not much active UL traffic – TCP ACKs for DL tests and ICMP echo requests for ping tests – the SA PUSCH power is either similar to or lower than the NSA PUSCH power. This highlights the operator's policy to use higher Tx power for SA in the presence of heavy UL traffic. This policy, combined with higher UL carrier aggregation for SA, explains the higher UL throughput observed in § 4.4.1 for 5G SA deployments compared to NSA.

4.7 Discussion

In this section, we discuss several unique aspects of our experimental methodology and explain why they have only minimal influence on our findings.

(i) Head-to-head comparison between SA and NSA. Our primary objective in this study is to understand and compare coverage and performance of the two modes (SA vs. NSA) *from the perspective of a general user*. To this end, we conducted our driving campaigns with a single phone, without locking it to a specific network mode – thus reflecting how a real user would experience the network. We acknowledge that this methodology does not fully capture the actual coverage of the two modes: in many locations, both NSA and SA may be available, and the mode selected by the UE depends on the operator's policies. To mitigate this limitation, we conducted two smaller *controlled* driving experiments with two S24 phones in parallel, explicitly locking one to NSA and the other to SA. The details of these experiments have been described in §4.3 and the results in §4.3 and §4.4.1 align with the findings from our main driving campaigns.

(ii) Driving routes. Our two driving routes shared the same endpoints (Boston, Chicago) and followed the same highways, but the second trip was in the reverse direction. While this could in theory cause minor variations in UE–base station distance (depending on which side of the highway the vehicle was on), a difference of a few meters has negligible effect on received power and performance, especially for low- and mid-band frequencies.[4] Routes within downtown areas also varied somewhat due to traffic conditions, signals, or road closures. These differences are unlikely to affect our findings, as downtowns of large cities typically have dense 5G coverage.

(iii) Duration and time of measurement campaigns. The 2023 campaign lasted four days, with the final morning dedicated to downtown Chicago. In 2024, we covered downtown Chicago on the first day before departure. The overall duration is not critical, as both campaigns followed the same route. The campaigns were conducted in different months (May vs. November), which could influence traffic load but not coverage, and traffic is inherently outside our control. We deliberately scheduled the 2024 campaign in November to capture a more mature stage of SA deployment. Weather effects are negligible for low- and mid-band frequencies, and all measurements were conducted under clear skies.

(iv) Generalization of the results. Our results are based on measurements from a single operator, T-Mobile, the only US carrier with SA deployed at scale during our study. At the same time, major European operators had only NSA deployments [58]. The study focuses on the East Coast and Midwest along one driving route; we acknowledge that results may differ in other regions such as rural areas. Nonetheless, we point out that this is the largest measurement study of SA networks to date, with the exception of the work in [92], which focuses exclusively on the impact of NSA/SA

[4] Recall that T-Mobile does not use mmWave bands, which have much shorter range.

on video streaming in China. As global SA rollouts accelerate, we expect our study to serve as a foundation for broader studies in other regions.

4.8 Related Work

5G Measurement Studies. Since the 5G rollout in 2019, a large number of studies [6, 7, 1, 2, 90, 55, 56, 8, 91, 67, 57, 45, 44, 46, 47, 60, 58, 62, 61, 3, 59, 66, 65, 105] have measured various aspects of operational 5G networks, including coverage, network performance, carrier aggregation, resource allocation, handovers, beamforming, and application QoE. Most of these studies are performed over commercial 5G NSA deployments. In contrast, 5G SA has received limited attention in the literature, particularly in terms of its large-scale evolution and comparative analysis with NSA.

Outside the US, only Yuan et al. in [92] study the performance of live video streaming over 5G SA and NSA networks in China from a content provider's perspective. They report that SA results in lower broadcast and viewer buffer delays and rebuffer proportions than NSA. In the US, Narayanan et al. in [1] and Hassan et al. in [3] conducted preliminary measurements with T-Mobile's SA deployments before 2022, when T-Mobile deployed 5G SA operating only at 600 MHz (5G low band) in a fragmented and limited capacity in the US. T-Mobile started deploying 5G mid band SA services as their primary deployments in late 2022. The works in [61, 93] are the only ones that performed measurement-based analysis of SA networks after 2022. However, [61] takes a snapshot of the SA performance in only two cities with just 10 hours of measurements in April 2023, when SA networks in the midband just started getting rolled out. As a result, it does not include any measurements over the n25 band or with 4-CC CA. It also reports a maximum aggregate bandwidth of only 160 MHz for SA with 3-CC CA, while our study (§4.4.2) reveals a much larger diversity of aggregate bandwidths – up to 180 MHz for both modes in 2023 and up to 190/230 MHz for NSA/SA in 2024. On the other hand, [93] only evaluates the latency of control plane procedures (attachment and registration, RRC connection, authentication and security, PDN/PDU connection) at three locations without studying data plane performance or coverage, and hence, it is complementary to our study. To the best of our knowledge, there is no other large-scale measurement work studying the coverage, performance, and network characteristics of operational SA networks and comparing them head to head with the existing NSA network across different regions of the US over an extended time period.

5G SA Evaluation over Simulations and Controlled Environments. There have been several studies [106, 107, 108, 109, 110, 111, 112] on different aspects of 5G SA based on theoretical

analysis, simulations, and measurements done in controlled settings. While these works offer valuable insights, they often fail to capture the full range of real-world scenarios, as different network operators implement unique policies and configurations that are difficult to replicate in controlled environments. Our study captures the real world performance of operational 5G SA networks via a large-scale measurement campaign comprising multiple cross-country trips.

4.9 Summary

This chapter presented the first large-scale measurement study characterizing the evolution of 5G Standalone (SA) deployments, performing a head-to-head comparison with the prevalent 5G Non-Standalone (NSA) architecture across two cross-country driving trips in the US (3,200+ km) 18 months apart. Our results show a significant improvement in 5G SA coverage from 2023 to 2024, with deployments expanding into both urban and non-urban areas. In terms of performance, 5G SA demonstrates clear advantages in both throughput and latency, driven by features such as high carrier aggregation and newer frequency bands that enable diverse intra-/inter-band and FDD/TDD configurations. In contrast, NSA's peak performance has plateaued, primarily due to its limited use of CA. Our analysis further reveals that the interplay between signaling complexity and security mechanisms plays an important role in governing handover duration for both SA and NSA. Overall, this study underscores the importance of continued investment in 5G SA networks to fully achieve the potential of 5G.

Together with the two preceding chapters, this chapter marks the conclusion of the first part of this thesis, which established the ground reality of commercial 5G networks through empirical measurement and longitudinal evolution studies. Having characterized what today's 5G networks actually deliver, we next turn to the second part of this thesis, which investigates the operator-specific policies that shape this behavior – beginning, in the next chapter, with an in-depth study of handover configurations.

Chapter 5

Too Many Reports, Too Few Handovers: A Large-Scale Study of Configuration Inefficiencies in Operational 5G Networks

Mobility management in cellular networks, and in particular the handover (HO) process, plays a critical role in providing seamless and ubiquitous Internet access. In operational networks, HO decisions are governed by a policy-based approach: each HO is triggered based on signal strength conditions, cell priorities, and a set of measurement events, all controlled by configuration parameters that operators set freely within the 3GPP framework [113].

The wide-scale deployment of 5G over the past six years has significantly increased the complexity of mobility management. 5G's support for diverse radio bands (low, mid, high), different architectures (non-standalone (NSA) vs. standalone (SA)), and the coexistence of LTE and 5G radio access technologies (RATs) make HO management in the 5G era far more complex than before. Prior measurement studies have shown that HOs can significantly degrade throughput and increase latency [7, 1, 3, 45, 114, 44, 90], including in our own measurements earlier in this thesis (§2, §3, §4). However, these studies focus on the *outcome* of the HO process — its impact on user performance — without examining its *root cause*: the configuration policies that govern when and how HOs are triggered. Whether current HO configurations are efficient, or whether they introduce unnecessary overhead and suboptimal decisions, remains largely unknown.

In this chapter, we revisit mobility management from a configuration-centric perspective and ask a fundamental question: *Are current HO configurations in operational 5G networks efficient and well-calibrated?* To answer this question, we conduct the first large-scale measurement study of HO configurations in the 5G era. Through four cross-country driving trips (Los Angeles to Boston in Aug. '22 and Nov. '24, Boston to Chicago in May '23, and Boston to Atlanta in Aug. '23) covering 15,000+ km over a 27-month period, we collect a dataset of 48K HOs and 239K HO-related reporting events spanning all three major US operators (Verizon, AT&T, T-Mobile), both RATs (LTE and 5G), both 5G architectures (NSA and SA), and all 5G frequency bands (low, mid, high). We analyze this dataset along four axes: the HO-triggering event landscape, the diversity and evolution of configuration parameters, the aggressiveness of those configurations and its role in generating overhead, and the impact of configurations on signal strength and ping-pong performance. Our key findings are:

• **HO-triggering events (§5.3).** The 5G HO landscape differs fundamentally from LTE: new inter-RAT events (B1, B2) have appeared while periodic events have been practically eliminated. Despite the large overall event volume, actual HO behavior converges on a single dominant event per HO type. The overhead is strongly bimodal — events are either largely decisive (A3, B1, A5) or almost entirely wasted (A1, A6, periodic) — and the conversion rate of the same event varies up to 10x across operators, confirming that overhead is a configuration choice rather than an inherent property of the event.

• **Diversity and evolution of configurations (§5.4.1).** LTE configurations have grown substantially more complex since prior work [52], with some parameters now taking 43–56 distinct values. 5G configurations have lower richness (fewer than 12 distinct values for most parameters) but still exhibit large dispersion, indicating that operators tune configurations differently across bands, area types, and deployment regions. LTE configurations have largely stabilized, while 5G configurations continue to diversify rapidly, with some parameters gaining more than 10 new values in two years.

• **Aggressiveness, overhead, and operational concentration (§5.4.2,§5.4.3).** We introduce a framework to classify each configuration as aggressive, moderate, or conservative. Our analysis shows that the overhead documented in §5.4.1 concentrates in a small number of event types whose configurations are needlessly aggressive. The decisive events are configured more tightly in 5G than LTE, showing that operators have learned to apply discipline where it matters most, but have not yet extended it to other events. Finally, despite the large and diverse configuration spaces operators provision, actual HO traffic concentrates in a tiny dominant subset. Most provisioned configurations are operationally irrelevant and this concentration is tighter in 5G than LTE.

• **Impact on performance (§5.5).** Optimizing HO configurations is fundamentally challenging: no single configuration simultaneously optimizes post-HO signal improvement, pre-HO signal degradation, and ping-pong rate, and a pervasive coverage confound means counter-intuitive outcomes across operators and configurations are frequently explained by deployment density rather than configuration quality. TTT is the most reliable lever for ping-pong control across all event types. Most strikingly, Verizon's 5G→LTE demotion strategy, while rational from a signal strength perspective, results in throughput degradation 80% of the time, a cost that RSRP-based optimization cannot capture.

Contributions. In summary, this study makes the following contributions: (1) We collect a large multi-carrier, multi-technology, multi-band dataset of cellular mobility management during four cross-country driving campaigns spanning 15,000+ km over a 27-month period. (2) We develop a revised methodology to accurately identify HO-triggering events in 5G NSA deployments, correcting a flawed assumption in prior work that affects 12% of HOs. (3) Using the collected dataset and revised methodology, we analyze the HO-triggering event landscape, the diversity and evolution of configuration parameters, the aggressiveness of those configurations and its role in generating overhead, and the impact of configurations on signal strength and ping-pong performance. (4) Our dataset will be open-sourced upon acceptance of the paper this work is part of.

5.1 HO Primer

A HO is used to transfer a user's network session from the cell it is attached to (serving cell) to another cell (neighboring cell), because either the user moves out of the serving cell's coverage area or there is another cell operating at a different RAT. HOs can be classified into idle-state HOs and active-state HOs, based on whether the UE is idle or sending/receiving traffic to/from the network. Idle-state HOs are performed by the device, which selects an appropriate cell for future access. Active-state HOs are initiated by the network; the serving cell migrates the UE to another neighboring cell. In this study, we focus on active-state HOs.

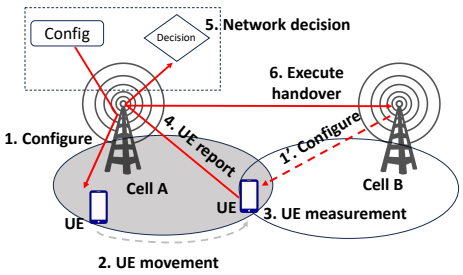


Figure 5.1: Basic HO procedure.

Fig. 5.1 shows a basic cellular HO procedure. When a UE connects to a new cell, the cell advertises a HO configuration (1) to the UE. Based on the configuration, the UE learns the criteria to trigger a HO, including whether to invoke measurement, what and when to measure, and when/what

Table 5.1: LTE/5G-NR HO-related reported events
(M = measurement, C = serving cell, N = neighboring cell)

Event Type	**Event Description**	**trigger Condition**
A1	Serving cell becomes better than a threshold	$M_S > \Theta_{A_1} + H_{A_1}$
A2	Serving cell becomes worse than a threshold	$M_S < \Theta_{A_2} - H_{A_2}$
A3	Neighboring cell becomes offset better than the serving cell	$M_N > M_S + \Delta_{A_3} + H_{A_3}$
A4	Neighboring cell becomes better than threshold	$M_N > \Theta_{A_4} + H_{A_4}$
A5	Serving cell becomes worse than a threshold and neighboring cell becomes better than another threshold	$M_S < \Theta^S_{A_5} - H_{A_5}$ $M_N > \Theta^N_{A_5} + H_{A_5}$
A6	Neighboring cell becomes offset better than the serving cell	$M_N > M_S + \Delta_{A_6} + H_{A_6}$
B1	Inter-RAT neighbor becomes better than threshold	$M_N > \Theta_{B_1} + H_{B_1}$
B2 (5G only)	Primary cell becomes worse than a threshold and inter-RAT neighbor becomes better than a threshold	$M_S < \Theta^S_{B_2} - H_{B_2}$ $M_N > \Theta^N_{B_2} + H_{B_2}$
P	Periodic reporting of serving and neighboring cell's radio signal quality.	N/A

to report. As the UE starts moving from the coverage area of Cell A to a new coverage area under Cell B, it starts measuring (3) different signal metrics (RSRP, RSRQ). 3GPP defines a set of measurement report (MR) mechanisms based on *events*, which are triggered when certain signal-related criteria are met. When an event is triggered, the UE sends an MR (4), which the network uses to decide (5) whether to perform a HO or not. Based on the network's decision, the HO is executed (6) from Cell A to B, followed by a new set of configurations (1') advertised from Cell B to the UE.

Intra-RAT events (A1–A6) are triggered by conditions on the serving or neighboring cell's signal strength: A1 is triggered when the serving cell becomes better than a threshold (typically used to cancel a pending HO), A2 when it becomes worse (used to prepare for a HO), A3 when a neighbor becomes stronger than the serving cell by an offset, and A5 when the serving cell drops below one threshold while a neighbor rises above another. Inter-RAT events B1 and B2 handle transitions between LTE and 5G. Each event is governed by three key parameters: a threshold or offset defining the required signal condition, a hysteresis margin preventing rapid back-and-forth triggering, and a time-to-trigger (TTT) specifying how long the condition must hold before a report is sent. Full details are in Tables 5.1 and 5.2. While events and parameters are standardized by 3GPP, operators are free to select different configurations (i.e., values for each parameter), implementing different HO policies — the central subject of this study.

Table 5.2: Description of common event parameters.

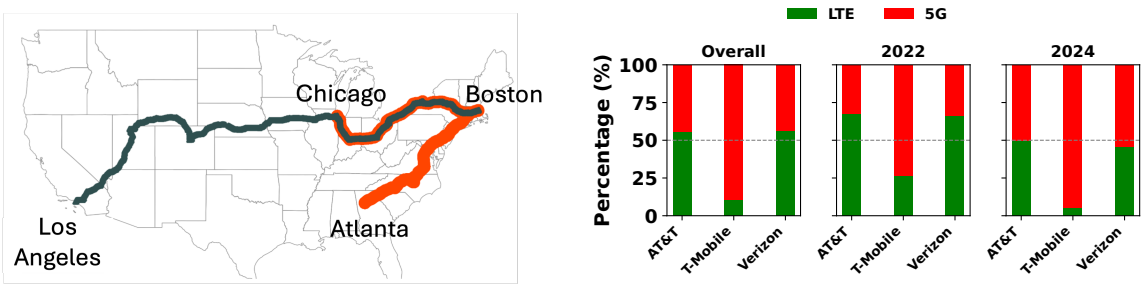
Parameter Name	**Description**
Hysteresis H	Prevents unnecessary and fast HOs. Sets a threshold below which the serving cell is considered better than a neighboring cell, even if the neighboring cell's signal strength is slightly stronger.
Threshold Θ	Defines the level at which a measurement is triggered. When the measured value crosses the threshold, a measurement report is generated.
Offset Δ	Added to the signal strength of a serving or a neighboring cell. Creates a margin in the comparison between the serving cell and the neighboring cell.
Time To Trigger TTT	Time during which specific criteria for the event need to be met in order to trigger a measurement report.
Trigger Quantity	Defines the measured parameter (RSRP or RSRQ) used to trigger an event.

5.2 Methodology

UEs and measurement tools. We used Samsung S21 Ultra (in 2022, 2023) and S24 (in 2024) phones as UEs, the state-of-the-art devices at the time of our measurement campaigns. Since we only focus on *active-state* HOs, which are controlled by the network, *our findings are independent of the UE model and can be generalized for any smartphone model*. We performed a mix of backlogged downlink/uplink throughput, ICMP-based ping RTT, and application tests in a round-robin fashion throughout the driving trips. Each UE was connected to an Accuver XCAL Solo device [14], which extracts and logs PHY layer KPIs and RRC layer signaling messages such as HOs, configurations, and MRs. We performed measurements *concurrently* over the three operators, using 3 UEs, each connected to a different operator.

Driving Tests. We drove 15500+ km during the four road trips covering all major cities on each route (Fig. 5.2a). In addition to driving on highways, we conducted additional downtown driving tests in select major cities on our routes to obtain data from urban environments.

5G Carriers. We purchased multiple unlimited data plans from all three major US service providers, Verizon, T-Mobile, and AT&T, to perform the experiments. Fig. 5.2b shows the overall cellular coverage landscape for all three operators in our dataset. The 5G coverage varies from 43.3% (for Verizon) up to 89.5% (T-Mobile). Further comparison of the coverage over the same route in 2022



(a) Measurement routes.

(b) Overall coverage.

Figure 5.2: Measurement route and cellular coverage.

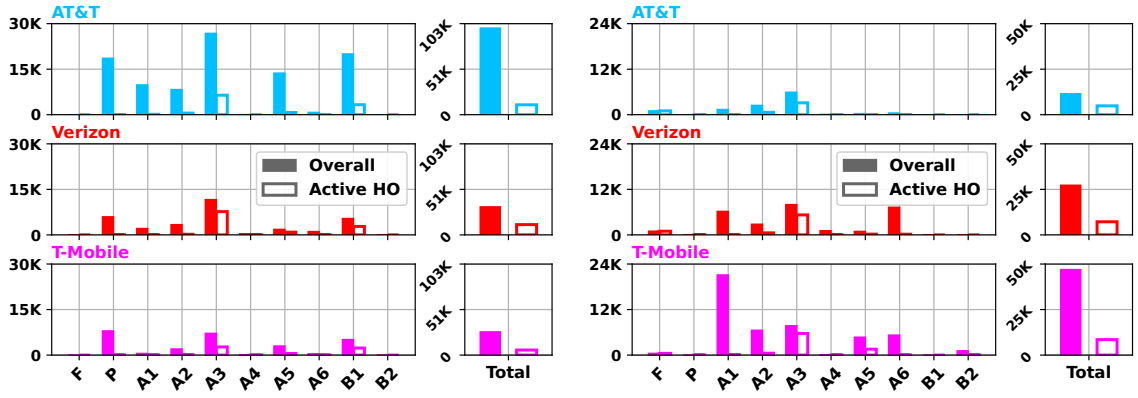
and 2024, shows an increase in 5G coverage for all 3 operators: from 32% to 50% for AT&T, from 73% to 94% for T-Mobile, and from 33% to 54% for Verizon. Notably, all three operators exhibit substantial coverage of *both* LTE and 5G throughout our measurement campaigns, allowing us to observe and study the full range of HO types.

Extracting HO-triggering events. Prior work [52] assumed the last reporting event before a HO is the one that triggered it. We found this assumption fails in 12% of HOs in our dataset, particularly in NSA deployments, and developed a revised methodology that inspects all MRs preceding a HO in both LTE and 5G UL DCCH messages to identify the one whose cell ID and frequency match the new cell. This approach uncovered several unusual cases: a B1 event, normally used for LTE→5G HOs, can sometimes trigger an LTE→LTE HO; certain A1/A2 reports include neighboring cell measurements, making them capable of triggering a HO directly; and for a subset of 5G HOs, complete configuration parameters must be traced back through multiple prior HOs due to the network sending only partial RRC reconfiguration messages when a UE reconnects to a previously visited cell.

5.3 HO Reporting Events: Diversity and Signaling Overhead

Distribution of HO-related reporting events. Figs. 5.3a and 5.3b break down the reporting events for each operator, for LTE and 5G respectively. For each event, we show both the total number of events reported by the UE (”Overall”) and the number of events that actually led to a HO (”Active HO”). The three operators generate vastly different overall numbers of events: 98K/32K/26K (AT&T/Verizon/T-Mobile) in LTE and 11.5K/27K/47K in 5G.

Fig. 5.3a shows that AT&T triggers the highest number of reporting events in LTE, followed



(a) LTE. (b) 5G.

Figure 5.3: Distribution of HO-related reporting events. F: 5G failures, P: periodic events.

by Verizon, and T-Mobile the fewest. The lower event count for T-Mobile is consistent with its significantly lower LTE coverage compared to the other two operators (Fig. 5.2b). The distribution of events also differs substantially across operators, suggesting the usage of different event parameter configurations. While A3 is the dominant event for all three operators in LTE (7K–26K), AT&T triggers a much larger number of A1, A2, A5, and B1 events (8K–20K) compared to Verizon and T-Mobile (0.5K–5K). Notably, AT&T and Verizon have comparable LTE coverage (Fig. 5.2b), yet their event distributions differ significantly, highlighting operator-specific diversity in event triggering policies, which we explore in detail in §5.4.

While a similar disparity in triggered event counts exists for 5G, the ordering among operators reverses: T-Mobile generates the most 5G events, consistent with its ˜90% 5G coverage, while AT&T generates the fewest. In terms of event distribution, however, we observe notable contrasts relative to LTE. While A3 remains the dominant event for AT&T and Verizon in 5G, consistent with their LTE behavior, T-Mobile's most frequently triggered event in 5G is A1, an event that was practically never triggered in LTE. A6 forms a significant fraction of total 5G events for both Verizon and T-Mobile compared to LTE. We also observe new event types absent in the pre-5G era: B1 (LTE→5G NSA) and B2 (5G SA→LTE), and various 5G failure events. Notably, periodic events (P) are practically absent in 5G across all operators.

Dominant triggering events. When we look at the events that actually triggered HOs in Figs. 5.3a and 5.3b, the picture is strikingly different from the overall distribution. The total number of HO-triggering events is very similar across operators (11K/11K/5K for AT&T/Verizon/T-Mobile in LTE and 4K/7K/8K in 5G), and only three events dominate: A3 (60–70% of all HOs), B1 (15–20%), and A5 (15% for T-Mobile, 5–6% for the other two). Table 5.3 summarizes the dominant triggering event

for each HO type. Despite the increased complexity of the 5G HO landscape, operators converge on a single dominant event for each HO type. The majority of 5G→LTE demotions are an exception to this pattern: 60–69% are caused by various 5G failure events rather than by a reporting event, meaning they lie outside the RSRP-based HO framework entirely, and cannot be addressed through configuration tuning alone.

Quantifying the event triggering overhead. Figs. 5.3a and 5.3b suggest that all three operators use aggressive configurations, forcing the UE to trigger a very large number of events, most of which are ignored by the network. We quantify this overhead as the ratio of reports that actually triggered HOs to the total number of reports, for each operator and technology and found only 11%/21%/36% (LTE) and 45%/18%/27% (5G) of reported events actually triggered HOs for AT&T/T-Mobile/Verizon. Fig. 5.4 breaks down the overhead by event type, revealing three important patterns.

Table 5.3: Dominant triggering event per HO type.

HO type	Dominant triggering event
LTE→LTE	A3 (80-86%)
5G→5G intra-band	A3 (79-96%)
5G→5G inter-band	A5 (52-72%)
LTE→5G	B1 (98-99%)
5G (NSA)→LTE	5G failures (60-69%)
5G (SA)→LTE	B2 (75%)

First, the conversion rate is strongly bimodal: events are either largely decisive (A3, B1, A5, A2, and A4 in 5G only) or almost entirely wasted. Periodic, A1, and A6 events convert at less than 3.5% across all operators and both technologies despite being reported in large volumes — these are the dominant source of overhead. The most extreme case is T-Mobile, which triggers 21K A1 events in 5G of which less than 1% actually trigger a HO. This is partly by design — A1 signals that the serving cell has recovered and cancels a pending HO, while A6 is responsible for carrier aggregation — but the large volume of these events represents a substantial and avoidable signaling cost. The absence of periodic events in 5G (Fig. 5.3b) suggests that operators are actually trying to reduce unnecessary signaling on the 5G interface, since periodic events do not typically lead to HOs for LTE (Fig. 5.3a); the same reasoning should be extended to A1 and A6.

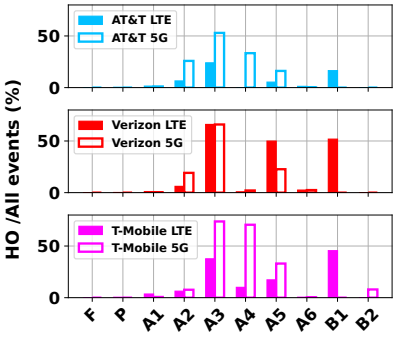


Figure 5.4: Event overhead breakdown.

Second, the conversion rate of the same event varies dramatically across operators, confirming that overhead is a configuration choice rather than an inherent property of the event. In LTE, an A5 event triggers a HO 49.9% of the time for Verizon but only 5.3% for AT&T, nearly a 10x difference. Similar gaps exist for B1 (52% vs. 16%) and A3 (66% vs. 24%). AT&T is the least

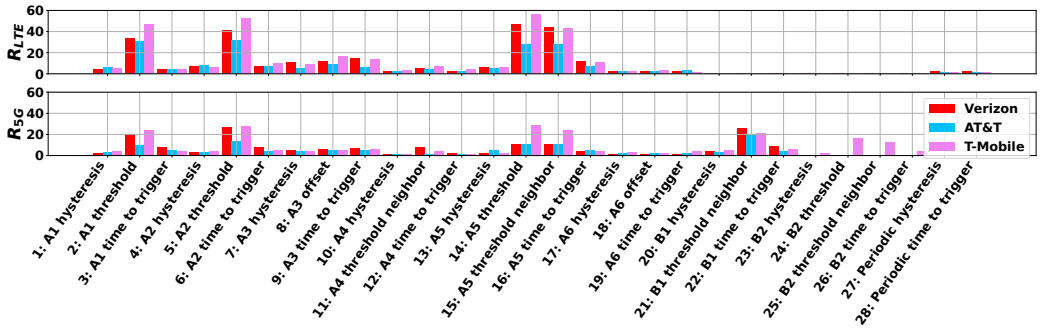


Figure 5.5: Richness of HO configuration parameters.

efficient operator overall: even its decisive events convert poorly in LTE, consistent with its lowest aggregate conversion rate (11%), whereas Verizon is the most efficient (36%).

Third, 5G configurations are generally tighter than LTE: for most decisive events, a larger fraction of reports lead to a HO in 5G than in LTE (e.g., A3 rises from 24% to 53% for AT&T and from 38% to 74% for T-Mobile), suggesting more targeted HO decisions in 5G. A notable exception is A5 for Verizon, whose conversion drops from 50% in LTE to 23% in 5G, consistent with the A4-equivalent A5 misconfiguration we identify in §5.5.1.2.

Summary. The 5G HO landscape differs fundamentally from the LTE landscape: new triggering events have appeared (B1, B2) while others have been practically eliminated (periodic events). Despite the large overall event volume, actual HO behavior is dominated by a single event per HO type, meaning much of the configuration complexity is underutilized. Most importantly, all three operators force the UE to report a very large number of events, the vast majority of which the network ignores, an inefficiency that is a direct consequence of configuration choices and that §5.4.2 examines in detail.

5.4 HO Configuration Parameters: Diversity and Aggressiveness

5.4.1 Diversity of HO configurations

Fig. 5.5 shows the richness R for 28 parameters related to different reporting events in LTE and 5G, which measures the number of unique values for a given parameter. We observe that most LTE parameters take multiple values. On the extreme end, A3 offset has 9–17 distinct values for each operator, and A1 threshold, A2 threshold, and the two A5 thresholds take 28–31 distinct values in AT&T, 34–44 in Verizon, and 43–56 in T-Mobile – even higher than those reported for AT&T in [52], where the largest richness observed was 21. Even parameters like A3 hysteresis, reported as

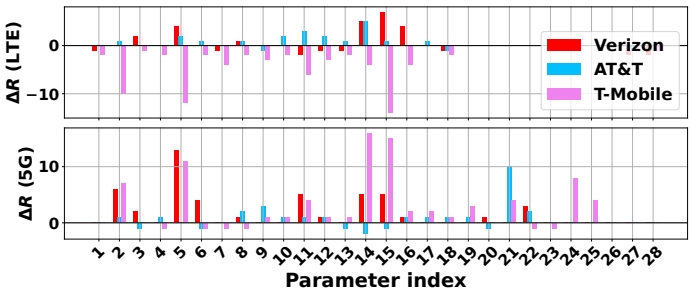


Figure 5.6: Richness comparison, R(2024) - R(2022).

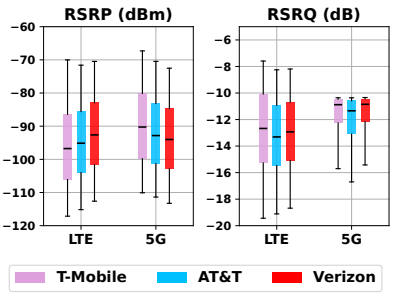


Figure 5.7: RSRP/RSRQ distributions.

single-valued in [52], now have 5–11 different values. The only parameters that exhibit low richness are those related to events used infrequently (A4, A6, and periodic). Interestingly, T-Mobile exhibits the highest richness for several parameters despite its lower LTE coverage, indicating that richness reflects operator policy rather than sampling volume. On the other hand, the richness of 5G values is much lower, with most parameters having fewer than 12 distinct values. This is true even for T-Mobile, despite its much higher 5G than LTE coverage (Fig. 5.2b), reinforcing our conjecture that observed richness is not an artifact of the number of encountered cells.

Evolution. Fig. 5.6 shows the difference in richness for each parameter in 2022 vs. 2024: $\Delta R = R(2024) - R(2022)$. For LTE, ΔR is zero or negative for most parameters: operators have largely stopped tweaking LTE configurations as the technology matures, and the negative values for T-Mobile reflect its sharply reduced LTE coverage in 2024. For 5G, by contrast, ΔR is mostly positive, with some parameters gaining more than 10 new values in two years, reflecting ongoing adaptation to expanding coverage and evolving traffic demands. Notably, the parameters of the most popular event A3 (indexes 7, 8, 9) show only minor variation over the same period, suggesting that operators have converged on stable A3 configurations while continuing to fine-tune less dominant events.

5.4.2 *Aggressiveness* of HO Configurations

Fig. 5.8 normalizes the reported events in Fig. 5.3 by each operator's LTE/5G event count. This removes differences due to coverage and highlights how heavily each operator relies on each event type.

Defining aggressiveness. A HO configuration governs how easily an event is triggered through two major components: *metric*, which is the threshold (or offset) that the measured signal must cross, and the *time-to-trigger* (TTT) for which the condition must hold. We classify each observed configuration along both components as *aggressive*, *moderate*, or *conservative*. A configuration is

aggressive if it makes the event trigger easily and quickly (an easily-satisfied threshold/offset and a short TTT) and conservative if otherwise.

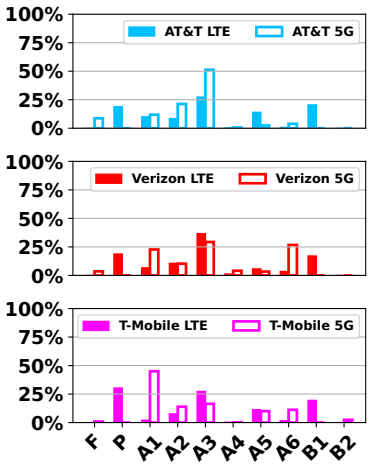


Figure 5.8: Normalized triggered events.

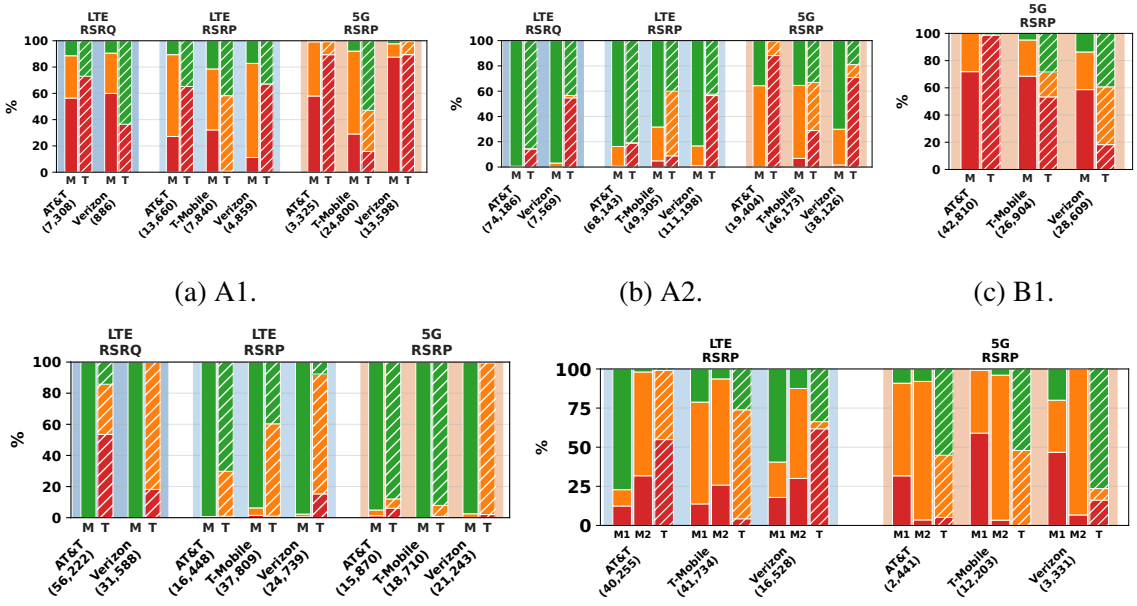
We set these cutoffs per event: threshold-based events (A1, A2, B1, and the two A5 thresholds) use absolute RSRP/RSRQ signal-quality bands chosen relative to the operating range we observe across our measurements (Fig. 5.7), while offset-based events (A3, A6) use offset cutoffs adopted in the spirit of the 3GPP reference parameter sets [115]. For events that trigger when a cell's signal strength rises *above* a threshold (A1, B1, and the neighboring cells threshold of A5), a *low* threshold is aggressive; for events that trigger when the serving cell's signal strength falls *below* a threshold (A2 and the serving cell threshold of A5), a *high* threshold is aggressive; and for offset-based events (A3, A6), a *small* offset is aggressive. Table 5.4 shows the cutoffs we use to classify the metric component of each HO event configuration as aggressive or conservative. For TTT, we label ≤ 200 ms as aggressive and ≥ 640 ms as conservative, following the discrete 3GPP TTT ladder [116, 117].

Fig. 5.9 shows, for each event, the breakdown of its configurations into aggressive, moderate, and conservative categories, separately for the metric M (threshold/offset) and the TTT (T). For 5G configurations, we show the metric with respect to RSRP, which was used exclusively as the trigger quantity. For LTE, we show the metric with respect to both RSRP and RSRQ, as AT&T and Verizon used both as trigger quantities.

Table 5.4: Aggressiveness classification.

Evt.	RSRP		RSRQ	
	Agg.	Con.	Agg.	Con.
A1	$\leq$-115 dB	$\geq$-90 dB	$\leq$-15 dB	$\geq$-12 dB
A2	$\geq$-105 dB	$\leq$-120 dB	$\geq$-13 dB	$\leq$-16 dB
A3	$\leq$2 dB	$\geq$5 dB	$\leq$1 dB	$\geq$2.5 dB
A5-S	$\geq$-105 dB	$\leq$-120 dB	$\geq$-12 dB	$\leq$-17 dB
A5-N	$\leq$-115 dB	$\geq$-90 dB	$\leq$-16 dB	$\geq$-12 dB
A6	$\leq$3 dB	$\geq$8 dB	$\leq$2 dB	$\geq$4 dB
B1	$\leq$-110 dB	$\geq$-95 dB	$\leq$-18 dB	$\geq$-12 dB

A1 — the dominant 5G overhead event. A1 is triggered when the serving cell exceeds a threshold, and is used primarily to cancel a pending HO. In LTE, A1 accounts for a modest share of events (2–10% across operators, Fig. 5.9a). AT&T triggers the most LTE A1 events, combining aggressive TTTs with aggressive RSRQ thresholds (56%), making A1 both easy and quick to trigger; Verizon's aggressive thresholds are concentrated mainly in RSRQ (60%), resulting in fewer A1 events than AT&T; T-Mobile, which relies almost exclusively on RSRP and uses moderate-to-conservative TTTs, triggers the fewest. In 5G, however, A1 becomes the single largest event type for T-Mobile, accounting for 45% of all its 5G events, more than any other event for any operator in either



(a) A1. (b) A2. (c) B1.

(d) A3. (e) A5.

Figure 5.9: Aggressiveness of an event to be triggered. M = Threshold/Offset Metric, M1 = Threshold Serving (A5), M2 = Threshold Neighbor (A5), T = TTT. Aggressive, Moderate, Conservative.

technology. Verizon and AT&T see lower but still notable 5G A1 shares (23% and 12%, respectively).

Strikingly, this ordering is the opposite of what configuration aggressiveness alone would predict: Verizon uses the most aggressive A1 configurations in 5G (88% aggressive thresholds, 89% aggressive TTTs), followed by AT&T (58%, 89%), while T-Mobile uses the least aggressive (29%, 16%). Aggressiveness explains why Verizon triggers more A1 events than AT&T, but not why T-Mobile (with the least aggressive configurations) triggers the most. The explanation lies in coverage: T-Mobile has a much stronger 5G RSRP distribution than the other two operators (Fig. 5.7), so even its strict thresholds are more easily satisfied by its strong serving cells, driving a large volume of A1 reports. The result is a large number of events that almost never lead to HOs; T-Mobile's 5G A1 conversion rate is under 1%. Since A1's role is to cancel pending HOs rather than trigger them, its high volume represents avoidable signaling cost, and as the T-Mobile case illustrates, even conservative A1 configurations cannot prevent excessive triggering when the underlying 5G signal is strong enough to satisfy the threshold frequently.

A6 — carrier aggregation (CA) overhead. A6 is used for secondary-cell management in CA rather than for actual HOs. In 5G, it accounts for 27% of Verizon's reported events and 11% of T-Mobile's (Fig. 5.8). The key distinction for A6 is not how aggressive the configurations are, but whether an operator uses it at all. Verizon actively uses A6 with an 8 dB offset and an aggressive 160 ms TTT,

while AT&T and T-Mobile largely disable it; 93% and 71% of their configurations set the offset to its maximum value with a 5120 ms TTT, so A6 rarely triggers. Since A6 does not trigger HOs, its entire contribution to event volume is overhead. Verizon's high A6 share reflects a deliberate CA management strategy, but at a significant signaling cost with no HO benefit.

A3 and B1 — the decisive events. A3 and B1 are the events that actually drive the majority of HOs (Table 5.3). Their configurations are notably tighter than A1: A3 offsets are overwhelmingly conservative ($\geq$5 dB for the majority of configurations, Fig. 5.9d and Table 5.4), and B1 thresholds are clustered tightly around -111 to -112 dBm (RSRP) across all three operators (Fig. 5.9c), reflecting a shared desire to promote UEs to 5G as soon as a viable cell is available. For A3 in LTE, AT&T uses RSRQ as the primary trigger quantity and sets RSRP offset and hysteresis to their maximum 60% of the time, effectively suppressing RSRP-based A3 triggering. For B1, operators use similar thresholds but differ substantially in TTT: AT&T uses a very aggressive median of 40 ms while Verizon uses a conservative median of 256 ms, with direct consequences for ping-pong HOs as we show in §5.5.1.3. Notably, 5G A3 configurations are tighter than their LTE counterparts; conversion rates are higher and configurations more targeted, suggesting operators have learned to apply discipline to their most-used intra-RAT event, though the same discipline has not yet been extended to A1 and A6.

A2. A2 is triggered when the serving cell falls below a threshold, making it an early indicator that the UE may need to be handed over. It accounts for a small but growing share of 5G events (8–21% across operators). In LTE, all three operators use a large fraction of conservative thresholds, which naturally limits A2 triggering; however, Verizon pairs these with aggressive RSRQ TTTs, triggering comparatively more A2 events than T-Mobile, which relies almost exclusively on RSRP. In 5G, all three operators relax their thresholds substantially (the conservative share drops from 68–99% in LTE to 35–70% in 5G) reflecting a need to trigger earlier fallback before the 5G link deteriorates too much. Operators differ mainly in TTT: AT&T uses an aggressive TTT 88% of the time in 5G, justifying its highest A2 share, while T-Mobile remains the most conservative, consistent with its goal of keeping UEs on 5G as long as possible given its strong 5G signal.

A5. A5 accounts for a small fraction of reported events (up to 14% in LTE, under 10% in 5G) and is not a significant overhead source. Its configuration is notable for a structural asymmetry: while the neighbor threshold is kept consistently moderate across all operators and technologies, the serving threshold is the real tuning lever. T-Mobile and Verizon frequently set it to -31 dBm in 5G (42% of configurations for Verizon, 16% for T-Mobile), a level that is almost always satisfied, making A5 functionally equivalent to an A4 event. This misconfiguration negates the key advantage of A5 over A3. Note that these figures reflect all deployed A5 configurations across the network. The subset that

actually triggers HOs, analyzed in §5.5.1.2, shows a different distribution: among HO-triggering configurations, AT&T's A4-equivalent rate rises to 25% and Verizon's to 43%, while T-Mobile's drops to 9%. This reflects which configurations are deployed in areas where A5 conditions are actually met in practice. Verizon is the most concerning operator on both counts, while T-Mobile's broad deployment of the -31 dBm threshold rarely translates into actual HOs.

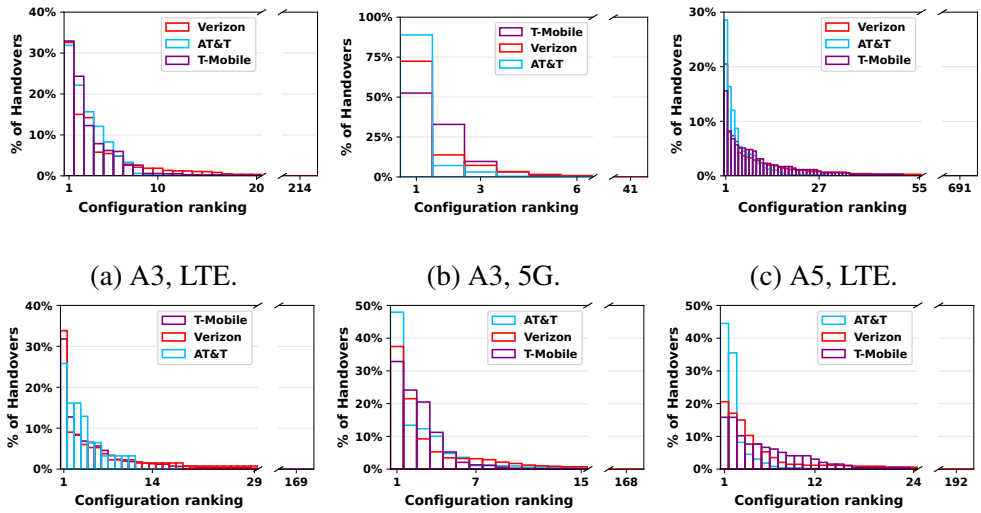
Periodic events. Periodic events, a significant source of overhead in LTE, are practically absent in 5G across all operators. Since periodic events rarely lead to HOs even in LTE, their elimination from 5G is a deliberate and effective reduction in unnecessary signaling, and a precedent that should be extended to A1 and A6.

Summary. The overhead documented in §5.3 is concentrated in a small number of event types: A1 (especially in 5G, where T-Mobile's 45% share is driven by strong coverage rather than aggressive configuration) and A6 (Verizon's 27% 5G share, entirely due to CA management with no HO benefit). The decisive events (A3 and B1) are configured more tightly, particularly in 5G, and the elimination of periodic events from 5G shows that operators can and do remove low-conversion events when the cost is recognized. Extending that discipline to A1 and A6, and correcting the A4-equivalent A5 misconfiguration deployed by T-Mobile and Verizon, represent the clearest opportunities for reducing overhead without affecting HO outcomes.

5.4.3 Operational Concentration of HO Configurations

We now examine what fraction of this large configuration space actually drives HOs in practice. For each event, we rank an operator's configurations (combinations of threshold/offset, hysteresis, and TTT) by the number of HOs they trigger and show each configuration's share of that operator's HOs (Fig. 5.10).

Across all events and operators, the distribution is steeply concentrated: a small set of configurations accounts for the majority of HOs, while the remaining ones form a long tail. For A3 in LTE (Fig. 5.10a), 5–14 configurations cover 90% of an operator's HOs, even though the total number of distinct configurations is 34/64/214 for AT&T/T-Mobile/Verizon. The concentration is even tighter in 5G: the number of distinct A3 configurations falls to 24–41 (Fig. 5.10b), and only 2–3 of them account for 90% of the HOs. AT&T's 5G A3 is the extreme case, with a single configuration driving 89% of its A3 HOs. A5 sharpens this contrast further. In LTE, A5 configurations form the longest tail with 267/691/569 distinct combinations for AT&T/T-Mobile/Verizon (Fig. 5.10c), yet only 11–34 of them contribute 90% of the HOs. As with A3, the A5 5G tail collapses relative to LTE, with T-Mobile



(a) A3, LTE. (b) A3, 5G. (c) A5, LTE.

(d) A5, 5G. (e) B1. (f) A2.

Figure 5.10: Concentration of HOs across different parameter configurations.

having the most at 169 combinations (Fig. 5.10d) and only 8–18 configurations contributing 90% of the HOs. B1 (Fig. 5.10e) is similarly steep, with its top five configurations covering 77–94% of LTE→5G HOs. The 5G A2 (Fig. 5.10f) is the most operator-dependent: AT&T's HOs collapse onto just four configurations (95% within its top five), while T-Mobile and Verizon spread theirs across 122 and 192 combinations, needing 12–15 to reach 90%. This is consistent with AT&T's highly standardized A2 TTT — 88% of its demotions use TTT = 160 ms (§5.4.2) — leaving little room for configuration diversity to matter in practice.

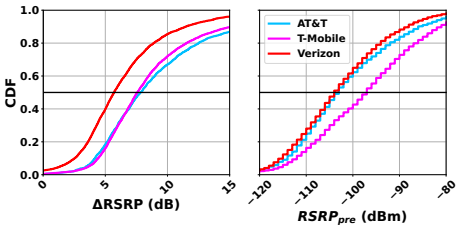
This concentration carries two implications: (1) Most of the configuration diversity is operationally irrelevant: although operators provision a large parameter space, real HO traffic flows through a tiny dominant subset, so tuning effort can focus on the few combinations that actually drive HOs; (2) It makes our analysis in § 5.5 principled rather than selective; by studying performance through these dominant configurations, we cover the configurations that matter in practice.

5.5 Impact on Performance

HO configurations are defined entirely in terms of RSRP thresholds and offsets — the levers operators control are signal strength conditions. The central question of this section is therefore whether the configurations operators have chosen actually achieve their intended goal: improving signal strength post-HO, limiting signal degradation pre-HO, and avoiding wasteful ping-pong HOs. We analyze the four major HO types in turn: A3 and A5 for intra-RAT 5G→5G HOs, B1 for

AT&T	T-Mobile	Verizon
(8, 640): 89%	(8, 640): 53%	(6, 256): 72%
(4, 320): 7%	(10, 640): 33%	(8, 256): 14%
Count: 2	(10, 320): 10%	(8, 480): 7%
	Count: 3	Count:3

(a) Dominant combinations $(\Delta_{A_3} + H_{A_3}, TTT)$.



(b) Performance.

Figure 5.11: Dominant parameter combinations and RSRP analysis for the A3 event. LTE→5G promotions, and A2 for 5G→LTE demotions.

§5.4.3 showed that, despite large provisioned configuration spaces, actual HO traffic concentrates in a tiny dominant subset. This has a direct consequence for performance analysis: the configurations that matter for HO outcomes are not necessarily the same as those operators deploy most broadly across their network. Our performance analysis therefore proceeds in two steps. First, we consider only configurations that actually triggered HOs. Second, among those HO-triggering configurations, we focus on the dominant ones that together account for at least 90% of the HOs for that event. These dominant configurations are the subject of analysis; they may differ from the most frequently provisioned configurations, a distinction that matters, as we show for A5 below.

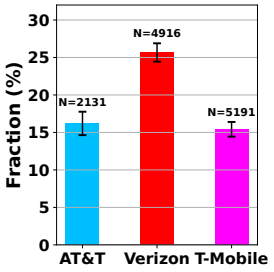
5.5.1 Impact on signal strength and ping pong HOs

We use three metrics to evaluate configuration effectiveness: $\Delta RSRP = RSRP_{post} - RSRP_{pre}$ quantifies post-HO signal improvement; $RSRP_{pre}$ quantifies how much signal quality has degraded before the HO executes; ping-pong rate (the fraction of HOs where the UE returns to the previous cell within a short amount of time (15 s in our analysis)) quantifies wasteful HO decisions. Together, these metrics capture the fundamental three-way tradeoff operators face: a larger offset or threshold produces better post-HO signal at the cost of worse pre-HO signal, while a longer TTT reduces ping-pong risk but delays the HO decision and worsens pre-HO signal. Optimizing all three simultaneously is extremely difficult, as we show below.

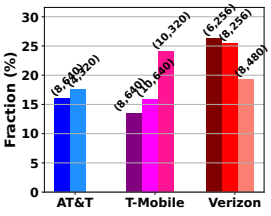
One important subtlety is worth noting before proceeding: unlike $\Delta RSRP$, which is a relative metric, $RSRP_{pre}$ is an absolute metric that depends not only on HO configuration choices, but also on the operator's overall deployment density and transmit power. An operator with a denser deployment may achieve higher $RSRP_{pre}$ than a competitor using a more aggressive configuration simply because its cells are closer together. This coverage confound means that counter-intuitive $RSRP_{pre}$ results across operators should not be interpreted as configuration failures without first checking whether deployment density explains them.

5.5.1.1 A3 HOs

An A3 event is triggered when the RSRP of a neighboring cell remains $\Delta_{A_3} + H_{A_3}$ dB higher than the RSRP of the serving cell for TTT ms. Fig. 5.11a shows the dominant configurations — tuples of $((\Delta_{A_3} + H_{A_3}, TTT))$ — for all three operators. AT&T has 2 dominant configurations responsible for 96% of its HOs, while T-Mobile and Verizon each have 3, responsible for 96% and 93%, respectively. The configurations differ across operators.



(a) Across operators.



(b) Across A3 configurations.

Figure 5.12: Fraction of A3 ping pong HOs.

RSRP analysis. Regardless of the A3 parameter choices, 98% or more of A3 HOs result in a positive $\Delta RSRP$ (Fig. 5.11b). This is a direct consequence of the A3 trigger condition, which requires the neighbor's RSRP to exceed the serving cell's by an offset before a HO is triggered. However, the magnitude of improvement and the degree of pre-HO degradation vary significantly across configurations.

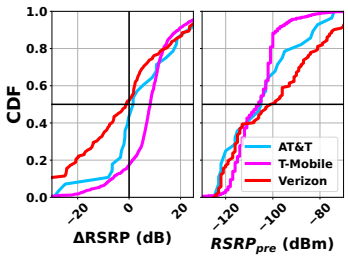
AT&T primarily uses one configuration (8, 640), while T-Mobile uses the same configuration 53% of the time and another configuration with higher $\Delta_{A_3} + H_{A_3}$ (10, 640) 33% of the time. In contrast, Verizon uses two configurations with lower TTT and/or offset 86% of the time ((6, 256) and (8, 256)). As a result, AT&T and T-Mobile achieve similar $\Delta RSRP$, outperforming Verizon by 2 dB at the median. The $RSRP_{pre}$ results, however, are counter-intuitive at first glance: T-Mobile has much higher $RSRP_{pre}$ than AT&T despite using the same primary configuration, and Verizon has the lowest $RSRP_{pre}$ despite using the most aggressive configurations, which should in principle trigger HOs earlier and thus, at higher signal levels.

The cross-operator comparison is explained by coverage differences rather than configuration choices. T-Mobile has the highest 5G RSRP and Verizon the lowest among the three operators (Fig. 5.7), which directly accounts for T-Mobile's higher $RSRP_{pre}$ compared to AT&T in spite of using the same configuration, and Verizon's lower $RSRP_{pre}$ in spite of its more aggressive configurations.

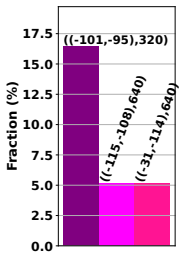
Ping pong analysis. Verizon experiences the highest A3 ping-pong rate (25%) compared to 15–16% for AT&T and T-Mobile (Fig. 5.12a), directly attributable to its low TTT values of 256 ms vs. 640 ms for the other two operators. Within each operator, longer TTT consistently produces lower ping-pong rates for T-Mobile and Verizon (Fig. 5.12b). The exception is AT&T, where the very

AT&T	T-Mobile	Verizon
((-31, -102), 640): 25%	((-101, -95), 320): 32%	((-31, -103), 640): 33%
((-104, -100), 320) : 16%	((-115, -108), 640): 13%	((-42, -118), 640): 10%
((-82, -110), 640): 16%	((-31, -114), 640): 9%	((-122, -111), 128): 8%
...	...	...
((-122, -118), 160): 3%	((-108, -108), 320): 2%	((-115, -111), 640): 1%
Count: 8	Count: 14	Count: 18

(a) Dominant combinations $((\Theta^S_{A_5} - H_{A_5}, \Theta^N_{A_5} + H_{A_5}), TTT)$.



(b) Performance.



(c) Ping pong HOs.

Figure 5.13: Dominant parameter combinations and performance analysis for the A5 event.

different configurations (8, 640) and (4, 320) produce similar ping-pong rates, suggesting that local deployment density can override the TTT effect when cell boundaries are sparse and unstable. TTT is thus the dominant, but not universal lever for ping-pong control — a finding we confirm across A5 and B1 below.

5.5.1.2 A5 HOs

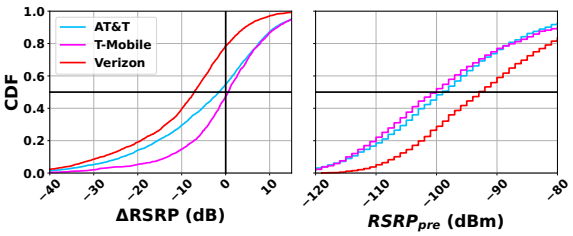
An A5 event requires two conditions to hold simultaneously for TTT ms: the serving cell's RSRP drops below $\Theta^S_{A_5} - H_{A_5}$ and a neighboring cell's RSRP rises above $\Theta^N_{A_5} + H_{A_5}$. The use of two independent thresholds rather than a single relative offset gives A5 finer-grained control than A3; operators can independently set how bad the serving cell must be and how good the neighbor must be before a HO is triggered. As we show below, this advantage is real, but requires careful configuration to realize. Fig. 5.13a shows that there are 8/14/18 dominant configurations for AT&T/T-Mobile/Verizon — substantially more than for A3 — with no single configuration exceeding a 33% share. The configurations differ again across operators.

A critical misconfiguration. The primary A5 configurations for both AT&T and Verizon are functionally equivalent to A4 events: the serving cell threshold $\Theta^S_{A_5} - H_{A_5}$ is set to -31 or -42 dBm, a level that is almost always satisfied in practice, meaning the HO is triggered whenever a neighbor satisfies the second condition regardless of the serving cell's signal quality. AT&T uses this A4-equivalent configuration 25% of the time and Verizon 43% of the time, compared to only 9% for T-Mobile. This is a misconfiguration: it negates the key advantage of A5 over A3, which is the ability to condition the HO on actual serving cell degradation rather than just neighbor strength.

RSRP analysis. The performance consequences of the A4-equivalent misconfiguration are clearly visible in Fig. 5.13a. Both AT&T and Verizon have negative $\Delta RSRP$ approximately 50% of the time, with Verizon achieving the worst outcomes. However, there is a compensating effect on

AT&T	T-Mobile	Verizon
(-112, 80): 48%	(-112, 80): 33%	(-114, 256): 37%
(-111, 40): 13%	(-115, 80): 24%	(-116, 256): 22%
(-121, 80): 12%	(-109, 256): 21%	(-105, 256): 9%
...	...	...
(-117, 80): 4%	(-103, 640): 5%	(-111, 1280): 2%
Count: 6	Count: 5	Count: 10

(a) Dominant combinations ($\Theta_{B_1} + H_{B_1}, TTT$)



(b) RSRP analysis for B1 event.

Figure 5.14: Dominant parameter combinations and performance analysis for B1 event.

$RSRP_{pre}$: by not waiting for the serving cell to degrade, these configurations ensure the HO is often completed while signal quality is still relatively high. Verizon achieves the highest $RSRP_{pre}$ among the three operators 60% of the time, and AT&T outperforms T-Mobile on this metric 35% of the time. This is the characteristic tradeoff of A4-equivalent A5 configurations; they sacrifice post-HO signal quality in exchange for executing the HO while the serving cell is still strong.

T-Mobile, which uses properly configured A5 events for the vast majority of the time, achieves the highest $\Delta RSRP$ (positive 80% of the time). It also achieves the highest $RSRP_{pre}$ 40% of the time, primarily due to the aggressive TTT in its primary configuration (320 ms vs. 640 ms for the other operators), but lags behind Verizon 60% of the time and behind AT&T 35% of the time.

Ping-pong analysis. Fig. 5.13c plots the fraction of A5 ping pong HOs for the top-3 configurations of T-Mobile. Consistent with the A3 findings, TTT is the dominant lever: (-101, -95), 320) produces a 16.3% ping-pong rate compared to 5% for configurations with TTT = 640 ms (Fig. 5.13c), despite having a higher neighbor threshold that should in principle ensure stronger post-HO signal. The higher threshold is insufficient to compensate for the low measurement confidence of a short TTT.

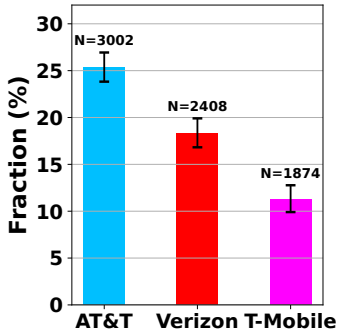
5.5.1.3 B1 HOs

B1 events are used to promote UEs from LTE to 5G. A report is sent when the RSRP of an inter-RAT 5G neighboring cell exceeds $\Theta_{B_1} + H_{B_1}$ for TTT ms. Notably, B1 does not condition on the serving LTE cell's signal quality; promotion to 5G is desired as long as the 5G cell is sufficiently strong, regardless of how good the LTE connection currently is.

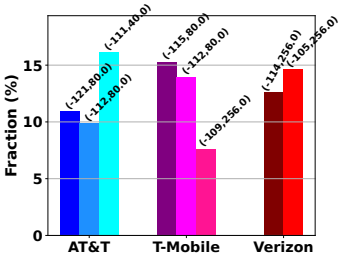
Fig. 5.14a shows the dominant configurations. All three operators use similar $\Theta_{B_1} + H_{B_1}$ values but very different TTT values, reflecting different promotion strategies. AT&T uses the shortest TTT (40–80 ms), aggressively promoting UEs to 5G at the first opportunity. T-Mobile uses a mix of short and long TTT values (80, 256, and 640 ms). Verizon uses a deliberately band-

differentiated strategy: very short TTT (128 ms) for mmWave, moderate TTT (160–512 ms) for 5G mid, and long TTT (1024–1280 ms) for 5G low. This reflects a rational policy — mmWave cells have limited coverage and must be seized quickly when available, while 5G low competes directly with Verizon's strong LTE signal, justifying conservatism.

RSRP analysis. Since B1 ignores the serving cell, RSRP improvement is not guaranteed. About 50% of B1-triggered HOs for AT&T and T-Mobile and 80% for Verizon result in a signal strength reduction (Fig. 5.14b). Verizon's high rate of signal-reducing B1 HOs is explained by its network characteristics: it has the highest LTE RSRP but the lowest 5G RSRP among the three operators (Fig. 5.7), meaning promotion to 5G almost always means accepting a weaker signal. Its long TTT for 5G low reflects awareness of this tradeoff. T-Mobile has the opposite profile — highest 5G RSRP, lowest LTE RSRP — and achieves the best $\Delta RSRP$ despite using more aggressive TTT values than Verizon, again illustrating that coverage context, not configuration alone, determines outcomes.



(a) Across operators.



(b) Across B1 configurations.

Figure 5.15: Fraction of LTE-5G ping pong HOs.

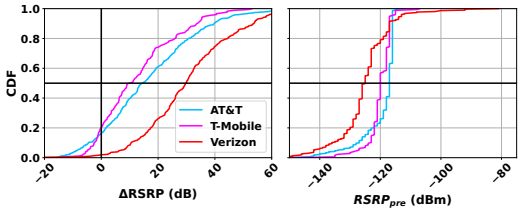
Ping pong analysis. We found the Verizon configuration (-116, 256) did not appear at all in 2024. Since the 2022 dataset contained a much smaller number of LTE-5G HOs and the 2023 dataset was collected only on the East Coast, our analysis is based on the 2024 dataset only for a fair comparison across configurations. T-Mobile has the lowest B1 ping-pong rate (10%), reflecting its strong 5G coverage (Fig. 5.15a). AT&T's aggressive short TTT produces the highest rate (18%). Verizon, despite using longer TTT than T-Mobile, experiences nearly as many ping-pong HOs as AT&T, because its weak 5G signal means that even carefully timed promotions frequently fail to hold. Within operators, longer TTT consistently reduces ping-pong rates for both AT&T and T-Mobile (Fig. 5.15b). Between configurations with the same TTT, a higher $\Theta_{B_1} + H_{B_1}$ reduces ping-pong only for AT&T, confirming again that TTT is the primary lever and threshold effects are secondary.

5.5.1.4 A2 HOs

A2 events are the primary mechanism for 5G NSA→LTE demotions. A report is sent when the serving 5G cell's RSRP remains below $\Theta_{A_2} - H_{A_2}$ for TTT ms. Since A2 does not consider

	AT&T	T-Mobile	Verizon
5G → LTE	(-117, 160): 44% (-115, 160): 36% (-125, 160): 8% (-116, 320): 4% Count: 4	(-118, 640): 16% (-121, 160): 16% (-118, 160): 10% . . (-82, 160): 3% Count: 12	(-120, 128): 21% (-124, 128): 17% (-126, 40): 15% . . (-96, 128): 1% Count: 15

(a) Dominant combinations ($\Theta_{A_2} - H_{A_2}, TTT$)



(b) RSRP analysis for A2 event.

Figure 5.16: Dominant parameter combinations and performance analysis for A2 event.

neighboring cell signal quality, RSRP improvement after the demotion is not guaranteed, but in practice all three operators use conservative enough thresholds (Fig. 5.16a) ensuring that $\Delta RSRP$ is positive 75–95% of the time (Fig. 5.16b).

Fig. 5.16a shows the dominant configurations. All three operators use similar, mostly conservative $\Theta_{A_2} - H_{A_2}$ values but differ substantially in TTT. AT&T uses TTT = 160 ms for 88% of its demotions — a standardized, moderately aggressive approach. T-Mobile distributes TTT broadly across 160–640 ms (86% of the time), with a small fraction using 40 ms (4%) — the most conservative overall, consistent with its goal of keeping UEs on 5G as long as possible given its strong 5G signal. Verizon uses the shortest TTT values: 128 ms alone accounts for 69% of demotions and TTT $\leq$ 160 ms covers 88% — the most aggressive strategy among the three operators.

These strategies make clear sense given each operator's network characteristics. Verizon has the lowest 5G RSRP and the strongest LTE RSRP, so it does not want to keep UEs on 5G when signal degrades, hence the short TTT and quick demotion. Fig. 5.16b shows indeed that Verizon has the lowest $RSRP_{pre}$ under 5G but switching to LTE results in the highest $\Delta RSRP$. T-Mobile has the opposite profile (highest 5G RSRP, lowest LTE RSRP) so it holds on to 5G connections as long as possible, accepting lower $\Delta RSRP$ as the cost of maximizing 5G time. AT&T sits between the two. Among the four event types analyzed in this section, A2 configurations are the most clearly well-matched to each operator's deployment context — a useful contrast to the misconfigured A5 events and aggressive B1 TTTs identified above. We omit ping-pong analysis for A2 since rapid return to 5G after a demotion is governed by B1 configurations, analyzed in §5.5.1.3.

5.5.1.5 Summary

Selecting optimal HO configurations is fundamentally challenging because the three performance metrics (post-HO signal improvement, pre-HO signal degradation, and ping-pong rate) cannot be simultaneously optimized, and two of them are substantially influenced by deployment

density in ways that are invisible when tuning parameters in isolation. Two findings emerge from the four event analyses. First, TTT is consistently the most reliable lever for controlling ping-pong rates across all event types, though coverage density can override this relationship in sparse deployments. Second, the coverage confound is pervasive: counter-intuitive performance outcomes across operators and across configurations within the same operator are frequently explained by deployment density differences rather than configuration quality, meaning operators cannot evaluate their configurations from aggregate performance metrics alone.

5.5.2 RSRP vs. throughput improvement

While the HO framework is defined entirely in terms of RSRP, a natural question is whether the signal strength improvements we analyzed in §5.5.1 translate to throughput improvements for users. We briefly examine this relationship to characterize the limits of RSRP-based optimization. We calculate the average throughput 1 s before (T_{pre}) and 1 s after (T_{post}) a HO and then define $\Delta T = T_{post} - T_{pre}$.

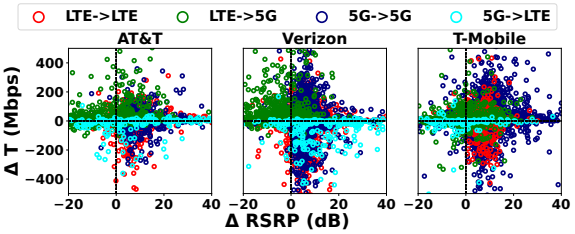


Figure 5.17: RSRP and throughput correlation.

Fig. 5.17 shows scatterplots of ΔT against $\Delta RSRP$ for all three operators and HO types. There is no meaningful correlation between signal strength improvement and throughput improvement across any operator or HO type. This confirms that RSRP is an appropriate lens for evaluating whether configurations achieve the signal-based objective of the existing HO framework, but it also highlights that RSRP alone is an inadequate proxy for user-perceived performance, since throughput is determined by factors including bandwidth, scheduling, and congestion that are largely outside the HO framework's control.

Three additional observations are worth noting. First, for intra-RAT LTE→LTE and 5G→5G HOs (mostly triggered by A3 events), the majority improve RSRP but ΔT is negative for 40–50% of cases across all operators, indicating that signal-improving HOs frequently harm user experience regardless of how well the RSRP outcome is optimized. Second, LTE→5G HOs result in signal strength reduction 45–75% of the time across operators, yet ΔT is negative for only 20–30%; aggressive 5G promotion often benefits users even when it worsens signal strength, due to the much higher bandwidth of 5G mid and high bands. Third, and most strikingly, Verizon's

5G→LTE demotions improve RSRP 85% of the time yet result in throughput degradation 80% of the time, with losses exceeding 150 Mbps in 18% of cases. AT&T and T-Mobile achieve a substantially better balance between RSRP and throughput for the same HO type. This suggests that Verizon's aggressive A2 demotion strategy, while rational from a signal strength perspective, comes at a significant throughput cost that RSRP-based optimization cannot capture.

5.6 Implications for Operators

For operators. The most immediate opportunity is reducing wasteful signaling from A1 and A6, which together dominate reported events while almost never leading to HOs. Operators already eliminated periodic events from 5G for exactly this reason; the same logic applies to A1 and A6. For A1, conservative configurations alone cannot prevent excessive triggering in networks with strong 5G coverage; operators should reconsider whether the current event triggering policy is necessary at current 5G deployment densities.

Operators should also audit their A5 configurations. Verizon and AT&T configure A5 in a way that makes it functionally equivalent to a simpler event in 43% and 25% of HO-triggering configurations respectively, negating its key design advantage. The performance consequences are clear: misconfigured A5 events produce negative post-HO RSRP approximately half the time, while properly configured ones achieve improvement the vast majority of the time. T-Mobile's HO-triggering A5 configurations provide a concrete reference point.

TTT is the most reliable lever for ping-pong control, but cannot be evaluated from aggregate metrics alone; deployment density confounds the relationship and must be controlled for. Finally, operators relying solely on RSRP to evaluate HO performance will miss throughput consequences: Verizon's 5G→LTE demotion strategy improves RSRP 85% of the time yet degrades throughput 80% of the time — a cost invisible to RSRP-based optimization.

For the research community. The coverage confound we document, where deployment density explains counter-intuitive performance outcomes across operators and configurations, poses a fundamental challenge for automated HO tuning systems (e.g., [118]). Ground-truth characterization of this confound, as we provide here, is a necessary input for any tuning system to define meaningful optimization targets.

The weak RSRP-throughput correlation suggests the 3GPP HO framework is optimizing the wrong objective from the user's perspective, particularly for inter-RAT HOs where bandwidth differences dominate. Designing HO policies that account for technology-specific characteristics is

an open and practically important problem. The operational concentration finding further suggests that data-driven optimization need not search the full parameter space; the dominant configurations we identify provide a compact, empirically grounded starting point.

Finally, while quantifying the precise cost of signaling overhead in terms of UE energy and network resources is beyond our scope, the rates we observe represent continuous unnecessary radio activity whose energy cost is well-established [119, 3]. The fact that operators have already eliminated periodic events from 5G to reduce overhead confirms the cost is real; our findings show A1 and A6 warrant the same treatment.

5.7 Related Work

HOs in the pre-5G era. Prior work studied LTE HOs in operational networks [120, 121, 122, 54, 53, 9, 123, 21, 124, 52], examining their performance impact [21, 9, 121, 122, 54, 53], mitigation solutions [21, 9, 54], and instability or unreachability issues [124, 123, 120]. Closest to ours is [52], the first large-scale characterization of LTE HO configurations; we provide the first such study in the 5G era.

HOs in the 5G era. Many experimental studies have evaluated 5G HOs [7, 1, 3, 91, 45, 114, 44, 90, 125, 126, 127, 128, 129, 105, 130, 131], most at small scale across a few km or 1–3 cities [7, 1, 45, 114, 44, 90, 125, 126, 129, 131]. The only large-scale 5G HO studies [91, 3, 127, 105, 130] characterize what happens during HOs: frequency, duration, and impact on transport and application performance from the user's perspective [91, 3, 105, 130], or geo-temporal dynamics and failure causes over a 4-week dataset from an operator's perspective [127] — but none examine the configuration parameters that cause HOs. We are the first to study these configuration policies at scale, and our 27-month longitudinal dataset lets us track their evolution over time.

Dependent misconfigurations and configuration diversity. Zhang et al. [126] identify dependent misconfigurations in 5G/4.5G RRC, where certain parameter combinations violate inter-dependency constraints in the 3GPP specification. We are orthogonal: rather than checking specification compliance, we ask whether operators' freely chosen values achieve good HO outcomes in practice — often they do not, most notably the A4-equivalent A5 misconfiguration deployed by AT&T and Verizon. A separate line of work [132, 133, 134, 118] proposes data-driven and learning-based approaches for automated cellular configuration selection; most relevant is Iridescence [118], which tunes 5G NSA configurations in the presence of deployment confounders — precisely the coverage confound we identify as a systematic obstacle to evaluating HO configurations. These operator-side tools address

how to tune configurations automatically across the full configuration space; we address the prior question of *what* is misconfigured and *why*, providing the ground-truth characterization such tuning systems need to define their optimization targets. The contributions are complementary.

5.8 Summary

We conducted the first large-scale, longitudinal study of HO configurations in the 5G era, collecting a 27-month dataset of 48K HOs and 239K reporting events across all three major US operators. Our analysis reveals that current configurations are often overly aggressive and inefficient, with overhead concentrated in a few event types operators can and should revisit; despite large provisioned configuration spaces, actual HO behavior concentrates in a tiny dominant subset, tighter in 5G than LTE. A pervasive coverage confound means counter-intuitive performance outcomes are frequently explained by deployment density rather than configuration quality. Finally, TTT is the most reliable lever for ping-pong control, a critical A5 misconfiguration deployed by two operators negates a key design advantage, and Verizon's aggressive 5G→LTE demotion incurs a significant throughput cost invisible to RSRP-based optimization. We expect our findings and released dataset to help operators reduce overhead and guide the research community toward more effective, adaptive HO policies for next-generation networks.

This chapter concludes the second part of this thesis, which investigated operator-specific configuration policies governing 5G mobility management. We now turn to the third and final part, where we translate our measurement-driven insights into a system-level solution. Specifically, the next chapter addresses the related problem of static, downlink-heavy resource allocation that contributes to the persistent DL/UL throughput asymmetry observed throughout this thesis (§2, §3, §4).

Chapter 6

Enabling Emerging Applications in 5G Through UE-Assisted Proactive PHY Frame Configuration

As previously discussed, emerging 5G "killer" applications – from extended reality to connected autonomous vehicles – often offload compute-intensive tasks such as object detection, recognition, and tracking to edge servers, demanding high uplink throughput and imposing stringent latency and reliability requirements. To ensure adherence to application QoS thresholds of these applications, 3GPP provides the flexibility of selecting a PHY frame configuration, comprising a specific number of downlink (DL) and uplink (UL) slots in the time domain, from a range of available options [135], [136], to aid time-domain resource allocation. This process is managed by the Network (NW), encompassing both the base station (gNB) and the core network (CN).

• **Bridging the gap in 5G:** Allocation of the DL/UL slots in a PHY frame among available UEs is achieved through time division duplexing (TDD). Here, the NW relies on UE reports of estimated wireless channel conditions, pending application traffic for the UEs, and their QoS requirements to determine the allocation of UEs to specific DL or UL slots in a PHY frame, along with UE transmission configurations like MCS [137]. This *reactive* approach may exhibit sluggish convergence to an optimal PHY frame configuration, along with optimal UE allocation to DL/UL slots, and MCS selection in dynamic environments characterized by high variance in channel metrics such as SINR, RSRP, RSSI. This potentially increases the likelihood of failing to meet the QoS thresholds for emerging applications. Commercial 5G networks get around this problem by having a *fixed* PHY

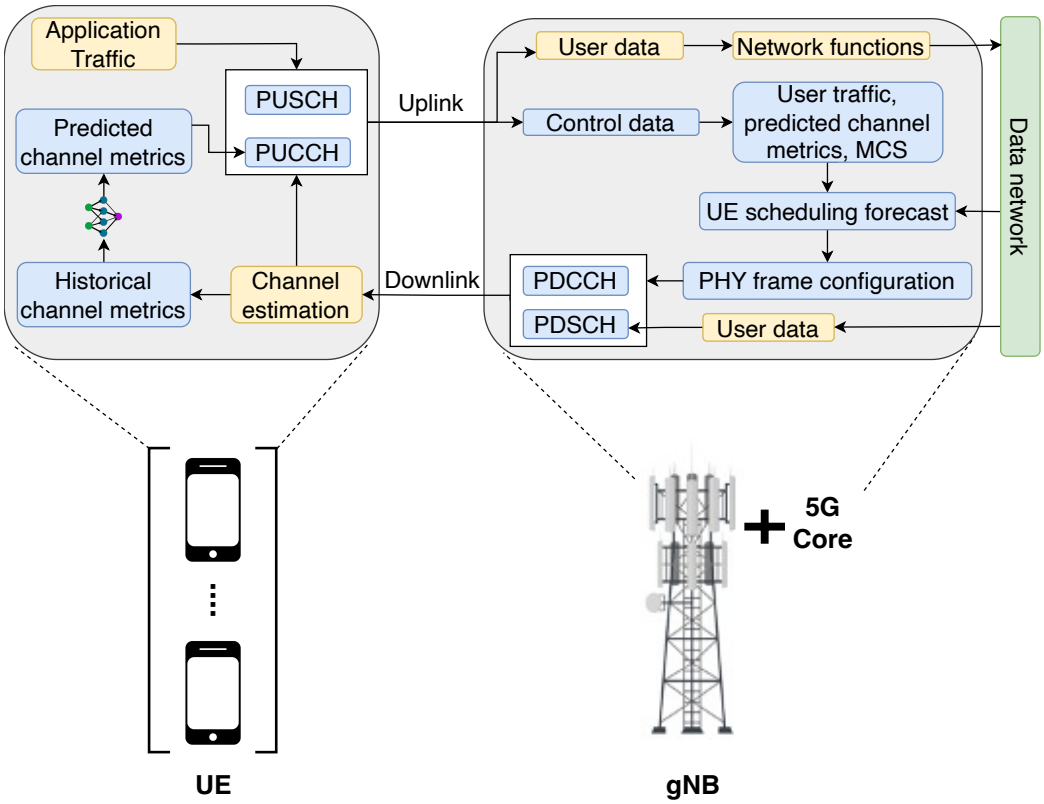


Figure 6.1: Proposed modules (in blue) integrated with 3GPP compliant 5G network modules (in yellow) for proactive UE time domain resource allocation.

frame configuration with more DL slots than UL [95], giving more priority to existing DL-heavy applications like video streaming, and resulting in a noticeable performance gap between the DL and UL in terms of throughput, as we showed earlier in this thesis (§2). While various factors contribute to this asymmetry, the reactive scheduling by the NW, particularly the lack of updates to DL and UL slot configurations in the PHY frame based on user demands and channel conditions, plays a significant role.

• **Proposed framework description:** We explore the possibility of addressing the aforementioned gaps in 5G by investigating *proactive* PHY frame reconfiguration. Here, the number of DL and UL slots in a PHY frame is decided based on the real-time application traffic demand and *predictions* of the UE channel metrics. We propose an adaptive and autonomous UE-based machine learning (ML)-enabled solution, that allows the UEs to predict their local channel metrics for future time instances, and report them to the NW. This *lookahead* from all the available UEs can enable the NW to proactively configure the DL and UL slots for future PHY frames and pre-allocate UEs to these slots to meet the diverse application requirements in dynamic channel conditions. While the NW

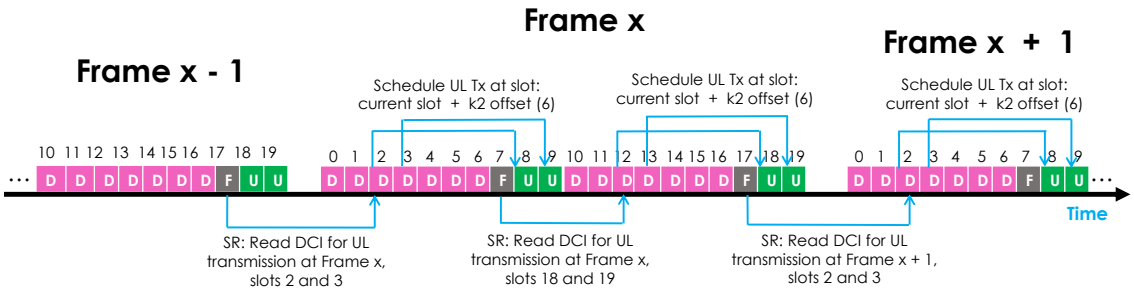


Figure 6.2: Uplink transmission process in legacy 5G PHY TDD frame configuration with 30 KHz sub-carrier spacing

can store channel metric data from all served UEs, executing ML algorithms on each UE's reported channel data for real-time forecasting is not scalable from the NW side. A more practical approach is for individual UEs to forecast their channel metrics and transmit them to the NW, facilitating proactive resource optimization.

Our solution is designed to operate with the existing 3GPP standard with added enhancements shown in Fig. 6.1: (i) Deployed UEs with trained machine learning (ML) modules for local channel metric prediction, that are reported to the NW along with the application traffic demand, and (ii) the optimization suite in the NW, which receives this forwarded information from the UEs and proactively reconfigures the PHY frame with an optimal number of DL and UL slots. This helps to allocate UEs to the best possible slots in the PHY frame and maintain the QoS requirement.

Our main contributions are as follows:

- We leverage 3GPP-compliant Matlab 5G toolbox [138], to characterize how legacy 5G allocates UEs in the PHY frame in the time domain based on UE reported channel metrics, and motivate how this approach can be a bottleneck for emerging UL-heavy applications.

- We introduce a proactive PHY frame configuration framework enabling dynamic allotment of DL and UL slots in a PHY frame through *look-ahead forecasts* of UE channel metrics and application demand, enhancing network QoS compared to reactive resource allocation. Crucially, our framework is straightforward and aligns with the 3GPP standard.

- We deploy our proposed method on the 3GPP-compliant OAI 5G testbed [139] to showcase its practicality and adherence to 3GPP standards. By comparing our solution with the baseline reactive legacy 5G TDD, through both testbed experiments and simulations, we demonstrate an enhanced equilibrium between DL and UL performance. These findings underscore the

critical role of proactive PHY frame reconfiguration in accommodating the diverse demands of emerging applications within dynamic channel environments.

6.1 Background and Motivation

3GPP employs dynamic scheduling to allocate resources to UEs in the time domain, representing the prevalent method in commercial cellular networks [95]. Our proposed solution builds upon this approach, aiming to enhance its efficacy.

6.1.1 Legacy 5G dynamic scheduling

Commercial 5G networks such as AT&T, T-Mobile, and Verizon all configure the 5G PHY frame of 10 ms duration, with the same *fixed* configuration that has more DL than UL slots – 7 DL-2 UL-1 F (7 DL 3 UL) [140], with a periodicity of 5 ms, as shown in Fig. 6.2, leading to imbalanced traffic distribution. With this frame configuration, the UE triggers the dynamic scheduling process by transmitting a Scheduling Request (SR) to the NW in a UL or F slot, indicating pending UL data, and providing channel estimates via the physical uplink control channel (PUCCH) [141]. Subsequently, the NW informs the UEs of resource allocation using downlink control information (DCI) in the downlink control channel (PDCCH). Here, among other things, the NW specifies slot assignments for UEs in the uplink service channel (PUSCH) and downlink service channel (PDSCH) for UL transmission and DL reception, respectively.

Existing drive-test studies on these commercial networks using 5G smartphones reveal impressive DL speeds, surpassing 3.5 Gbps, while UL throughput remains consistently lower, as we showed earlier in this thesis (§2). The fixed DL-heavy PHY frame configuration may strain 5G networks, particularly with the rise of uplink-heavy applications. The use of this fixed PHY frame configuration persists due to the prevalence of existing DL-heavy applications and the 3GPP-supported *reactive* scheduling method based on UE-reported channel metrics. The lack of channel forecast knowledge provides little incentive for the NW to modify PHY frame configuration relying solely on UE-reported channel metrics as this process may take time to converge and can result in sub-optimal NW performance.

6.1.2 Preliminary Experiment and Observation

To further quantify the limitations of different fixed PHY frame configurations on UL-heavy applications, we conducted controlled experiments with the 3GPP-compliant Matlab 5G

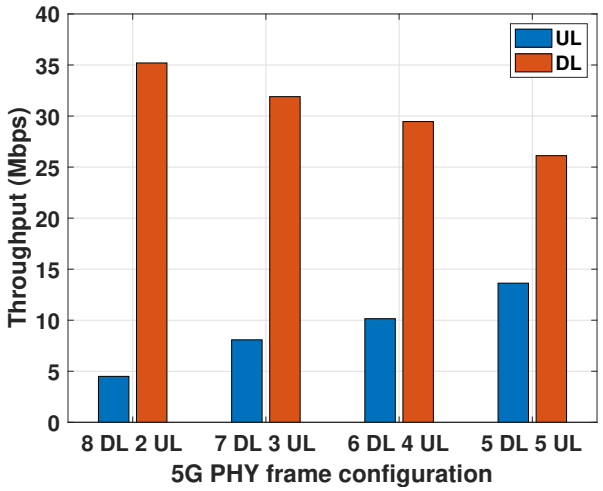


Figure 6.3: Imbalanced throughput distribution between UL and DL with different 5G PHY frame configurations

toolbox. The experiment featured a gNB and a UE connected via the 900 MHz frequency in SISO configuration, with 40 MHz BW and 30 KHz subcarrier spacing (SCS). We simulated typical UL and DL communication using on-off application traffic with no mobility, thereby ensuring stable channel conditions to specifically assess the impact of PHY frame configurations on UE throughput. Utilizing flexible (F) slots as UL slots, as per 3GPP standard, we measured throughput in UL and DL with various PHY frame configurations: 8 DL 2 UL, 7 DL 3 UL, 6 DL 4 UL, and 5 DL 5 UL.

The results from the experiments are shown in Fig. 6.3. Here, we observe UL throughput of ∼8 Mbps for the 7 DL 3 UL configuration, ∼10 Mbps for the 6 DL 4 UL configuration, ∼4.5 Mbps for the 8 DL 2 UL configuration and ∼14 Mbps for 5 DL 5 UL configuration. DL throughput ranged between 25 - 35 Mbps in these frame configurations. These results underscore a clear inverse relationship between DL-heavy PHY frame configurations and UL throughput. Note that the UL to DL throughput ratio does not exactly follow the slot allocation ratio for two reasons: First, for UL transmissions, the UE utilizes one whole F/mixed slot to send a scheduling request to notify the NW regarding incoming UL data transmission in the upcoming frame cycle. Second, in 3GPP for a given MCS, the maximum data transport block size for DL is higher compared to UL [137], leading to such asymmetry in throughput.

6.2 Proactive PHY frame configuration

We now elaborate on how our proposed solution facilitates the proactive selection of PHY frame configuration with active UE assistance. The overall system diagram for this process is depicted in Fig. 6.4. Here, the UE reports the most recent channel estimates to the NW (legacy method) and also predicts/forecasts the channel related metrics such as SINR, RSRP, RSSI, etc., through local AI/ML models, by using historical channel metric data as input and transmits it along with the application traffic status to the NW in the UL. The NW on receiving this information from multiple UEs, computes the preferred resource allocation metrics (e.g., slot/symbol, MCS, etc.), and proactively configures future frames with slots that cater to the needs of all the UEs, while minimizing interference to neighbor cells, thus optimizing the overall performance of the network. This information is communicated to the UE via legacy time domain resource allocation. We next describe how UE channel metric prediction helps in time domain resource allocation, through proactive PHY frame configuration.

6.2.1 UE Channel Metric Prediction

In UEs that support channel metric prediction, the process is as depicted in Fig. 6.5. Here, each UE keeps a record of the historical channel metric data in a local buffer. At the beginning of time $t1$, the UE uses this historical channel metric data as input for the channel metric prediction module, which forecasts channel metrics for a desired number of time slots in the future by leveraging Long Short Term Memory (LSTM) neural networks, a type of recurrent neural network (RNN) specifically designed to capture and analyze sequential data. This forecast data is then shared with the NW by each UE through the UL. The NW can use this forecast knowledge to proactively allocate network resources in the time and frequency domains to the UEs. This resource allocation information is then communicated back to the UEs through the DL channel. At time instant $t2$, to predict the next batch of channel metrics, the UEs follow a sliding window mechanism, as shown in Fig. 6.5. This allows the UEs to use the most recent historical data, for the prediction module. At the next cycle of channel metric prediction (time $t2$), this window moves forward by the number of time slots for which the channel metrics are predicted (e.g., 3 slots in Fig. 6.5). At every time slot, the UEs verify how far the predicted channel metric values fall from the actual estimated value for that time slot. This prediction accuracy is important since lower accuracy means false channel metric prediction, which can affect the UE QoS. For this reason, the UEs can revert to legacy reactive resource allocation and re-train the ML model, if the predicted value crosses a predefined accuracy threshold metric (Fig. 6.4).

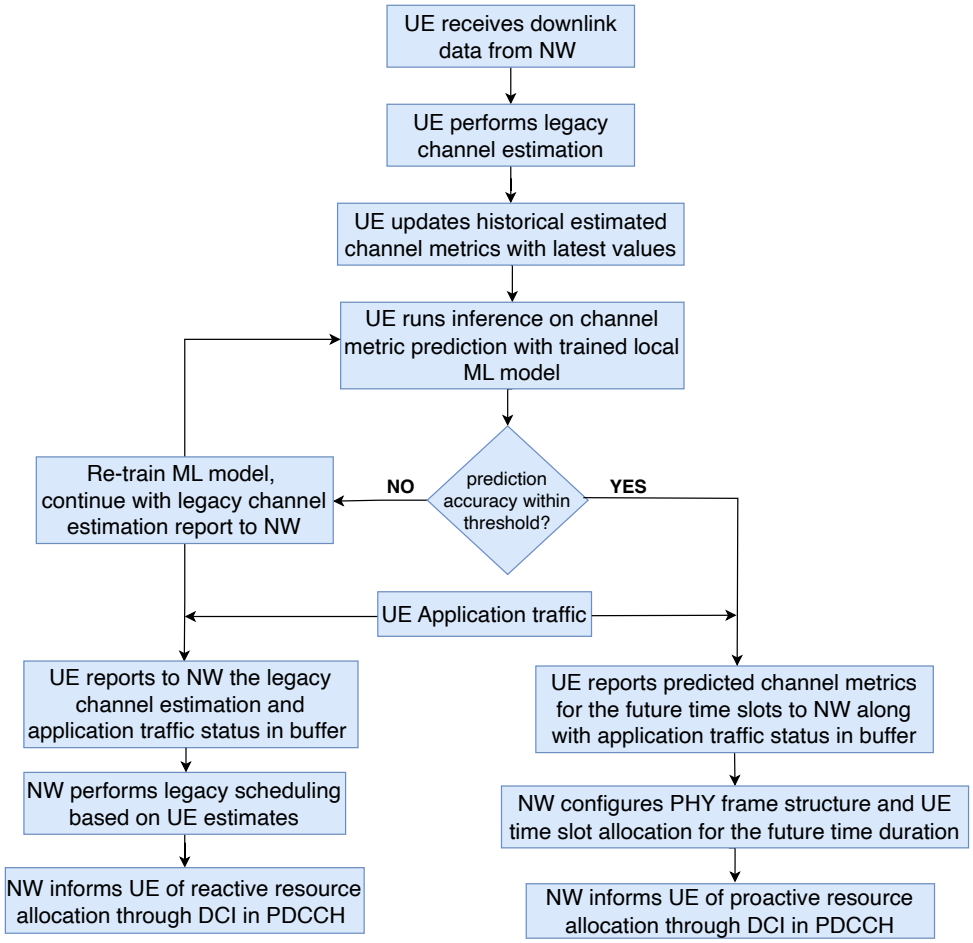


Figure 6.4: Dynamic PHY frame reconfiguration at NW aided by UE forecast of local channel metrics.

• **LSTM based prediction:** The capability of deep learning methods to identify complex, non-linear patterns in data makes them ideal for time series forecasting [142], [143]. Among the available methods, Long Short-Term Memory (LSTM) has been successfully applied in a range of different domains, including aspects of wireless network infrastructure management [144]. Despite the widespread adoption of machine learning architectures like Transformers for time series prediction tasks, their extensive parameterization, substantial memory footprint, and computational demands render them unsuitable for deployment on resource-constrained low-power devices such as UEs [145]. Given our study's focus on leveraging accurate time series predictions to guide proactive design strategies for 5G PHY frame configurations and support emerging UL-intensive applications, we select LSTM networks as an exemplary, demonstrative, and cost-effective approach.

The LSTM architecture that we use for UE channel metric prediction is depicted in Fig. 6.6. Given

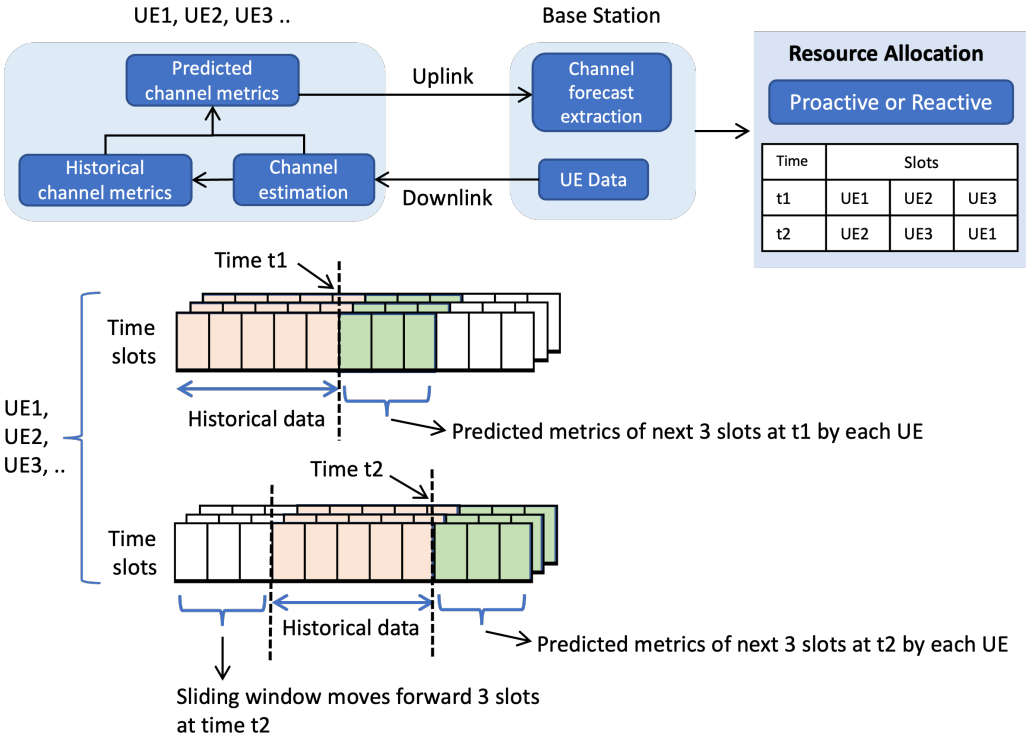


Figure 6.5: Sliding window mechanism at UE in using historical channel metrics (in orange) to predict the channel metrics of future time slots (in green).

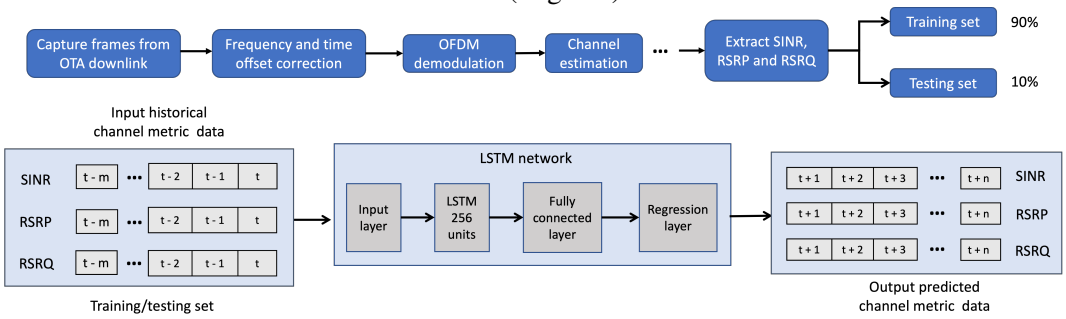


Figure 6.6: Tensor forming, training and testing processes in the proposed LSTM network for UE channel metric prediction. The SINR, RSRP and RSRQ sequences are divided into training, and testing sets, which are used for channel metric prediction.

an input of historical time series or sequence ($t-m$ to t) of channel metric data, the LSTM network forecasts the channel metric values of future time steps $t+1$ to $t+n$, with $t+n$ being the forecast time horizon. We measure the accuracy of the predictions through Root Mean Square Error (RMSE) $\mathcal{E}$, which shows how far the predictions fall from measured true values using Euclidean distance, and it is expressed as $\mathcal{E} = \sqrt{\frac{\sum_{t=1}^{\mathcal{N}} \|x_t - \hat{x}_t\|^2}{\mathcal{N}}}$ where, $\mathcal{N}$ is the number of data points, x_t is the measurement

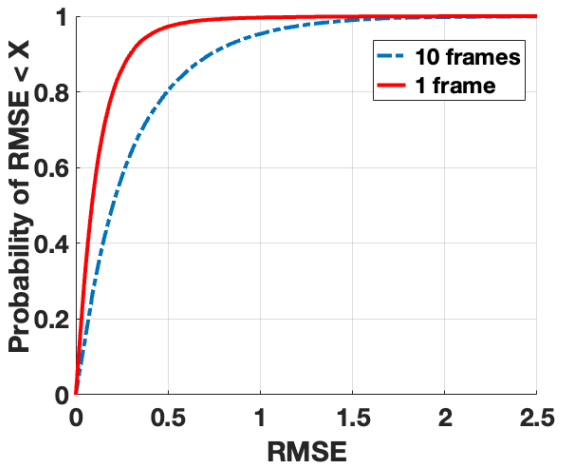
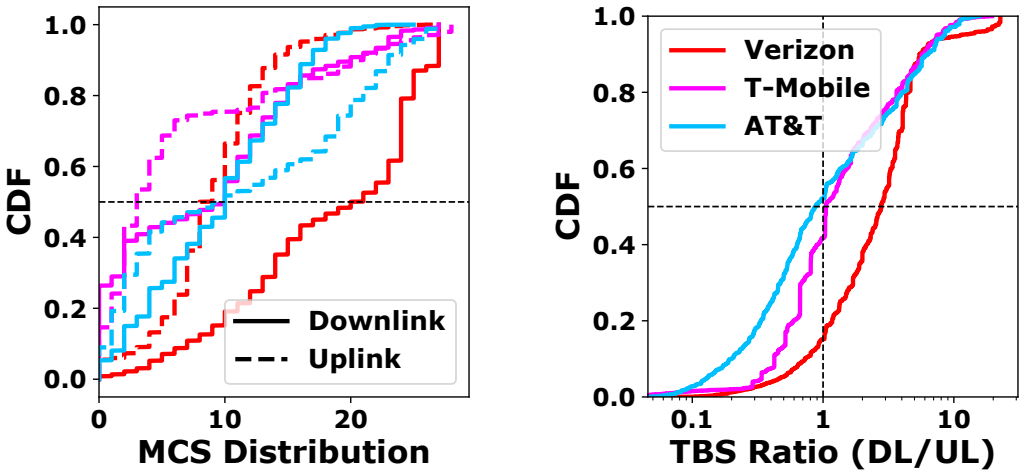


Figure 6.7: RMSE of LSTM for predicting channel metrics for duration of 1 and 10 frames with input historical channel metric data of 400 previous frames.

at time t, and $\hat{x}_t$ is its corresponding prediction. For accurate prediction, we set the RMSE threshold to 1. Fig. 6.7 shows the performance of the LSTM network in predicting the channel metrics for 1 future frame and 10 future frames with input channel metric data of 400 previous frames collected from over-the-air drive test experiments, with the channel metrics having a granularity of 10 ms.

6.2.2 Proactive Time Domain Resource Allocation

We next describe the algorithm designed to address the proactive time domain resource allocation between two UEs, aided by the UE channel metric prediction in a 5G network with stringent latency requirements. The resource allocation algorithm tries to choose a frame configuration per TDD PHY frame which maximizes the total throughput of a network. On the UE side, once the channel metric prediction module provides a series of SINR values for the next prediction window, the UE maps it to an MCS from the channel quality indicator (CQI) table and fetches the corresponding maximum achievable Transport Block Size (TBS), per slot in each frame, for the duration of the forecast time. The forecast time corresponds with the considered latency deadline (e.g., 20 ms, 100 ms). These predicted channel metrics, application buffer status, and the TBS values are then sent to the NW via the PUCCH at the beginning of every resource allocation period. The resource allocation is triggered by the arrival of application traffic at the UE buffer. The NW, upon receiving the list of TBS from the two UEs, runs the PHY frame configuration algorithm to perform the proactive UE resource allocation. Since our goal in this study is to demonstrate the benefits of proactive PHY frame configuration and not to propose an optimal scheduling algorithm, we describe a simple algorithm assuming that there are two UEs in the network (UE1 and UE2), and they are



(a) 5G MCS distribution

(b) Ratio of DL to UL TBS per slot

Figure 6.8: MCS and DL to UL TBS ratio distribution in mobility scenarios for all 3 operators.

using UL-centric and DL-centric applications respectively.

We initialize each UE's traffic demands in the current prediction window and assign the frame configurations for the number of frames that cover the prediction window, with each frame having 10 ms duration. For each frame, our algorithm calculates the sum of DL (UE2) and UL (UE1) TBS, for all possible allowed PHY frame configurations ranging from (1 DL 9 UL), (2 DL 8 UL), to (9 DL 1 UL). Note, when both the UEs' traffic demands cannot be satisfied within the latency deadline due to poor channel conditions, the algorithm never chooses an extreme frame configuration like (10 DL 0 UL) or (0 DL 10 UL), even if that frame configuration maximizes the network throughput. This ensures none of the UEs get starved until their traffic demands are satisfied. The sum TBS of the two UEs acts as a proxy to the total network throughput. The algorithm then chooses the frame configuration with the maximum sum TBS value and assigns that configuration to the current frame. The allocated TBS per UE is then deducted from the traffic demands of both the UEs respectively. The algorithm then checks if the UEs' traffic demands are satisfied in the current frame configuration cycle. If both the UEs' traffic demands are satisfied, then the remaining frames are assigned an equal distribution of DL and UL slots (5 DL 5 UL). If either of the UE's requirements gets satisfied first, then for the remainder of the frames within the latency deadline, the other UE is allocated all the slots needed to finish its task; (10 DL 0 UL) for the DL UE or (0 DL 10 UL) for the UL UE. After allocating the slots in each PHY frame for the duration of the prediction window (latency deadline), the NW sends the indication of allocated slots to the UEs via legacy DCI in PDCCH.

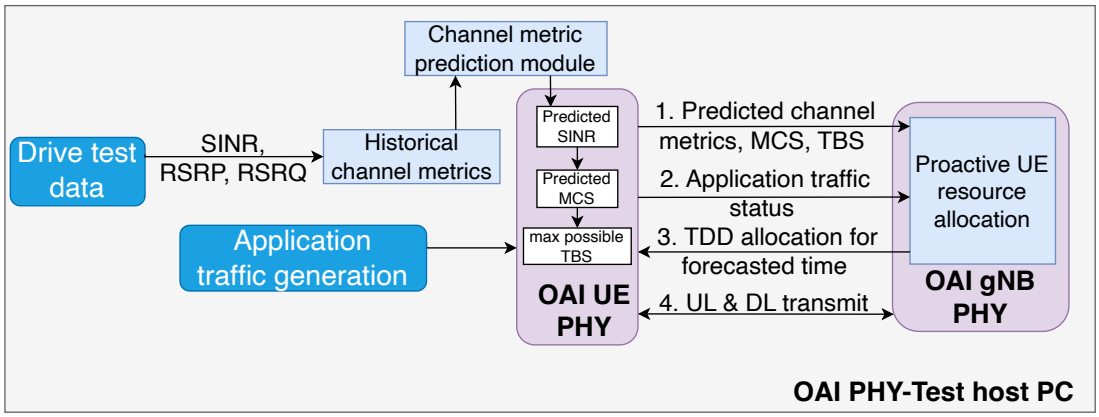
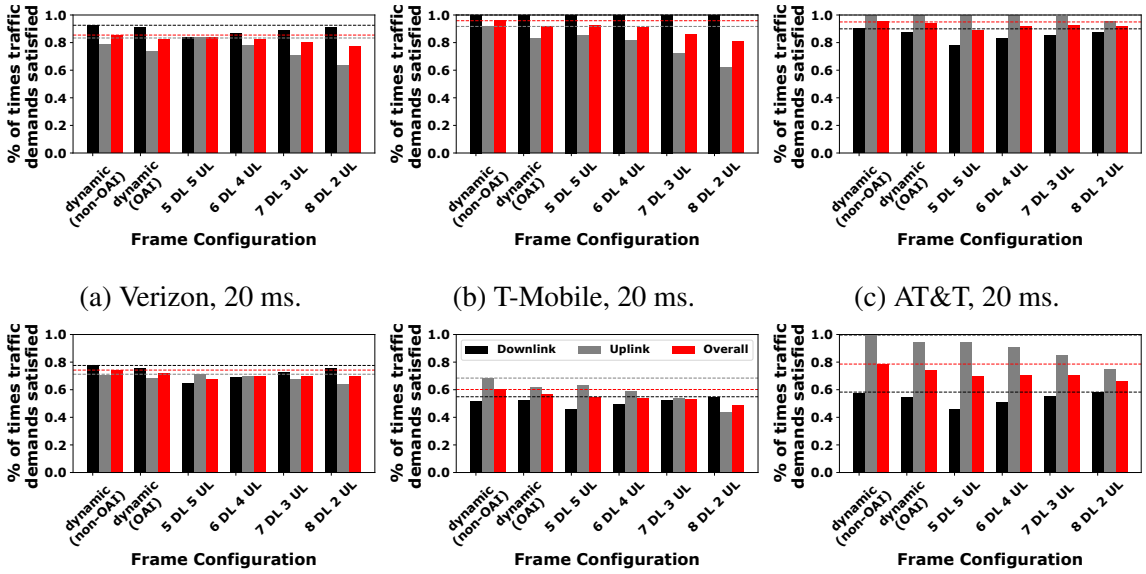


Figure 6.9: Schematic diagram of proactive PHY frame configuration in OAI by leveraging drive test data



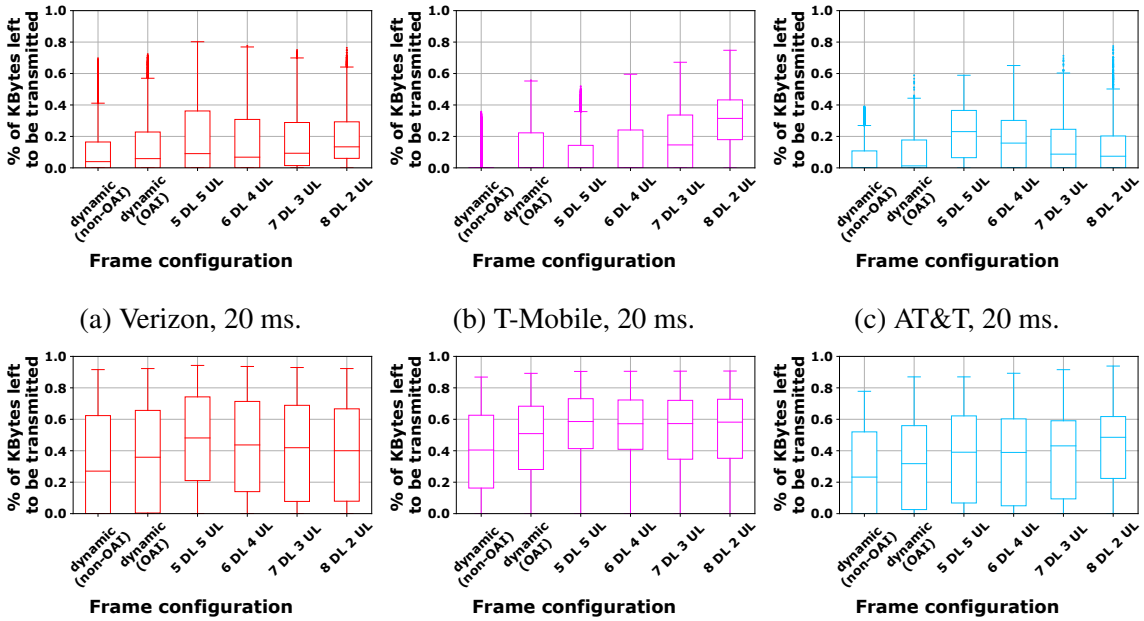
(a) Verizon, 20 ms. (b) T-Mobile, 20 ms. (c) AT&T, 20 ms.

(d) Verizon, 100 ms. (e) T-Mobile, 100 ms. (f) AT&T, 100 ms.

Figure 6.10: % times the DL and UL traffic demands were satisfied with different PHY frame configurations for 20 ms and 100 ms latency deadlines.

6.3 Performance Analysis

We explore the performance gain of our proposed solution by leveraging channel metrics collected from outdoor drive tests using 5G-enabled commercial smartphones.



(a) Verizon, 20 ms. (b) T-Mobile, 20 ms. (c) AT&T, 20 ms.

(d) Verizon, 100 ms. (e) T-Mobile, 100 ms. (f) AT&T, 100 ms.

Figure 6.11: % of total DL + UL untransmitted traffic with respect to the traffic demand in a 20 ms and 100 ms interval for the cases where the traffic demand was not satisfied within its deadline in at least one direction.

6.3.1 Experiments

The outdoor drive test dataset is a collection of cellular channel metrics and packet traces collected from Samsung S21 smartphones (UEs) during different outdoor drive test settings. Each of the UEs run the XCAL Solo tool [146], which provides low-level access to detailed cellular signal metrics and protocol information. Each of the UEs was equipped with SIM cards of one of the major US network operators (Verizon, T-Mobile, AT&T) and we conducted extensive drive test campaigns in a major city in the US. During each of these tests, the UEs continuously ran backlogged traffic using iperf. The dataset obtained here is a time series of channel metrics (SINR, CQI, RSRP, etc.) having a granularity of 10 ms. The goal of this dataset is to capture *realistic*, *mobile* settings in which a typical UE might operate as seen in Fig. 6.8a which shows how drastically the 5G MCS can fluctuate under mobility for all three operators. We observe that with the exception of AT&T, where 50% of the time the UL MCS was higher than the DL MCS and vice-versa, the other two operators had higher DL MCS dominating most of the time. We also calculated the ratio between each 10 ms DL and UL max TBS sample extracted at the same time instant and show it in Fig. 6.8b.

Unsurprisingly, for Verizon, due to its extremely DL-heavy MCS distribution (9 vs. 21 median UL and DL MCS respectively), the TBS ratio is more than 1 most of the time. For AT&T and T-Mobile we see a much more balanced distribution of the DL to UL TBS ratio.

We integrate the drive test channel metric data (SINR, MCS) to the OAI testbed, running in PHY-test mode [139], with added customizations as explained in Sec. 6.2, to enable the dynamic PHY frame configuration without requiring the assistance of the upper layers of the network stack. The schematics for this experiment are shown in Fig. 6.9, where the forecasted channel metrics are transmitted to the NW via the UE for proactive PHY frame configuration. Following the gNB indication of UE resource allocation in the DL, the UE starts the UL and DL transmission. The PHY layer logs capturing the UE and gNB interaction are used for performance evaluation.

We ran the OAI-PHY test mode using each of the traffic traces collected from the drive tests for 2 different PHY frame configurations: (a) Previously explained static configurations, which remained constant from start to end of a run: (5 DL 5 UL), (6 DL 4 UL), (7 DL 3 UL), and (8 DL 2 UL), and (b) Dynamic configuration, which lets the NW choose different frame configurations for each frame depending on the traffic requirements and predicted channel quality. For the latency thresholds, we choose the emerging applications of intelligent transportation (100 ms) and machine-type communication (20 ms) [147], with traffic arrival rate at 100 ms and 20 ms respectively. In this scenario, even though our system can dynamically configure each frame with different DL and UL slots, due to OAI constraints we are only allowed to choose among the 4 configurations mentioned in the static scenario. To fully realize the potential of dynamic PHY frame configuration with DL and UL slots, we also run offline simulations where the OAI constraints do not restrict the frame configuration algorithm and it selects the per-frame configuration (among all possible configurations) that maximizes the throughput. We define this as the *dynamic-nonOAI* scenario and the previous dynamic scenario bounded by OAI constraints as *dynamic-OAI*. In the experiments, we perform channel metric prediction and dynamic PHY frame configuration for a window of 100 ms (10 frames) and 20 ms (2 frames) emulating the behavior of emerging applications. Accordingly, based on the application type and the maximum capacity of our network, we choose the maximum traffic for each of the cases. For the application with 100 ms latency, we set the maximum traffic generated per 100 ms cycle as 62.5 KB, and for the application with 20 ms latency, we set the maximum traffic to be 9.5 KB for 20 ms.

6.3.2 Results

For each of the frame configuration types: *static*, *dynamic-OAI*, and *dynamic-nonOAI*, we evaluate the efficacy of each type using two metrics: a) % of times the traffic demand in a 20 ms or 100 ms latency deadline cycle was satisfied using a specific frame configuration type (the higher the better), and b) % of remaining bytes that could not be transmitted within the latency deadline for the cases where the traffic demand was not satisfied within its deadline in at least one direction (the lower, the better).

Fig. 6.10 shows the % of times the traffic demand was satisfied within a 20 ms and 100 ms latency deadline in each direction separately and in total for all three network operators. In general, we observe the dynamic-nonOAI system overall outperforms all other PHY frame configurations. Unlike static configurations, which tend to satisfy the traffic demand requirements in only one direction depending on whether the configuration is DL-heavy or UL-heavy, the dynamic-nonOAI system strikes a balance between both traffic directions by dynamically selecting the best configuration based on predicted traffic demands and channel conditions. As a result, its performance is (i) always better overall than the best static configuration and (ii) either better than or as good as the best static configuration in each direction separately most of the times. Even in the rare cases where a static configuration outperforms the dynamic-nonOAI system in one direction (5DL 5 UL in the UL direction for Verizon with 20 ms latency deadline in Fig. 6.10a and 8 DL 2 UL in the DL direction for T-Mobile with 100 ms deadline in Fig. 6.10e), the dynamic nonOAI system offers better overall performance without starving either direction.

We find that these rare cases stem from the channel condition being sometimes substantially poor in a particular traffic direction forcing the resource allocation algorithm to allocate more slots to the other UE with better channel condition. Recall that the resource allocation algorithm tries to maximize the network throughput, thus an asymmetric channel condition will favor the UE with better MCS and finish the task for that UE faster. However, once it notices that a particular UE's task is finished within the expected latency deadline, our algorithm allocates all the remaining resources to the other UE with poor channel condition to complete its task within the latency deadline. This is evident in Fig. 6.11, which shows that the dynamic-nonOAI is always left with the lowest % of un-transmitted bytes, both in the median and the worst case, in comparison to other frame configurations. In fact, the most symmetric static configuration, 5DL 5 UL, which often achieves comparable performance with dynamic-nonOAI in terms of meeting the latency deadline (Fig. 6.10), is left with the maximum number of untransmitted bytes in most of the cases (Fig. 6.11).

In summary, *our dynamic PHY frame configuration satisfies the overall traffic demands more times than any static configuration in every scenario (combination of operator and latency deadline) and, at the same time, it is left with the lowest percentage of un-transmitted bytes in all scenarios.*

6.4 Summary

Motivated by the persistent DL/UL throughput asymmetry we observed in commercial 5G networks earlier in this thesis (§2, §3), the final thrust of our research set out to tackle this asymmetry at its root. We uncovered limitations in the existing time domain resource allocation approach, which relies on fixed PHY frame configurations derived from UE channel estimates. While effective for prevalent DL-heavy applications, this method struggles with the dynamic and diverse traffic demands of emerging applications spanning both DL and UL. Through extensive experimentation, we demonstrated the superiority of employing proactive PHY frame configurations based on look-ahead forecasting of UE channel metrics to meet application QoS requirements. Our proposed 3GPP-compatible framework maximized network capacity in both DL and UL, surpassing the performance of legacy DL-heavy PHY frame configurations. This enhancement underscored the importance of precise channel metric prediction in enabling the coexistence of emerging UL-heavy applications with existing DL-heavy applications through efficient time domain resource allocation, leveraging knowledge of current and future channel conditions.

Together with the preceding chapters, this concludes the third and final part of this thesis, which turned our measurement-driven insights into a system-level solution. We synthesize the overall lessons and recommendations from all three parts of this thesis in the concluding chapter.

Chapter 7

Conclusion

This thesis set out to answer three questions: what is the ground reality of commercial 5G networks today, what operator-controlled policies shape the experience users actually get, and how can we begin to close the gap between 5G's promise and its ability to support next-generation, latency-critical applications. Across four large-scale, real-world measurement studies and a measurement-driven system design – spanning more than 15,000 km of cross-country driving measurements collected over a 27-month period – we distill three lessons about the state of operational 5G today and two broader lessons for measurement-based networking research, which we present below along with the concrete recommendations they motivate for operators, application developers, and researchers.

7.1 Lessons Learned

5G in the wild. Coverage and throughput improved substantially for all three major US operators between our first cross-country drive in 2022 and our replication of the same route in 2024 (§2, §3). Median downlink throughput improved by 2×–40× and uplink throughput by 2×–6×, and high-speed 5G coverage expanded well beyond dense urban cores. Yet this progress remains uneven: two of the three operators still cover less than half of our driving route with 5G, even five years after the initial rollout, and continue to rely on LTE for roughly half of the trip. Most of the observed throughput gains are attributable not to new spectrum alone, but to operators aggressively expanding carrier aggregation (CA) levels and aggregate channel bandwidth in the 5G mid band – a configuration choice rather than an inherent property of the technology.

Operational reality. 5G Standalone (SA) is expanding rapidly and increasingly displacing Non-Standalone (NSA) as the dominant architecture: SA coverage grew from 22.3% to 61.9% in just 18

months, overtaking NSA, while NSA's peak performance has largely plateaued (§4). At the same time, the policies that govern mobility management remain aggressive and inefficient: only 13–31% of handover-triggering measurement reports actually result in a handover, and this signaling overhead is concentrated in a small number of event types that are needlessly aggressive by configuration, not by necessity (§5). Finally, terrestrial and non-terrestrial networks are best understood as complementary rather than competing: Starlink offers near-uninterrupted availability and RTT competitive with (or better than) cellular, but its peak downlink and overall uplink throughput remain well below what commercial 5G already delivers (§3).

Application support. Better network-side key performance indicators (KPIs) do not always translate into better application-level quality of experience (QoE). Signal strength (RSRP) improvements are only weakly correlated with throughput (§5), while our application studies show that improvements in network-level KPIs do not necessarily translate directly into application QoE (§2, §3); operators optimizing handover policy for RSRP alone can therefore make decisions that improve their own KPI while degrading what the user actually experiences. More fundamentally, today's commercial 5G networks still rely on static, downlink-heavy resource allocation that assumes homogeneous, download-dominated traffic – a poor fit for uplink-heavy emerging applications such as AR and CAVs, which is precisely the gap our proposed proactive, UE-assisted PHY frame configuration framework is designed to close (§6).

Studying operational networks. Beyond these 5G-specific findings, the methodology in this thesis generalizes beyond any single network generation. Standard-compliance and network-reported metrics are not reliable proxies for what a network actually delivers: 5G's DL/UL throughput asymmetry, the weak RSRP-to-throughput correlation, and the gap between provisioned handover configurations and their actual operational behavior were all invisible without direct, in-the-wild measurements. Equally, no single measurement snapshot is sufficient: it is only by returning to the same routes and operators over time – two years apart for coverage and performance, 18 months apart for SA/NSA evolution, 27 months for handover configurations – that this thesis could distinguish genuine progress from stagnation, and configuration choices from inherent technical limits.

Building measurement infrastructure. The findings in this thesis were possible because of a purpose-built, cross-layer measurement testbed: professional diagnostic hardware tapping into the PHY-layer signaling and control-plane events of commercial, off-the-shelf smartphones, combined with custom application-layer test suites and cloud/edge servers, and spanning multiple carriers and technologies simultaneously. Realizing this testbed required solving a problem that is easy to underestimate: different tools and software layers log data at different sampling rates, in different

timestamp formats, and across different timezones, and fusing them into a single, temporally-aligned, cross-layer view of network behavior requires dedicated parsers and synchronization pipelines built specifically for this purpose. This infrastructure – and the discipline of designing for cross-layer synchronization from the outset – is itself a transferable contribution: it is a prerequisite for any future study that wants to move beyond single-layer, network-reported, or application-only measurements towards a genuinely holistic picture of operational network behavior.

7.2 Recommendations

Building on the lessons above, we offer targeted recommendations for the three communities most affected by them.

For operators. Our results suggest two concrete levers for improving user-perceived performance without necessarily requiring new spectrum or infrastructure. First, operators should prioritize expanding mid-band coverage and, in particular, uplink capacity, which we repeatedly find to be the binding constraint for emerging latency-critical applications. Second, operators should audit their handover configurations: reducing reporting overhead for measurement events that rarely convert into handovers, retiring configuration parameters that are provisioned but never used in practice, and shifting optimization targets from signal strength alone toward metrics that better reflect end-user QoE.

For application developers and content providers. Despite the substantial improvements in 5G performance we observe between 2022 and 2024, application-level optimizations developed in the pre-5G era – such as frame compression and on-device local tracking – remain necessary to achieve acceptable QoE, and are unlikely to become unnecessary in the near future. We therefore recommend that developers continue to design for the network conditions 5G actually delivers rather than the ones it promises, and, where possible, move compute closer to the end user via edge servers to reduce end-to-end latency for offloading-based applications.

For researchers. Operational cellular networks evolve quickly and in ways that are largely invisible without direct measurement: coverage, throughput, deployment architecture, and configuration policy all shifted meaningfully within the two-to-three-year windows we studied. We recommend that the research community continues to conduct repeated, longitudinal, and cross-sectional studies of real-world networks under mobility, rather than relying on one-off snapshots, since it is precisely this kind of longitudinal comparison that surfaced several of the findings in this thesis.

Chapter 8

Future Research Directions

This thesis has focused on characterizing *public* commercial 5G networks, from cross-country measurement campaigns (§2, §3, §4) to the configuration policies that govern mobility management (§5) and a system-level solution for static resource allocation (§6). Yet 5G itself is past its midlife (§3) and continues to evolve. Private deployments are emerging as a distinct and increasingly important complement to public networks, and standardization efforts are already looking beyond 5G toward a unified terrestrial and non-terrestrial 6G. We outline three directions that build directly on the findings and methodology of this thesis.

Continued large-scale measurement of 5G and next-generation networks. The methodology developed throughout this thesis – repeated, longitudinal, cross-operator, driving-based measurement – is what surfaced findings that a single snapshot or a controlled testbed could not. It showed that most of 5G's throughput gains between 2022 and 2024 were driven primarily by expanded carrier aggregation (§3), that SA is overtaking NSA as NSA's peak performance has already plateaued (§4), and that handover configurations remain needlessly inefficient years into deployment (§5). As operators continue rolling out 5G-Advanced features – and as the industry looks ahead to 6G, for which bands such as the upper-mid FR3 are still under active 3GPP study rather than commercially deployed – this same methodology should be extended to characterizing interference as networks densify and reuse spectrum more aggressively, to low-level scheduling and resource-allocation policies more broadly, and to emerging bandwidth-hungry applications such as ultra-HD and volumetric video streaming under mobility. As this thesis repeatedly shows, the gap between a new generation's promise and its operational reality is only visible through direct, real-world measurement – a gap that will not close on its own as the industry moves from 5G to 6G.

From public to private 5G. Every measurement study in this thesis characterizes *public* 5G, shared

across a large, heterogeneous user base under macro-scale mobility. Private 5G (P5G) deployments – dedicated, single-tenant networks operating in enterprise, industrial, and campus settings – are gaining traction as an alternative that offers operators (in this case, the enterprise itself) full control over spectrum, configuration, and coverage. P5G is gaining traction precisely because it offers a degree of customizability that public networks cannot: a single tenant can dedicate spectrum and compute to its own traffic, apply network slicing to isolate and prioritize different classes of traffic (e.g., control signals for robotics versus best-effort data), dynamically reallocate resources to match a small, known set of applications rather than an unpredictable public user base, and enforce tighter, site-specific security and access policies. An important direction building on this thesis is to characterize how much of this customizability is actually exploited in practice, and to extend the configuration-analysis and proactive resource-allocation techniques developed here (§5, §6) to fully exploit the control a private operator has but a public one does not. Because P5G networks typically coexist with enterprise Wi-Fi covering the same physical space, a second direction is convergence: today, the two operate as independent, uncoordinated networks, and an application or device has no principled way to choose between them. Building on the measurement and configuration-analysis techniques developed in this thesis, a natural next step is a unified control plane that selects or steers traffic between P5G and Wi-Fi based on real-time, measured performance rather than static association policy.

Integrating terrestrial and non-terrestrial networks. Our head-to-head comparison of cellular and Starlink's LEO satellite network (§3) shows that the two are complementary rather than competing: Starlink offers near-uninterrupted availability and RTT competitive with (or better than) cellular, while commercial 5G retains a substantial advantage in peak downlink and, especially, uplink throughput. 6G standardization already envisions unifying terrestrial networks (TNs) and non-terrestrial networks (NTNs) – LEO, MEO, and high-altitude platforms – under a single, harmonized architecture rather than the disjoint, separately-managed networks we measured. The complementary performance profile we observe suggests a concrete path toward this goal at the transport layer: a multipath scheduler that steers latency-sensitive or control traffic toward the network with better RTT and bulk or uplink-heavy traffic toward the network with higher throughput, dynamically, based on the kind of continuous, real-world measurement this thesis relies on throughout, rather than a static, single-network association decided once at connection time.

Ultimately, this thesis argues for a simple but consequential principle: commercial cellular networks should be continuously measured, understood, and improved through real-world, measurement-driven inquiry; not assumed to work as specified.

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